



# Balancing Act: Complementary Strategies for Indonesia's Sustainable Nickel Downstream Industry



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# TERMS, ABBREVIATIONS, AND NOTATIONS

|                                                                              |                                                                                                                                                                                                                        |
|------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>ADB (Asian Development Bank)</b>                                          | A regional development bank that provides financial and technical assistance for economic growth in Asia.                                                                                                              |
| <b>AEER (Asia Energy and Environmental Research)</b>                         | A research institute focused on energy and environmental policies in Asia.                                                                                                                                             |
| <b>AMDAL (<i>Analisis Mengenai Dampak Lingkungan</i>)</b>                    | Indonesia's Environmental Impact Assessment (EIA) required for projects with significant environmental impacts.                                                                                                        |
| <b>ASEAN (Association of Southeast Asian Nations)</b>                        | A regional organization promoting economic and political cooperation in Southeast Asia.                                                                                                                                |
| <b>ATR (Agraria dan Tata Ruang) or Agrarian Affairs and Spatial Planning</b> | A ministry that organizes government affairs in the field of agrarian/land and spatial planning.                                                                                                                       |
| <b>B3 (<i>Bahan Berbahaya dan Beracun</i>)</b>                               | Hazardous and Toxic Materials; a category for dangerous waste in Indonesia.                                                                                                                                            |
| <b>B3 (<i>Bahan Berbahaya dan Beracun</i>)</b>                               | Hazardous and Toxic Materials; a category for dangerous waste in Indonesia.                                                                                                                                            |
| <b>BKPM (<i>Badan Koordinasi Penanaman Modal</i>)</b>                        | Indonesia's Investment Coordinating Board, now part of the Ministry of Investment.                                                                                                                                     |
| <b>BPS (<i>Badan Pusat Statistik</i>)</b>                                    | Indonesia's Central Statistics Agency, responsible for national data collection and analysis.                                                                                                                          |
| <b>BRICS (Brazil, Russia, India, China, South Africa)</b>                    | A bloc of emerging economies formed to promote economic cooperation, political dialogue, and development among these fast-growing countries.                                                                           |
| <b>CFPP (Captive Coal-Fired Power Plant)</b>                                 | A coal power plant built exclusively to supply electricity for industrial operations such as nickel smelters.                                                                                                          |
| <b>CBAM (Carbon Border Adjustment Mechanism)</b>                             | A European Union policy imposing carbon tariffs on imported goods to prevent carbon leakage.                                                                                                                           |
| <b>CO<sub>2</sub>e (Carbon dioxide equivalent)</b>                           | A standardized metric for comparing the emissions or impacts of different greenhouse gases.                                                                                                                            |
| <b>CRI (Climate Research Initiative)</b>                                     | A research organization focused on climate impact analysis and policy.                                                                                                                                                 |
| <b>DBH (Dana Bagi Hasil)</b>                                                 | Fiscal transfer to local government from the state revenue, which is allocated based on two main principles: by origin and based on actual revenue. DBH revenues are obtained from tax and natural resources revenues. |
| <b>Deglobalization</b>                                                       | A shift away from globalization, marked by increased economic protectionism and declining international cooperation.                                                                                                   |
| <b>Downstream Processing</b>                                                 | The industrial process of refining raw materials (e.g., nickel ore) into higher-value products such as nickel sulphate for EV batteries.                                                                               |

|                                                                                        |                                                                                                                                                                                                        |
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| <b>ESG (Environmental, Social, and Governance)</b>                                     | A regional development bank that provides financial and technical assistance for economic growth in Asia.                                                                                              |
| <b>EV (Electric Vehicle)</b>                                                           | A vehicle powered by electricity, commonly using nickel-based batteries.                                                                                                                               |
| <b>FPIC (Free, Prior, and Informed Consent)</b>                                        | A principle recognizing the right of indigenous people to make decisions about land use and development projects affecting their territories.                                                          |
| <b>gCO<sub>2</sub>eq/kWh (Grams of CO<sub>2</sub> equivalent per kilowatt-hour)</b>    | A measure of carbon intensity in electricity generation.                                                                                                                                               |
| <b>GEM (Global Energy Monitor)</b>                                                     | A research group tracking energy projects, including fossil fuel and renewable energy developments.                                                                                                    |
| <b>GFC (Global Forest Change)</b>                                                      | A dataset monitoring deforestation and land-use changes worldwide.                                                                                                                                     |
| <b>GHG (Greenhouse Gas Emissions)</b>                                                  | Gases that contribute to global warming, including CO <sub>2</sub> , CH <sub>4</sub> (methane), and N <sub>2</sub> O (nitrous oxide).                                                                  |
| <b>GR (Government Regulation) or PP (Peraturan Pemerintah)</b>                         | A set of rules, laws, and standards imposed by the Indonesian Government                                                                                                                               |
| <b>GtCO<sub>2</sub>eq (Gigatons of Carbon Dioxide equivalent)</b>                      | A unit of measurement used in large-scale climate impact assessments.                                                                                                                                  |
| <b>HPAL (High-Pressure Acid Leach)</b>                                                 | A hydrometallurgical process that uses high-temperature sulfuric acid to extract nickel from laterite ores.                                                                                            |
| <b>IEEFA (Institute for Energy Economics and Financial Analysis)</b>                   | A think tank that analyzes energy market, trends, and policies with a focus on a diverse, sustainable energy economics.                                                                                |
| <b>IESR (Institute for Essential Services Reform)</b>                                  | An Indonesian research institute focused on energy transition and policy.                                                                                                                              |
| <b>IMIP (Indonesia Morowali Industrial Park)</b>                                       | A major nickel mining and processing industrial zone located in Central Sulawesi.                                                                                                                      |
| <b>IUP (<i>Izin Usaha Pertambangan</i>) / Mining Business License</b>                  | A legal permit that allows a company or business entity to carry out mining activities, from exploration, mining, processing, to nickel refining.                                                      |
| <b>IUPK (<i>Izin Usaha Pertambangan Khusus</i>) or Special Mining Business License</b> | A legal permit to conduct mining business in the Special Mining Business Permit Area (WIUPK). IUPK is given to BUMN, BUMD, or private business entities to conduct mining activities in certain areas. |
| <b>IWIP (Indonesia Weda Bay Industrial Park)</b>                                       | A major nickel and smelting industrial zone in North Maluku.                                                                                                                                           |
| <b>KLHK (<i>Kementerian Lingkungan Hidup dan Kehutanan</i>)</b>                        | Indonesia's Ministry of Environment and Forestry (MoEF).                                                                                                                                               |

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| <b>kWh (Kilowatt-hour)</b>                                             | A unit of energy consumption.                                                                                                                                                                                                                  |
| <b>LCOE (Levelized Cost of Electricity)</b>                            | The average cost of generating electricity over a power plant's lifetime, expressed in USD/kWh.                                                                                                                                                |
| <b>LCR (Local Content Requirement)</b>                                 | Regulations requiring companies to use a certain percentage of local materials, labor, or components in the production of goods or services.                                                                                                   |
| <b>MODI (Minerba One Data Indonesia)</b>                               | Indonesia's mining permit and operational data platform.                                                                                                                                                                                       |
| <b>MoEF (Ministry of Environment and Forestry)</b>                     | The Indonesian government agency overseeing environmental regulations.                                                                                                                                                                         |
| <b>MoEMR (Ministry of Energy and Mineral Resources)</b>                | An Indonesian ministry responsible for providing assistance to the President and Vice President in performing government's affairs in the field of energy and mineral resources.                                                               |
| <b>Multilateral Agreement</b>                                          | A treaty or framework signed by multiple countries to address global issues like climate change and trade.                                                                                                                                     |
| <b>MW (Megawatt)</b>                                                   | A unit of power capacity, commonly used for electricity generation projects.                                                                                                                                                                   |
| <b>NPI (Nickel Pig Iron)</b>                                           | A low-grade ferroalloy used in stainless steel production, primarily made from laterite nickel ore.                                                                                                                                            |
| <b>Nickel-rich Region</b>                                              | A region with at least one nickel mining concession.                                                                                                                                                                                           |
| <b>OECD (Organization for Economic Cooperation and Development)</b>    | An international body promoting policies for economic growth and stability.                                                                                                                                                                    |
| <b>OSH (Occupational Safety and Health) standards</b>                  | Legally mandated rules designed to protect workers from hazards in the workplace. They are developed and enforced by agencies like OSHA (Occupational Safety and Health Administration) and the ILO (International Labour Organization).       |
| <b>PAD (Pendapatan Asli Daerah)</b>                                    | Local own-source government revenue, including local tax and local retribution.                                                                                                                                                                |
| <b>PD (Presidential Decree) or Keppres (<i>Keputusan Presiden</i>)</b> | Legal norms made by the President to regulate matters that are concrete, individual, and completed once. Presidential Decrees usually regulate matters that are specific and apply to parties or objects mentioned in the Presidential Decree. |
| <b>PLN (<i>Perusahaan Listrik Negara</i>)</b>                          | Indonesia's State Electricity Company, responsible for electricity generation and distribution.                                                                                                                                                |
| <b>Populism</b>                                                        | A political ideology that emphasizes national interests over international cooperation, often opposing globalization.                                                                                                                          |
| <b>PPP (Public-Private Partnership)</b>                                | A collaboration between government and private entities to develop infrastructure and services.                                                                                                                                                |

|                                                                                                                                       |                                                                                                                                                                                                                                                                      |
|---------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>PR (Presidential Regulation) or Perpres (Peraturan Presiden)</b>                                                                   | A legal regulation stipulated by the President to implement the orders of higher legal regulations or to exercise governmental powers.                                                                                                                               |
| <b>PSN (<i>Proyek Strategis Nasional</i>)</b>                                                                                         | Indonesia's National Strategic Projects, designated for rapid development.                                                                                                                                                                                           |
| <b>REDD+ (Reducing Emissions from Deforestation and Forest Degradation Plus)</b>                                                      | A global climate initiative to reduce emissions from deforestation.                                                                                                                                                                                                  |
| <b>Regional Governance</b>                                                                                                            | A governance model where decision-making and policies are managed at a regional level rather than centrally.                                                                                                                                                         |
| <b>Reshoring &amp; Friendshoring</b>                                                                                                  | Relocating industrial production back to domestic markets (reshoring) or to politically aligned countries (friendshoring).                                                                                                                                           |
| <b>RPTKA (Rencana Penggunaan Tenaga Kerja Asing) or Foreign Workers Utilization Plan</b>                                              | A document required by the Indonesian government for companies intending to employ foreign workers.                                                                                                                                                                  |
| <b>SCC (Social Cost of Carbon)</b>                                                                                                    | The estimated economic damage caused by emitting one ton of CO <sub>2</sub> into the atmosphere.                                                                                                                                                                     |
| <b>Smelting</b>                                                                                                                       | The process of extracting metals from ores through heating and melting.                                                                                                                                                                                              |
| <b>TSM (Towards Sustainable Mining)</b>                                                                                               | A global initiative to improve the sustainability of mining practices.                                                                                                                                                                                               |
| <b>UMD (University of Maryland)</b>                                                                                                   | A research institution producing global datasets on deforestation and climate change.                                                                                                                                                                                |
| <b>USD/kWh</b>                                                                                                                        | The cost of electricity per kilowatt-hour in USD, often used in LCOE calculations.                                                                                                                                                                                   |
| <b>UU PPLH (<i>Undang-Undang Perlindungan dan Pengelolaan Lingkungan Hidup</i>) or Law on Environmental Protection and Management</b> | A law that regulates systematic and integrated efforts to preserve environmental functions and prevent environmental pollution or damage.                                                                                                                            |
| <b>WIUPK (Wilayah Izin Usaha Pertambangan Khusus) or Special Mining Business Permit Area</b>                                          | Areas given to holders of IUP (Mining Business Permit) or SIPB (Rock Mining Permit) to carry out mining business activities. WIUP is determined by the government and can cover land, sea or seabed. This permit is usually awarded through an open auction process. |
| <b>WACC (Weighted Average Cost of Capital)</b>                                                                                        | A financial metric that calculates the average cost of financing investments. It helps companies in calculating LCOE.                                                                                                                                                |
| <b>WTO (World Trade Organization)</b>                                                                                                 | The international body regulating trade between nations.                                                                                                                                                                                                             |

# EXECUTIVE SUMMARY

## IMPACT ASSESSMENT

### ■ Economic Aspect

- Current Nickel Downstream Offers Limited National Economic Contribution:** Indonesia has implemented two significant nickel ore export bans. Despite the substantial growth in nickel-related industries following the second ban, our preliminary analysis suggests that the current nickel downstream activities have not generated sufficient value-added to meet Indonesia's need for higher growth needs. Relying solely on nickel downstream will not be enough to ensure sustainable growth for Indonesia's economic transition.
- Current Suboptimal Gains from Nickel Downstream:** Indonesia's downstream activities are still primarily focused on lower value-added process, such as steel-making manufacturing, rather than higher value-added process like electric vehicle (EV) manufacturing. In 2023 Indonesia's ferronickel exports (steel-making process) were approximately three times the combined exports of nickel mattes and nickel sulfate (EV-making process). In addition, since many nickel mining and industrial firms are not domestically owned, it is estimated that around 90% of the value-added creation of nickel is not distributed to the people of Indonesia.
- Nickel Downstream Has Larger Local Economic Impact:** Nickel mining and industry activities have significantly boosted the local economy. Between 2021-2023, several nickel-producing provinces, such as North Maluku and Central Sulawesi, experienced high economic growth compared to the period from 2014 to 2019. North Maluku's growth increased from 6.73% (2014-2019) to 20.07% (2021-2024), while Central Sulawesi's growth rose from 11.17% (2014-2019) to 12.94% (2021-2024). The high growth of metal ore mining and metal industry, especially in North Maluku, supported the overall economic growth.
- Nickel Downstream Does Not Guarantee Local Structural Transformation:** Central Sulawesi and North Maluku have seen a significant increase in their share of metal ore mining and basic metal manufacturing, which now account for approximately 40% of their total GDRP. However, the agriculture sector in these nickel-rich provinces continues to grow, albeit at a declining rate since 2021, particularly in Southeast Sulawesi and North Maluku.
- Nickel Downstream Improves Local Fiscal Revenue:** PAD revenue has more than tripled from 2011 to 2023 in the nickel-rich provinces of Central Sulawesi, Southeast Sulawesi, and North Maluku. Additionally, DBH revenue in these nickel-rich provinces has increased significantly, rising 6-8 times from 2019 to 2023, largely due to mineral and coal DBH.



- **North Maluku Experiences a Sharp Employment Increase:** The surge in manufacturing employment in North Maluku indicates a positive correlation between nickel manufacturing activities and increased regional employment. This significant expansion in manufacturing employment is likely driven by the operations of the Indonesian Weda Bay Industrial Park (IWIP) in North Maluku, which employed over 24,000 workers in 2021.

## Environmental Aspect

- **Deforestation and Habitat Loss:** Nickel mining has led to the clearing of 76,031 hectares of forest, with concessions covering 1,037,435 hectares, 68% of which overlap with forests in 2022. A study by the UMD (University of Maryland) estimated that 153,364 hectares were lost between 2001 and 2022. Indonesia accounts for 26.4% of global mining-related deforestation (369,356 hectares from 2001 to 2020), with 41.5% attributed to nickel mining.
- **Unregulated Mining and Smelter Expansion:** Illegal mining exacerbates deforestation, with at least ten unregistered mining sites in East Halmahera. Smelters are also accelerating forest loss, with deforestation near smelters doubling from 33 km<sup>2</sup> to 63 km<sup>2</sup> per smelter, a 234% rise in a decade. Weda Bay Nickel (IWIP) operates in a 54,874-hectare concession, with 65% in protected mangrove forests.
- **Carbon Storage and Biodiversity Risk:** Nickel expansion poses a significant threat to High Carbon Stock (HCS) forests, resulting in the loss of 2.04 million metric tons of carbon storage in Halmahera alone. Additionally, six of the top 25 nickel deforestation concessions impact 27,693 hectares of Key Biodiversity Areas (KBAs), endangering species such as the Sulawesi hornbill.
- **Marine Ecosystem Damage:** Coastal nickel smelting is causing significant destruction to mangroves and coral reefs, with land reclamation at IWIP exacerbating biodiversity loss. Indonesia, as part of the Coral Triangle, houses 30% of the world's coral reefs and 76% of global coral species, making nickel smelting a major threat to marine ecosystems.
- **Captive Coal Power Dependency:** 67% of nickel smelter energy comes from captive coal-fired power plants (CFPPs), making Indonesia the fifth largest globally in captive CFPP capacity (9,815 MW). Sulawesi and Maluku host the highest CFPP concentrations, emitting 47 million tons of CO<sub>2</sub> annually, with an estimated social carbon cost of USD 522 million.
- **Nickel Smelting Emissions:** Antam, MBMA, TBP Harita, and Vale produced 353,000 tons of nickel, emitting 15.3 million tons of CO<sub>2</sub> in 2023. Vale, using hydropower, had the lowest intensity (28.7 tCO<sub>2</sub>/ton), while coal-powered smelters emitted 56.9–69.9 tCO<sub>2</sub>/ton. With production set to triple by 2028, emissions could reach 38.5 million tons (4.5% of Indonesia's total emissions). Switching to renewables could cut emissions by 43%, but limited PLN grid access forces reliance on captive CFPPs.

- **Severe Air Pollution Risks:** Smelters and captive power plants in Central Sulawesi, Southeast Sulawesi, and North Maluku are projected to exceed WHO air quality limits by 2030. NO<sub>2</sub> levels are expected to surpass 30 µg/m<sup>3</sup>, while PM<sub>2.5</sub> and SO<sub>2</sub> concentrations will rise beyond safe thresholds. Mercury and heavy metals from coal combustion pose severe threats to marine biodiversity, with 3,700 kg of mercury deposition expected annually, affecting protected areas like Aketajawe Lolobata National Park.
- **Water Pollution Crisis:** Water contamination has surged since the 2020 export ban in many villages in resource-rich nickel regions. Industrial pollution has risen sharply in nickel-smelting areas, from 0% to 71% in North Konawe, 59% in Morowali, and 61% in Central Halmahera. Groundwater depletion is severe in Wawonii and Gebe Islands, while Akejira, Ake Kobe, and Sageyan Rivers show heavy pollution from erosion and deforestation.
- **Toxic Waste to Ocean:** The High-Pressure Acid Leach (HPAL) process generates 32 million m<sup>3</sup> of toxic wastewater annually in IWIP, containing hexavalent chromium and high temperatures, threatening coral reefs, mangroves, and seagrass beds. Coastal waters in Boedingi, Obi Island, and Morowali have turned reddish-brown and warmer, harming local fisheries and marine ecosystems.

## Social Aspect

- **Limited Formal Employment Generation:** Despite claims of local prosperity through job creation, nickel mining's capacity to shift employment towards the formal sector remains doubtful. Historically, the mining and quarrying sector has been a major source of informal work in Indonesia; and employment increases were observed following nickel export restrictions. However, nickel manufacturing, particularly within the Indonesian Weda Bay Industrial Park (IWIP) in North Maluku, shows promise for expanding formal manufacturing employment.
- **Increase in Informal Non-agricultural Employment:** North Maluku's manufacturing employment surpassed Sulawesi provinces, driven by nickel manufacturing and value addition policies. However, the establishment of the Indonesian Weda Bay Industrial Park (IWIP) in North Maluku has led to a rise in informal employment in non-agricultural sector, which raises concerns. Central Sulawesi has shown an increase in informal non-agricultural employment, challenging the expectation that new manufacturing firms would lead to more formal job opportunities. This paradox suggests a labor reallocation from agriculture to non-agriculture sectors due to the adverse effects of mining and manufacturing.
- **High Unemployment Rate:** Despite the growth in nickel mining, unemployment remains high in these regions. The mining sector attracts many workers from other areas, increasing the labor pool. This influx saturates the market, limiting job availability and maintaining high unemployment despite mining expansion.

- **Limited Impact of Nickel Mining on Poverty:** The initial export restriction had a greater impact on poverty in mining-dominated regions, while the second ban compelled domestic manufacturing firms to sustain operations and employment, especially in regions with both nickel mining and manufacturing. Regions with substantial nickel mining and manufacturing outputs, such as Morowali, North Morowali, and Central Halmahera, showed a decline from median poverty rates within three years of the second policy. Areas with little to no nickel manufacturing activity did not exhibit similar improvements, except for East Halmahera. This suggests that nickel mining alone is insufficient to alleviate poverty due to negative externalities in agriculture.
- **Rising Agrarian Conflicts:** Nickel mining and processing in Indonesia raise concerns about social costs related to unethical and unsafe practices. Land-grabbing infringes on Indigenous communities' rights through displacement and lack of free consent. Reports reveal inequitable land sales, intimidation, and compulsory excavation in Halmahera, with communities uninformed about land acquisition purposes. Agrarian conflicts in nickel extraction areas are rising, impacting thousands of families due to regulatory modifications and the classification of nickel mining companies under National Strategic Projects, leading to contentious land procurement.
- **Neglected Labor Welfare:** Unsafe working conditions within nickel manufacturing companies are a major concern, with numerous workplace incidents causing fatalities and injuries between 2015 and 2023. Accidents such as toxic gas leaks highlight systemic vulnerabilities in occupational safety practices and regulatory audits. Problematic labor arrangements, including inadequate wages, excessive hours, and arbitrary dismissals, worsen the situation. Discrimination against female employees, such as barriers to menstrual leave and workplace harassment, further undermine worker welfare, leading to ethical concerns and the withdrawal of investors from nickel projects in Indonesia.

## POLICY ANALYSIS

### Economic Aspect

- **Ease of Doing Business:** Indonesia needs to attract productive investment, which requires a regulatory streamlining to reduce investment barriers and bureaucratic hurdles. This initiative has been attempted with the formulation of Job Creation Law (Law No. 6 of 2023). This law has regulated several areas to attract more investment: investment climate and business activities; research and innovation; land procurement; economic zones; and national strategic project. This job creation law also affects the ease of doing business in nickel mining and industries.

- **Nickel Ore Export Ban:** In 2014, Indonesia has imposed the first export ban on nickel ore based on Law No.4 of 2009 on Mineral and Coal. This ban was then relaxed in 2017. However, at the end of 2019, the government implemented a stronger measure by imposing Law No.3 of 2020 on Mineral and Coal. Since the second export ban, nickel-related sectors have generated significant economic value-added creation. Despite this, challenges remain in the implementation, including unfair market structure, trade retaliations and disputes, and suboptimal value-added creation.
- **Resource-based Industrialization Strategies:** Beyond the export ban, Indonesia has formulated local content requirements (LCR) to develop its domestic electric vehicle (EV) ecosystem. The Presidential Regulation (PR) No. 79/2023 (*Peraturan Presiden 79/2023*), which revises the more ambitious targets set in the PR No. 55/2019 (*Peraturan Presiden 55/2019*), mandates a 40% of LCR for EV cars in 2026, 60% in 2029, and 80% thereafter. Additionally, to provide more business certainty, Indonesia has established special economic zones (SEZ) and national strategic projects (*Proyek Strategis Nasional/PSN*) with special incentives and sound institutional mechanisms to attract more investors to nickel-based industries.
- **Revenue Sharing Funds:** Indonesia has institutionalized the revenue sharing mechanism to regional governments through the Law No. 1/2022 (*Undang-Undang No.1/2022*). With a high revenue sharing rate to provincial and cities/regencies-level government (80%), these additional revenues should be utilized productively. Effective revenue management is necessary, considering the significant rise in local fiscal revenues from nickel-related activities. The use of revenue should be regulated for several allocations, including the expansion of productive sectors, compensation for affected people and/or sectors due to downstream process, and preparation for future transition.

## Environmental Aspect

- **Permission and Environmental Assessment (AMDAL):** Indonesia's regulatory framework, particularly through the Omnibus Law No. 6/2023, has significantly weakened environmental safeguards, allowing nickel mining projects to proceed with minimal oversight. Centralizing risk assessments, permitting, and environmental approvals under the central government has reduced local government authority, limited public participation, and allowed for accelerated business permits, despite non-compliance with environmental regulations.
- **Forest Protection:** The Omnibus Law and its supporting regulations have significantly weakened forest protection in Indonesia, particularly for nickel downstream activities. Key provisions, such as the removal of DPR approval for forest designation changes, the simplification of permits into a single "business permit", and the centralization of authority in the central government, have facilitated easier access for both domestic and foreign companies to secure forest concessions. These regulatory changes, along with expedited permitting and reduced financial obligations, prioritize economic expansion over environmental sustainability, leading to increased deforestation and biodiversity loss.

- **Emission and Pollution Control:** Indonesia's regulatory framework for pollution control remains weak, particularly in the nickel downstream sector, where coal-fired power plants and Deep-Sea Tailings Placement (DSTP) are still permitted despite their environmental risks. Environmental standards are significantly lower compared to international benchmarks, and weakened laws such as the removal of criminal sanctions for hazardous waste handling reduce accountability. Additionally, insufficient monitoring and the lack of a precautionary approach to marine pollution exacerbate the environmental damage, particularly in high-risk regions like Obi Island and IMIP.

## Social Aspect

- **Foreign Worker Permit:** To streamline the employment of foreign labor, the Indonesian government eliminated the need for companies to obtain the IMTA (Permit to Employ Foreign Workers) and instead required only the legalization of the RPTKA (Foreign Workers Utilization Plan). This shift could weaken the regulatory structure, as the RPTKA is merely an administrative procedure compared to the IMTA's function as a tangible permit and inspection document. The absence of a mandatory IMTA may increase the number of low-skilled and undocumented foreign workers, intensifying competition for local jobs and potentially exacerbating social tensions.
- **Land Dispute:** The Indonesian government listed nickel processing facilities and industrial parks such as IMIP, IWIP, and smelters as national strategic projects (PSN) under Presidential Decree No. 109 of 2020, providing them with state facilitation, including land acquisition dispute resolution. However, numerous land disputes suggest inconsistent implementation. Furthermore, GR No. 64 of 2021 on Land Banking is being criticized for expanding government control over land and easing foreign corporate access, risks exacerbating land conflicts, displacing local communities, and undermining equitable land governance.
- **Occupational Safety and Health Standard:** Despite the state mandating adherence to MEMR Regulation No. 26 of 2018, which requires safety equipment, safety governance, and regular audits in both nickel mining and processing, there are persistent workplace hazards and substandard work environments. This indicates a critical deficiency in implementing Occupational Safety and Health (OSH) standards, alongside inadequate regulatory enforcement and monitoring.

## POLICY RECOMMENDATION

### Economic Aspect

- **Supporting higher value-added resource-based industrialization:** Indonesia needs to promote industrial policies in developing an EV supply chain ecosystem. This industrial strategy, however, have to consider the short, medium, and long-term feasibility.
- **Improving domestic benefits from downstream activities:** Indonesia—along with other nickel-rich economies—needs to establish a comprehensive international cooperation to mitigate unfair monopolistic market structure. Additionally, Indonesia should reevaluate and diversify its nickel industry ownership by considering more domestic ownership or diversification with investment from other countries.
- **Preparing for future transition:** Indonesia needs to develop a comprehensive transition framework to address the depletion of nickel reserves and the technological advancements that may replace nickel. Nickel-rich regions should also diversify their economies to reduce reliance on nickel activities.
- **Managing subnational fiscal revenue and spending:** Indonesia needs to establish a local investment fund to mitigate the commodity price volatility and the depletion of nickel reserves. Resource revenues should be allocated to long-term intergenerational savings, compensation for affected people and sectors due to downstream process, future transition preparation, and productive spending.

### Environmental Aspect

- **Improving environmental governance in Indonesia's nickel industry:** The government should consider shortening mining concession durations, ensuring transparent bidding processes, and strengthening AMDAL procedures with rigorous screenings, public participation, and legal protections for environmental defenders. Additionally, improving data transparency for permits and environmental assessments, empowering local governments for better oversight, and collaborating with China to implement green practices in Belt and Road Initiative (BRI) projects, will help ensure sustainable development in line with global environmental standards.
- **Strengthening forest protection:** The government should repeal corporate amnesty provisions in the Omnibus Law, restore local government oversight and public participation, and reinstate limits on forest conversion, especially for protected and High Carbon Stock areas. Additionally, enforcing reclamation and post-mining restoration with stricter penalties for non-compliance is essential to ensure sustainable forest management and prevent environmental degradation.



- **Supporting higher value-added resource-based industrialization:** Indonesia needs to promote industrial policies in developing an EV supply chain ecosystem. This industrial strategy, however, have to consider the short, medium, and long-term feasibility.
- **Improving emission and pollution control:** The government should advocate for cleaner energy sources for smelters, prohibit Deep Sea Tailings Placement (DSTP), and enforce stricter environmental standards that align with global benchmarks. This includes bolstering the monitoring and enforcement of pollution controls, mandating PROPER participation for high-risk industries, and increasing transparency in corporate environmental practices to ensure accountability and reduce environmental harm.

## Social Aspect

- **Preventing local job displacement:** To address the influx of low-skilled foreign workers, complementary provisions, such as quotas and monitoring mechanisms, should be added to the RPTKA. These measures could prevent worsening social tension between local and foreign workers and encourage firms to prioritize entry for higher skilled workers.
- **Enhancing transparency in land dispute resolution:** The government should more effectively facilitate resolutions between nickel manufacturing facilities, especially those listed as PSNs, and landowners. A publicly accessible dashboard providing real-time updates on ongoing conflicts, resolution status, affected households, and outcomes is recommended to boost accountability.
- **Enforcing occupational safety and health (OSH) standards:** The government should establish clear mechanisms to incentivize compliance in nickel mining and manufacturing. This includes impartial violation investigations, escalating penalties for non-compliance, and restricting companies from engaging in image-improvement activities when OSH violations or cover-ups of workplace accidents are evident.

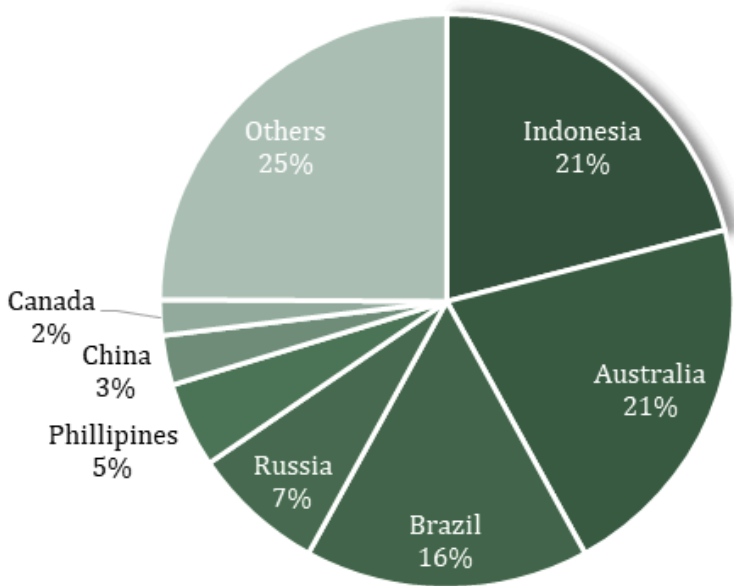
# CHAPTER 1

## I. Introduction

### 1.1. Background

Indonesia is rich in geological and mineral resources, making it a significant player in the global mining trade. The country boasts the world’s largest reserves of nickel and tin, and ranks sixth globally for coal and bauxite reserves, and seventh for copper (MEMR, 2021). This wealth of minerals positions Indonesia as a crucial contributor to the growing global demand for these commodities (Figure 1.1). Nickel, in particular, has seen a surge in demand, largely driven due to its essential role in the rapidly expanding electric vehicle (EV) industry (Figure 1.2). Over the past five years, global nickel demand has grown at an annual rate of 10%, increasing from 2.44 million tons in 2019 to 3.61 million tons in 2023 (Statista, 2022).

**FIGURE 1.1** Share of World Nickel Reserves



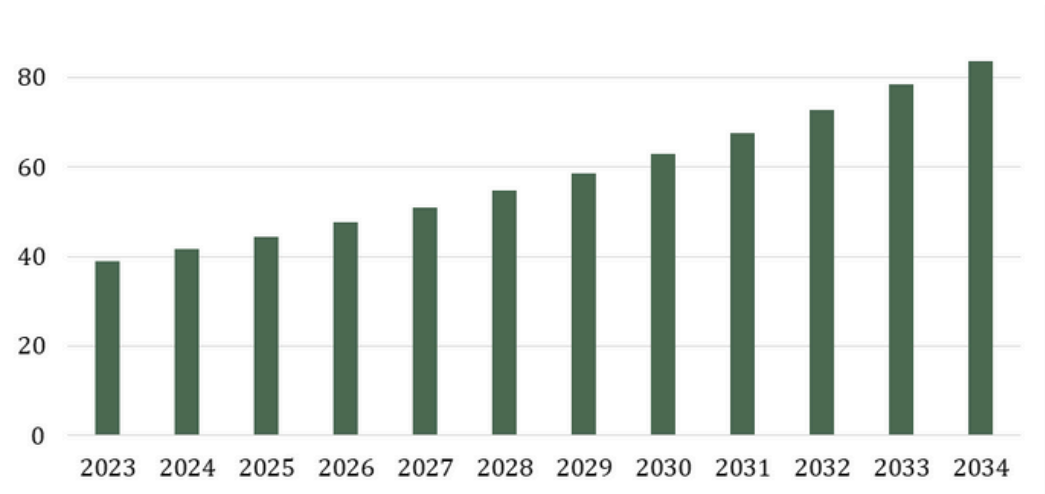
Source: Statista (2024)

Recognizing the strategic importance of nickel, Indonesia has prioritized downstreaming to maximize its mineral wealth and support the Golden Indonesia 2045 Vision (Kominfo, 2023). Under Jokowi’s administration, key initiatives included industrial zones like Morowali Industrial Park and regulatory frameworks such as Law No. 3/2014 on Industry, Law No. 3/2020 on Mineral and Coal Mining, and Law No. 6/2023 on Job Creation. The 2020 nickel ore export ban further solidified Indonesia’s commitment to domestic processing, enabling production of Nickel Pig Iron (NPI), ferronickel, and battery-grade nickel (Yayasan Indonesia CERAH, 2024). Prabowo’s administration continues this momentum, targeting 8% economic growth by reinforcing industrialization and investment. The rebranding of Badan Koordinasi Penanaman Modal (BKPM) or Indonesia Investment Board into the Ministry of Investment and Downstreaming highlights this strategic focus.

Strengthening its role in the EV battery supply chain, Indonesia aims to leverage its nickel resources for sustainable and inclusive economic growth (Sangadji & Ginting, 2023; CREA & CELIOS, 2024; Deloitte Indonesia, 2022).

Looking at economic impact, nickel downstream strategy has significantly boosted exports, generating over USD20 billion annually despite losing USD1 billion in potential ore exports (Guberman et al., 2024). However, the economic benefits are unevenly distributed. Despite high regional growth—20.49% in North Maluku and 11.91% in Central Sulawesi (2023)—poverty reduction and employment gains remain limited, with only a 2% poverty rate drop in Central Sulawesi since 2015 (BPS Sulawesi Tengah, 2023). Meanwhile, agriculture and fisheries face projected losses of IDR6 trillion over 15 years due to pollution and land degradation (CREA & CELIOS, 2024). With nickel reserves potentially depleting within 12 years (Yayasan Indonesia CERAH, 2024), the long-term sustainability of this strategy is uncertain. To ensure long-term economic sustainability, it is essential to enhance the downstream process into a higher value-added stage.

**FIGURE 1.2** Global Demand Nickel Value Projection (USD Billion)



Source: Precedence Research (2024)

In the other side, nickel downstreaming has led to significant environmental damage, including deforestation, river pollution, and high carbon emissions. Between 2000 and 2023, over 76,000 hectares of forest were lost in Central and Southeast Sulawesi, largely due to illegal mining activities (Prakarsa, 2024). Smelting operations have contaminated rivers, degrading water quality and disrupting ecosystems (CREA & CELIOS, 2024). The industry’s reliance on coal-fired power plants makes it a major carbon emitter, with nickel matte production emitting 59 tCO<sub>2</sub>e per ton, exceeding the global benchmark of 48 tCO<sub>2</sub>e (Sani, 2024). Pollution from the sector is projected to cause 3,800 premature deaths and USD2.63 billion in health-related costs by 2025 (CREA & CELIOS, 2024).

Socially, nickel mining and downstream activities in Indonesia involve unethical land acquisition and unsafe working conditions, imposing high social costs. Companies engage in land grabbing, forced displacement, and intimidation, violating indigenous rights and neglecting Free, Prior, and Informed Consent (FPIC) (Climate Rights International, 2024). In North Maluku, locals were pressured into unfair land sales, often unaware of the acquisition's purpose. Meanwhile, poor Occupational Safety and Health (OSH) standards have led to 68 workplace accidents (2015–2022), 47 deaths, and 76 injuries from toxic gas exposure, excluding recent smelter explosions (Solidar, 2024). Adding to the problem is the general concern about work arrangements that compromises workers rights such as lack of decent wage, extra-long working time and arbitrary dismissal from the company (INKRISPENA, 2022). Without reforms, social and human rights violations will continue to undermine the industry's sustainability.

Given the rapid development of Indonesia's nickel industry and its prominence on the national agenda, it is crucial to assess its impact on economic, social, and environmental aspects. This evaluation will provide a foundation for formulating well-crafted policies that enhance sustainable practices, maximize value-added production, and ensure the protection of environmental and social welfare.

## 1.2. Objectives

Given the Indonesian government's strong commitment to advancing downstreaming, particularly in the nickel industry, this study aims to achieve the following objectives:

- **To analyze the current impact of nickel downstream on the triple bottom line:** This involves evaluating its economic, environmental, and social outcomes, assessing how well the industry contributes to economic growth, limits environmental degradation, and promotes social equity.
- **To assess the risk (potential disadvantage) and opportunity (potential benefit) of current government policies:** This includes examining whether existing policies effectively support sustainable nickel downstream by reviewing their frameworks and enforcement mechanism.
- **To identify necessary policy improvements to enhance the sustainability of nickel downstream activities:** By highlighting deficiencies in current policies, the research will propose policy recommendations to ensure that nickel downstream not only drives economic growth but also aligns with environmental protection and social well-being.

## 1.3. Research Scope

The analysis focuses on Southeast Sulawesi, Central Sulawesi, and North Maluku, which host around 90% of Indonesia's nickel reserves (Naryono, 2023) and serve as key hubs for nickel mining and smelting operations. This research narrows its scope to nickel-rich regions, which is further split into nickel mining areas<sup>1</sup> and nickel mining and manufacturing areas.<sup>2</sup> The categorization of the region into these areas is based on the presence of nickel mining and manufacturing companies, and the magnitude of regional GRDP in mining and manufacturing. Regions are grouped

into nickel mining areas when there is a presence of nickel mining companies operating in this region based on MEMR database.<sup>3</sup> These regions exhibit higher GRDP in mining and quarrying compared to regions with no presence of nickel companies, although this may not always be the case when other regions have mining of other minerals. Regions are grouped into nickel mining and manufacturing category if there is a presence of nickel mining and nickel processing firms. Additionally, this category exhibits higher manufacturing output compared to regions with nickel mining only, which is indicative of the intensity of the nickel processing activities.

Due to the data availability, this study will use the data from metal category from the Statistics Indonesia database (BPS) to proxy for the impact of nickel in the economic section. Additionally, while Indonesia's nickel industry involves various policy dimensions, this study will not cover all aspects of nickel-related policymaking. Instead, it focuses on specific laws and regulations that directly impact the economic, social, and environmental issues examined in this analysis.

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1 Southeast Sulawesi: Konawe, Kolaka; Central Sulawesi: Banggai, Tojo Una-una; North Maluku: East Halmahera, North Halmahera

2 Southeast Sulawesi: South Konawe, Bombana, North Kolaka, North Konawe; Central Sulawesi: Morowali, North Morowali; North Maluku: Central Halmahera, South Halmahera

3 The MEMR's geoportal database is accessible through <https://geoportal.esdm.go.id/minerba/>

# CHAPTER 2

## II. Impact Assessment

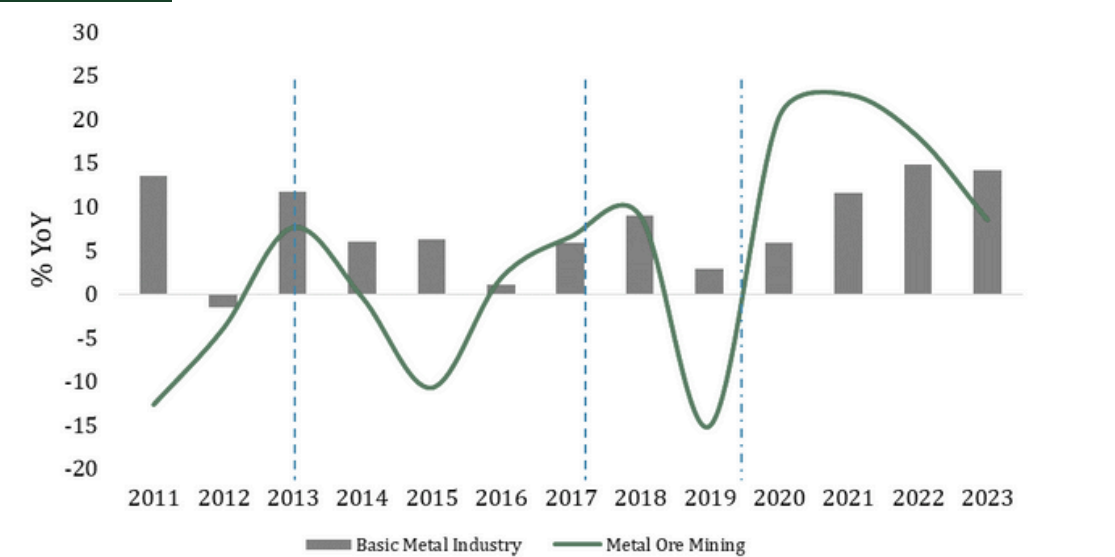
This chapter explores the impact assessment of nickel downstream across three crucial dimensions: economic, social, and environmental. Given the wide-ranging effects of nickel downstream, it is not feasible to cover every potential impact in these areas. Instead, the focus will be on the most significant and relevant impacts that are likely to influence the future trajectory of Indonesia's nickel industry. By concentrating on these key impacts, this chapter provides an in-depth understanding of how the nickel downstream sector interacts with and affects various aspects of society, the economy, and the environment.

Additionally, the impact assessments serve as the foundation for identifying the critical policy gaps that need to be addressed. Through a detailed exploration of the most pertinent impacts in each of the three dimensions, this chapter sets the groundwork for discussing the necessary policy reforms and strategies needed to mitigate risks and enhance the positive outcomes of nickel downstream for Indonesia.

### 2.1. Result and Analysis of Economic Impact

In the last two decades, Indonesia has implemented two significant nickel-export bans. Driven by the ambition to secure more domestic value-added from its resource endowment, Indonesia

**FIGURE 2.1** Sectoral National Growth

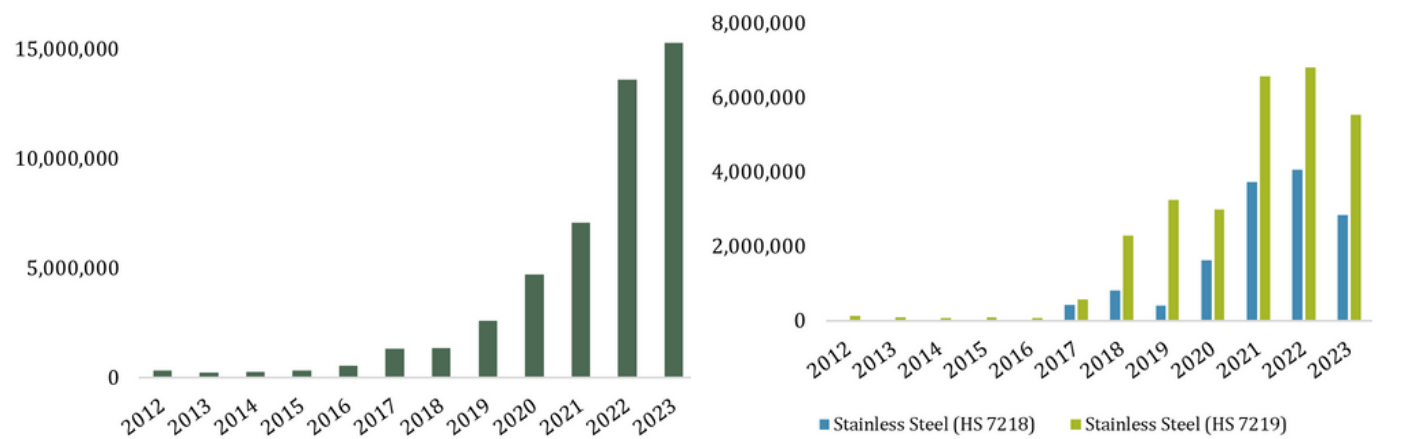


Source: BPS (2024)

imposed stricter rules for its nickel-export policies at the end of 2019. This measure has successfully boosted the growth of nickel-related mining and industries (Figure 2.1). Despite the rapid surge in nickel-related industries since the second nickel ore export ban, downstream mineral activities contribute less than 5% of the domestic economy. This suggests that relying on Indonesia's current progress in nickel downstream will not be sufficient to transition the country's economy.

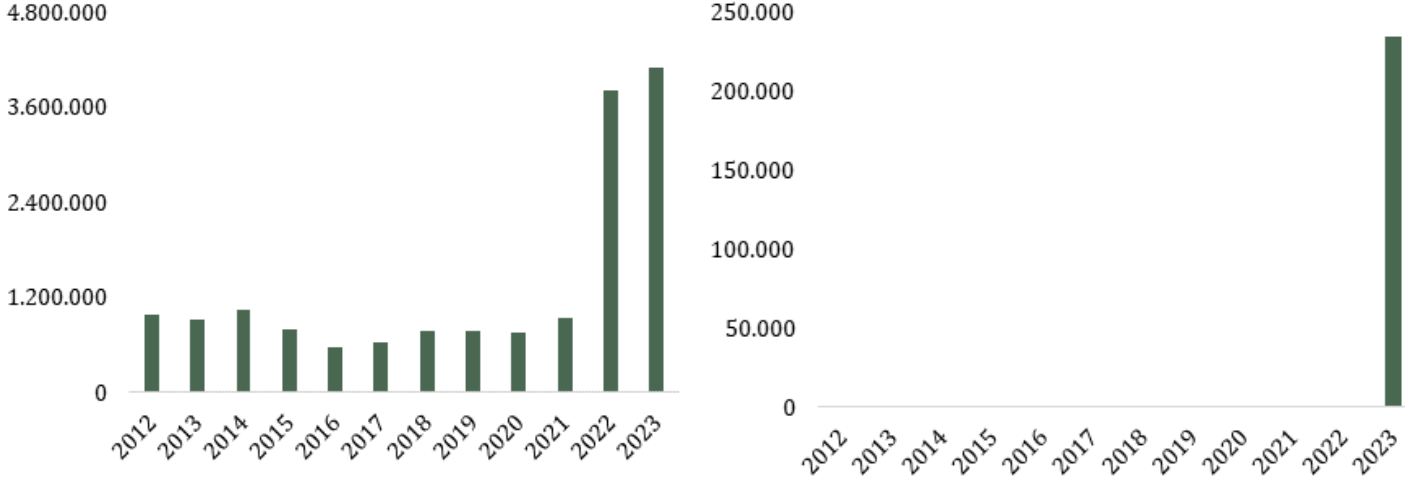


**FIGURE 2.2** Nickel in Stainless Steel Supply Chain (USD Thousand); Ferronickel (left hand side) and Stainless Steel (right hand side)



Source: Trade Map (2024)

**FIGURE 2.3** Nickel in EV Supply Chain (USD Thousand); Nickel Mattes (left hand side) and Nickel Sulfate (right hand side)

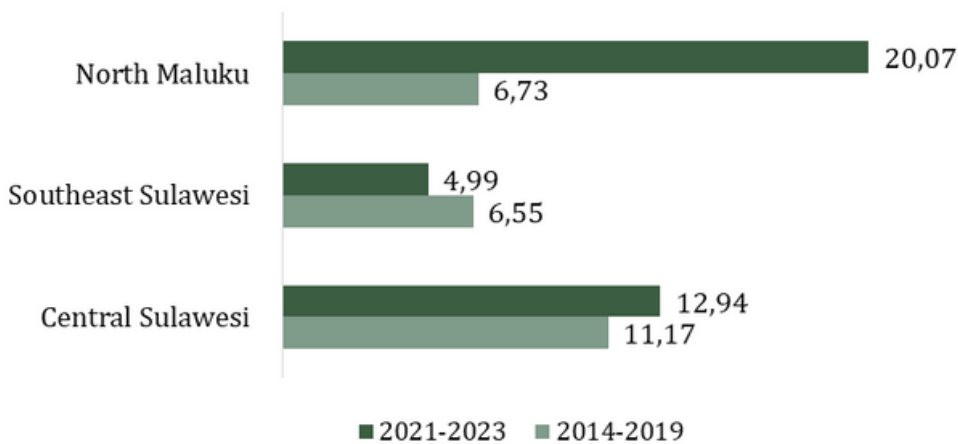


Source: Trade Map (2024)

Indonesia has not been maximizing its potential revenue from nickel resources extraction. Currently, the country 's current downstream activities is still heavily focused on lower value-added products. Figures 2.2 and 2.3 illustrate that Indonesia exports more on lower value-added products (such as stainless steel-related products) relative to higher value-added products (such as EV battery-related products). In 2023, ferronickel exports were approximately three times the combined exports of nickel mattes and nickel sulfate. In addition, since most of Indonesia's nickel downstream products are produced by foreign investors, it is estimated that around 90% of the value-added creation does not remain in Indonesia (CNN Indonesia, 2023). Indonesia's reliance on a few countries (e.g. China) for its nickel downstream process and market necessitates further diversification of foreign investment partners to enhance competition and transparency in the nickel commodity market and partnerships.

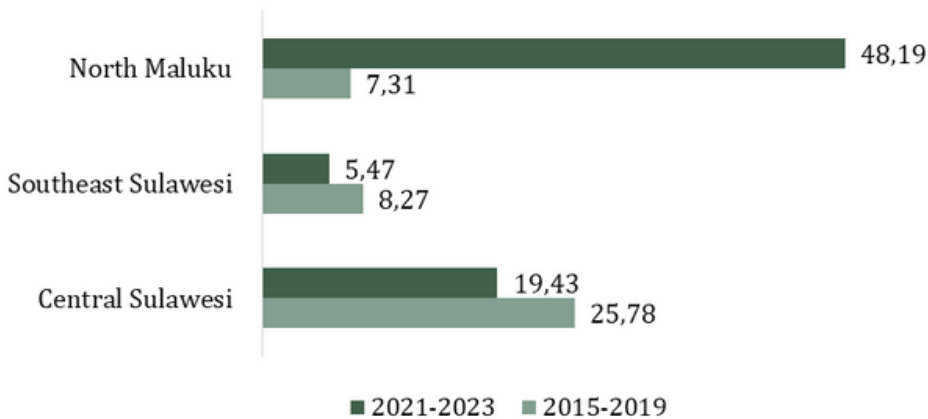
Although nickel downstream has a relatively small impact on the national economy, it plays a significant role at the regional level. On a macro level, nickel mining and industry activities have greatly expanded the local economy. Since 2021, several nickel-producing provinces (North Maluku and Central Sulawesi) had performed a high economic growth relative to the previous period (Figure 2.4). This growth, particularly in North Maluku, is primarily driven by the metal ore mining and metal industry sectors (Figure 2.5 and Figure 2.6). The second export ban of nickel ore (2019/2020) has led to the rapid growth of nickel-based industries in North Maluku, including in Weda Bay and Obi Island.

**FIGURE 2.4** Gross Domestic Regional Product Growth of Nickel-Rich Provinces in Indonesia (% average)



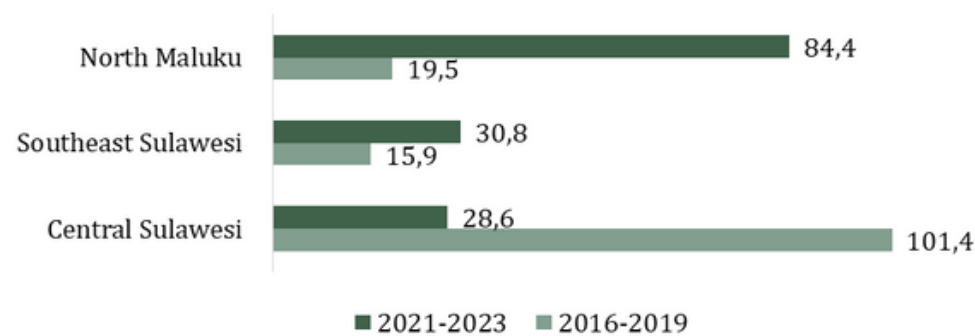
Source: BPS (2024)

**FIGURE 2.5** Iron Ore Mining Growth of Nickel-Rich Provinces in Indonesia (% average)



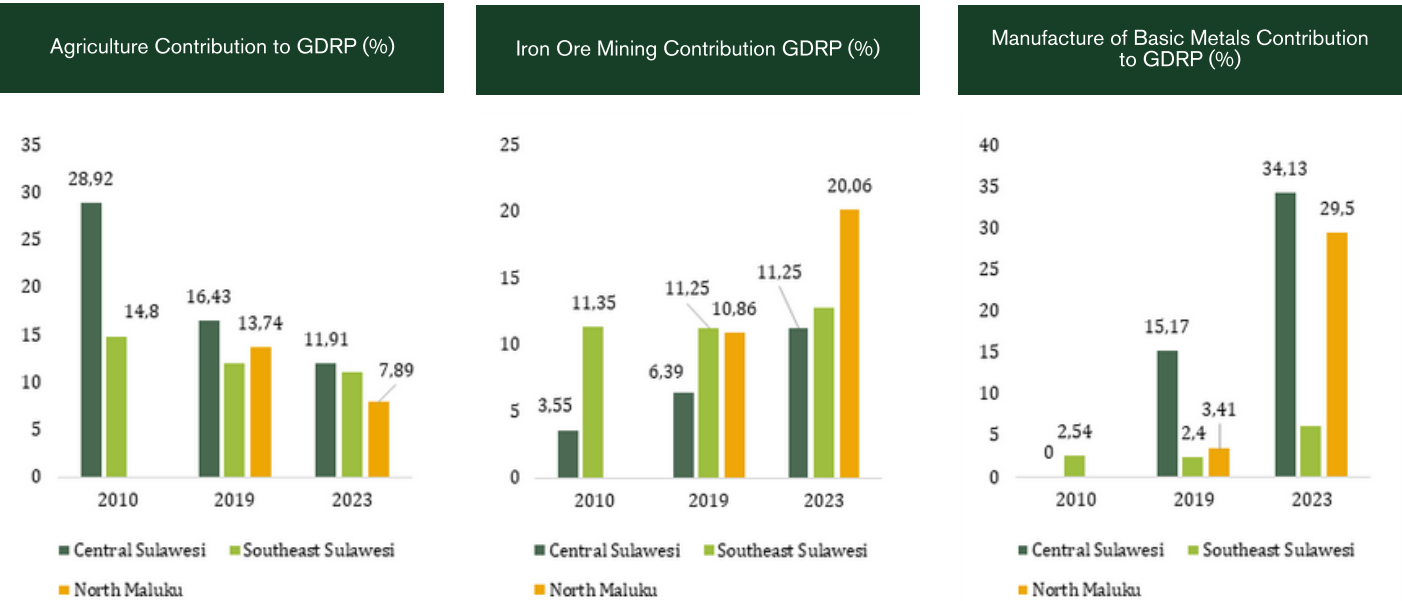
Source: BPS (2024)

**FIGURE 2.6** Metal Industry Growth of Nickel-Rich Provinces in Indonesia  
(%, average)



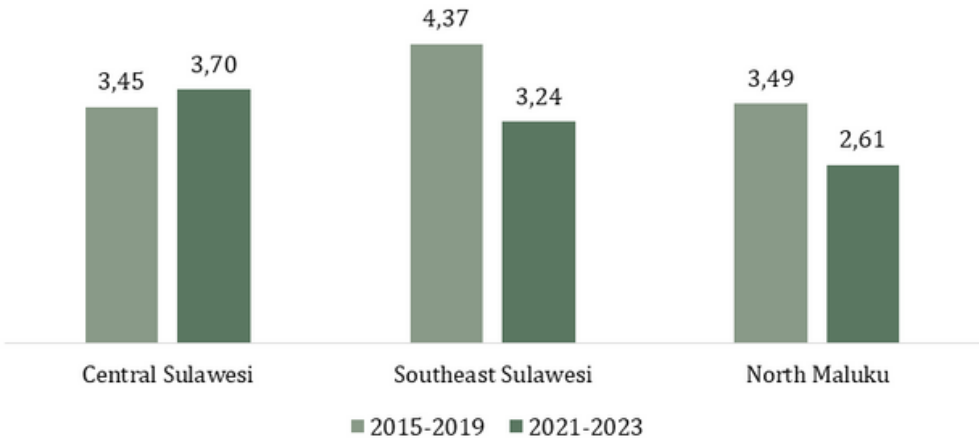
Source: BPS (2024)

**FIGURE 2.7** Selected Sectoral Contribution to Nickel-Rich Provinces in Indonesia



Source: BPS (2024)

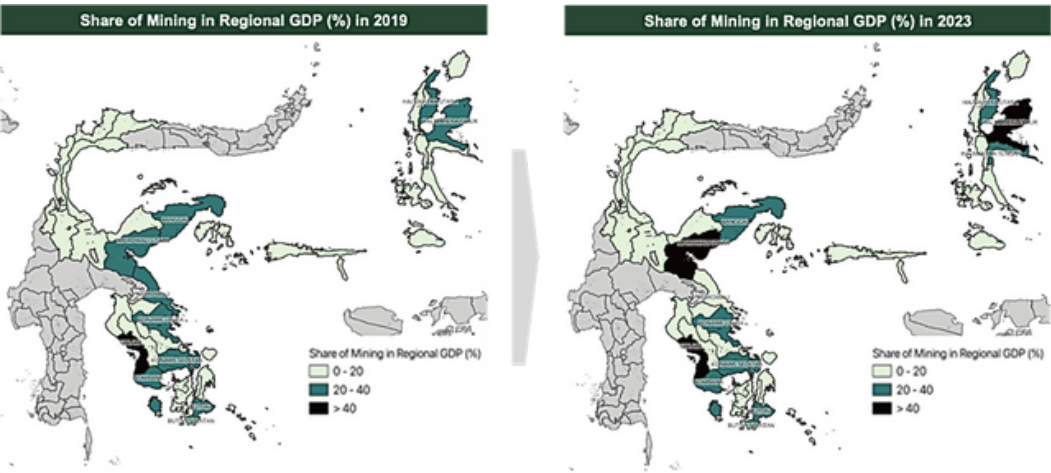
**FIGURE 2.8** Agriculture Growth of Nickel-producing Provinces in Indonesia  
(%, average)



Source: BPS (2024)

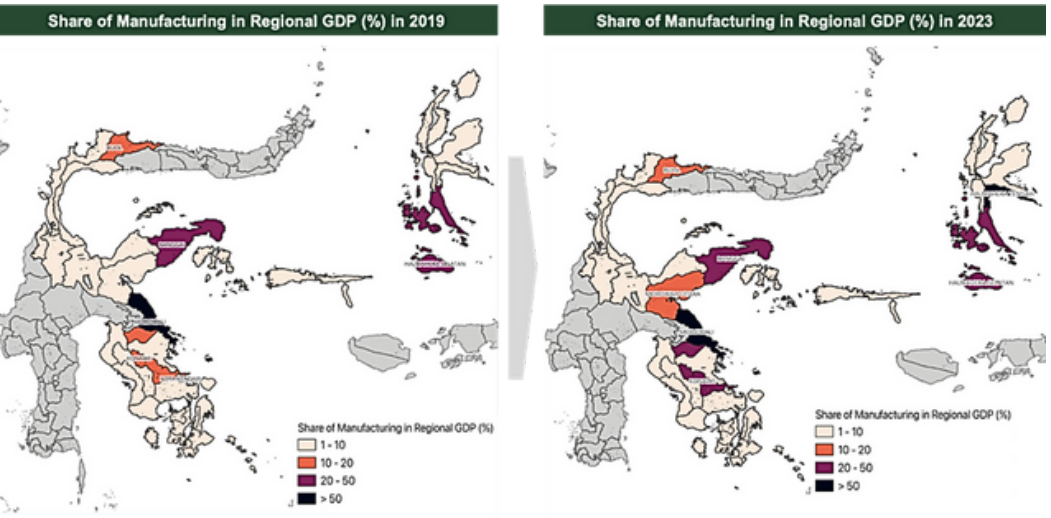
This rapid growth of nickel mining and related industries in the recent years have also shifted the economies of nickel-rich provinces, reducing the agricultural sector’s contribution, which was previously more important (Figure 2.7). Among these provinces, Central Sulawesi and North Maluku, have seen a substantial increase in their dependence on nickel-related processes (metal ore mining and the manufacture of basic metals), which now account for approximately 40% of their total GDRP, along with other interrelated sectors. Although there has been a shifting in sectoral contribution due to nickel downstream, it has not yet structurally transformed the local economy. The agricultural sector in these provinces is still growing, but at a declining rate since 2021, especially in Southeast Sulawesi and North Maluku (Figure 2.8).

**FIGURE 2.9** Share of Mining in Regional GDP, 2019 & 2023



Source: BPS (2024), author's calculation

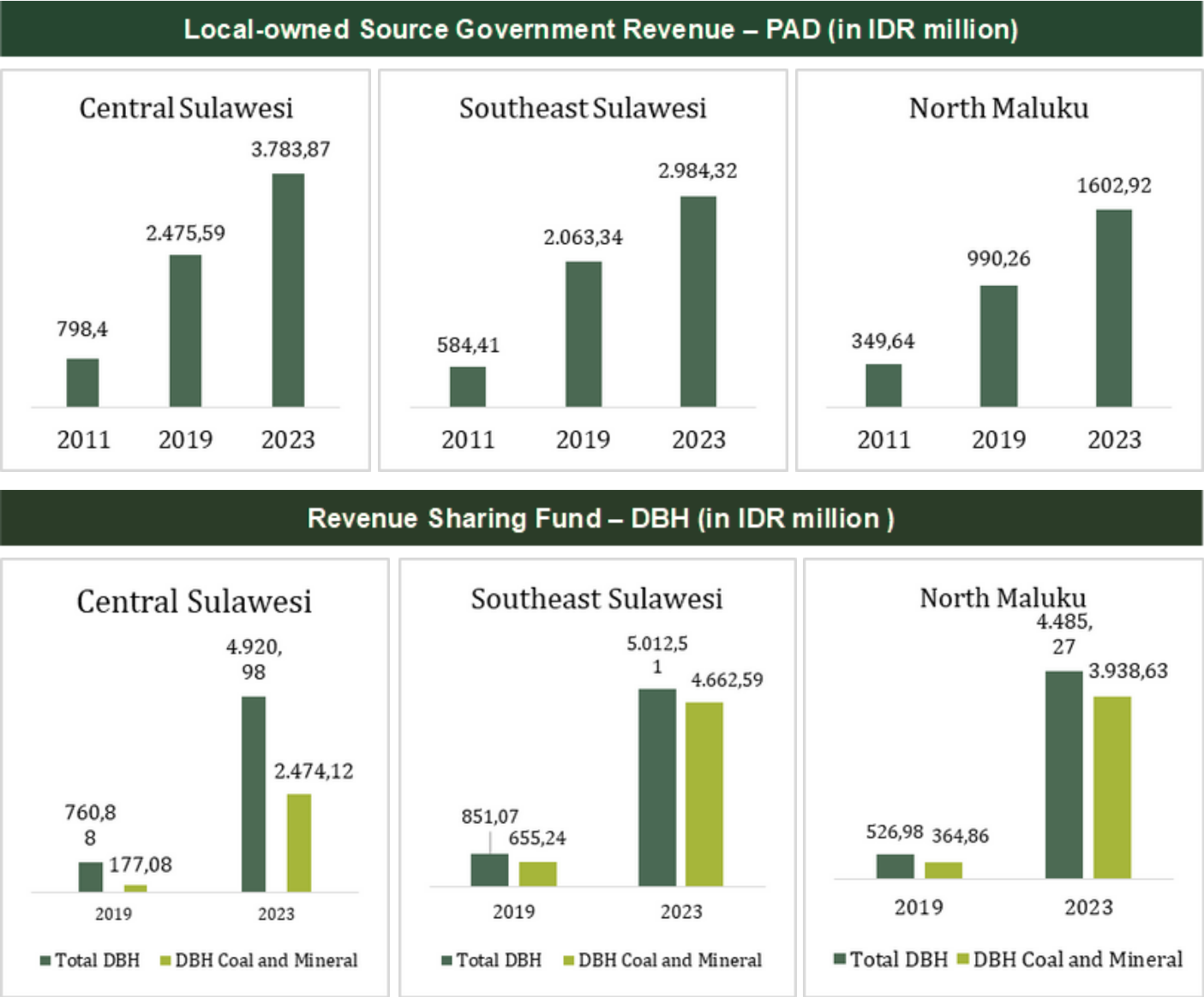
**FIGURE 2.10** Share of Manufacturing in Regional GDP, 2019 & 2023



Source: BPS (2024), author's calculation

Although current nickel regions have gained several benefits from the nickel downstream, their overdependence on nickel-related sectors may pose risks to their economic sustainability. Figure 2.9 shows that nickel-rich regencies, except for Morowali, have been increasingly reliant on mining, where Figure 2.10 also shows the increasing manufacturing share of nickel-rich regions, specifically Central Halmahera, with a rapid share increase to more than five times. These mining and industrial activities may decline in the future, considering nickel’s reserves depletion and the emergence of its substitutes. Local and national governments should initiate transition framework for these regions to mitigate potential economic slowdown in the future.

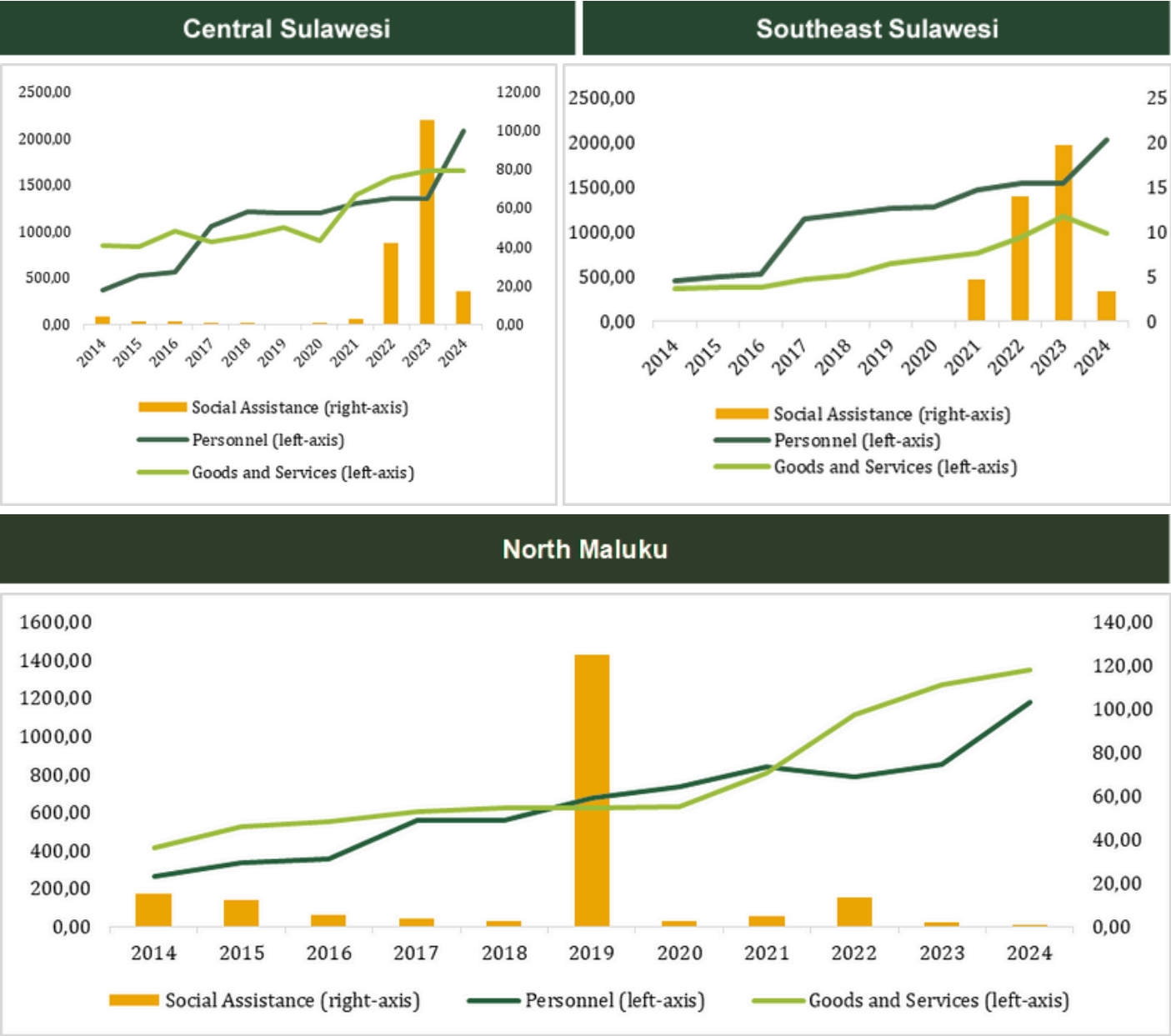
**FIGURE 2.11** Local-owned Revenue (Top) and Revenue Sharing Fund (Bottom) in Nickel-rich Provinces



Source: Ministry of Finance of Indonesia (2024)

From a regional fiscal perspective, current downstream activities have significantly increased regional government revenue. Figure 2.11 illustrates the regional fiscal revenue in nickel-rich provinces, including the local own-source revenue (Pendapatan Asli Daerah/ PAD) and revenue sharing fund (Dana Bagi Hasil/DBH). The revenue from PAD has more than doubled from 2011 to 2023 in these provinces. PAD comprises regional government revenue from local economic value-added, such as local tax, local user fees, revenue from separated regional government assets, and others. DBH in nickel-rich provinces has also increased significantly, rising 9-10 times from 2019 to 2023, largely due to mineral and coal DBH. DBH consists of regional fiscal transfer, including the share of these three provinces' DBH combined to the total national DBH, which increased from 2.06% in 2019 to 7.12% in 2023. This additional fiscal revenue provides substantial benefits by offering more fiscal space to fund their development agenda.

**FIGURE 2.12** Regional Government Current Expenditure in Nickel-rich Provinces (IDR Billion)

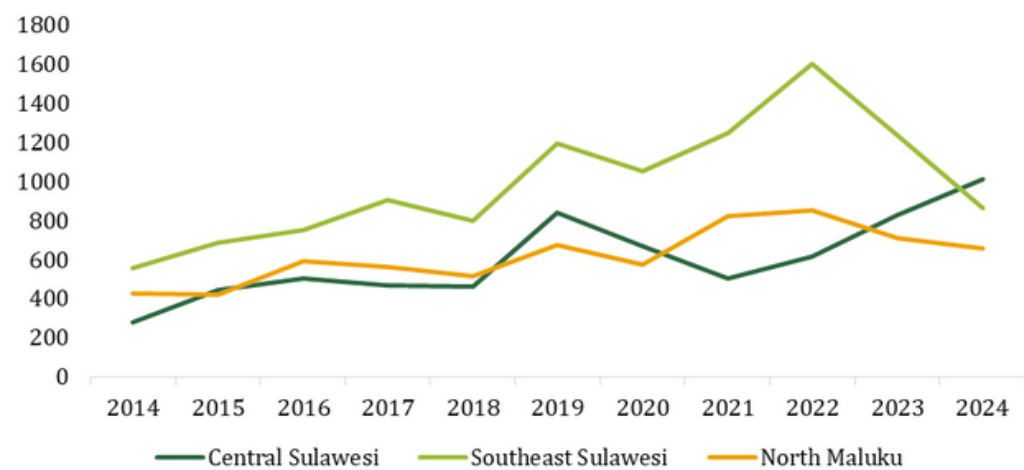


Source: BPS (2024)



The increase in regional fiscal revenue due to the nickel downstream activities has led to varied spending trends among nickel-rich provinces. Current expenditure has also increased in these provinces, with most of the spending in 2024 allocated to personnel costs, except for North Maluku (Figure 2.12). The allocation for social assistance remains minimal across all regions. Capital expenditure characteristics differ among nickel-rich provinces, with only Central Sulawesi increasing its capital expenditure in recent years (Figure 2.13). While it is unclear whether capital expenditure is allocated productively, the allocation for capital goods may provide a more lasting benefit compared to current expenditure.

**FIGURE 2.13** Regional Government Capital Expenditure in Nickel-rich Provinces, 2014 -2024 (IDR Billion)

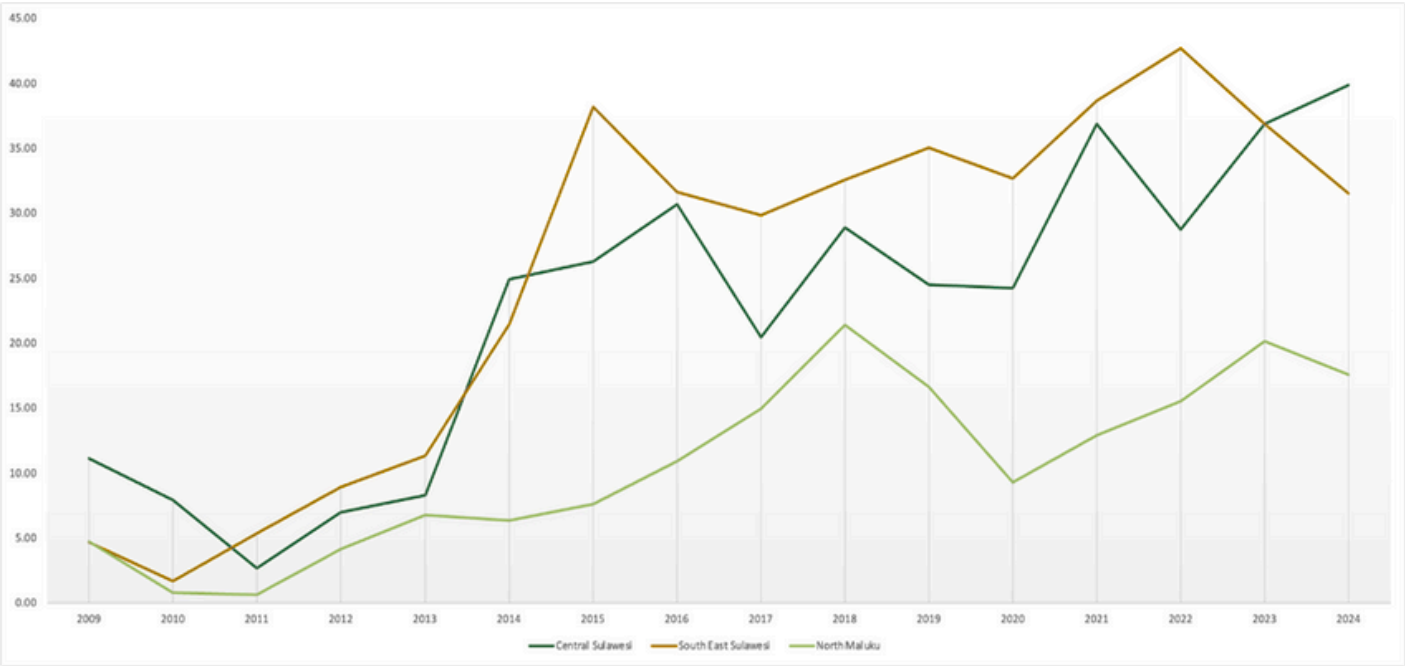


Source: BPS (2024)

**2.1.1. Employment**

Employment gains in mining and manufacturing have shown significant improvement in nickel-rich provinces. The mining and quarrying sector in Central and Southeast Sulawesi soared in 2013, just a year before the first nickel export restriction took place in 2015 (Figure 2.14). Subsequently, the re-implementation of the nickel export ban in 2020 sparked employment expansion across North Maluku, Central Sulawesi, and Southeast Sulawesi. Nevertheless, considering that the mining and quarrying sector historically stands as the third largest employer of informal workers in the Indonesian economy (Asian Development Bank, 2009), reliance on it for achieving formal employment transformation appears imprudent.

**FIGURE 2.14** Trend of Mining Employment in 3 Intensive Nickel-Mining Regions (in Thousands of Workers)



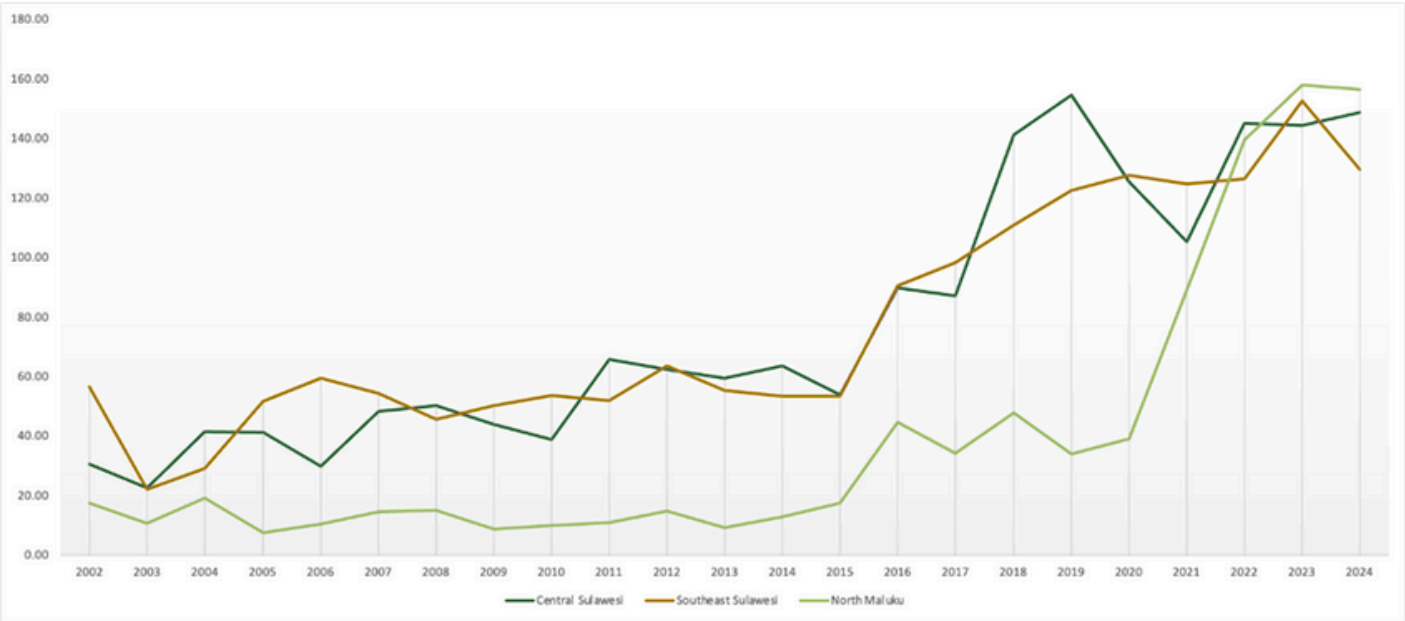
Source: Data from CEIC (2025)

Nickel processing offers significant potential for expanding formal manufacturing employment. Following the initial export restriction in 2014, nickel-intensive regions saw a synchronous increase in manufacturing outputs, likely driven by the domestic processing of nickel ores. Initially, Sulawesi provinces led in manufacturing outputs after the surge in 2015. However, in 2020, North Maluku experienced a sharp increase coinciding with the reintroduction of the second export ban (Figure 2.15). This substantial expansion in manufacturing employment is likely due to operations within the newly established Indonesian Weda Bay Industrial Park (IWIP) in North Maluku. By 2021, IWIP reported employing over 24,000 workers<sup>4</sup> since its establishment in 2018.

Within five years, North Maluku's manufacturing employment surpassed that of both Southeast and Central Sulawesi, establishing it as the leading nickel-producing region in Indonesia. This increase in manufacturing employment in North Maluku indicates a positive relationship between nickel manufacturing activities, value addition policies, and regional employment growth.

4 IWIP Website. Accessible through <https://iwip.co.id/>

**FIGURE 2.15** Trend of Manufacturing Employment in 3 Intensive Nickel-Manufacturing Regions 2002 – 2024 (in Thousands of Workers)



Source: Data from BPS (2024)

Unlike North Maluku, Central Sulawesi experienced a decline in manufacturing employment during the 2020 ban period, although a rise was noted between 2021 and 2022. While local economic gains suggest potential improvement, it remains to be determined whether these gains will correlate with positive changes in social indicators like poverty rates. We will discuss this in the social impact section.

**2.2. Result and Analysis of Environmental Impact**

The rapid expansion of the nickel industry is crucial for economic growth and meeting global demand for electric vehicles, has significant environmental consequences. Both nickel mining and smelting—the key components of downstream—contribute to environmental degradation through distinct pathways. Mining operations lead to extensive land clearing, habitat destruction, and water contamination, while smelting activities primarily result in air pollution and the discharge of hazardous waste into water bodies. These environmental impacts highlight the need for a more sustainable approach to nickel production, balancing industrial development with the preservation of ecosystems and community well-being.

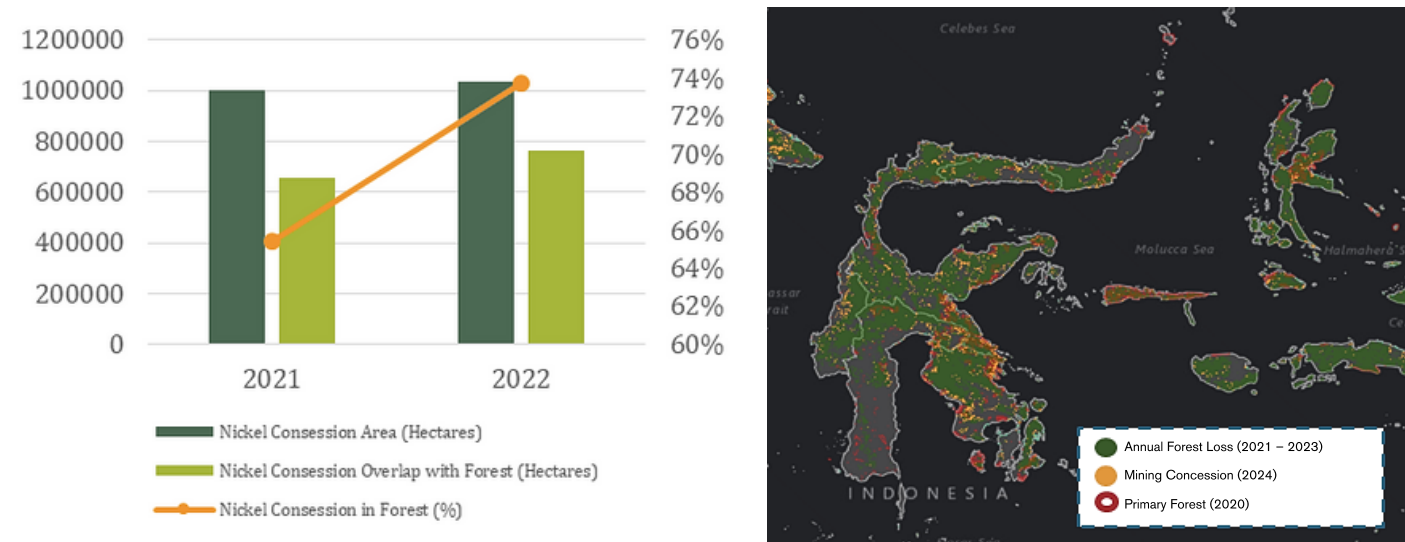
**2.2.1. Deforestation and Biodiversity Loss**

Nickel mining on land is a significant cause of large-scale deforestation, posing a threat to Indonesia’s climate commitments and forest conservation efforts. The extraction process involves removing topsoil, destroying vegetation, and disrupting wildlife habitats, leading to severe ecological damage. Extensive land clearing is projected to make nickel mining surpass palm oil as the leading cause of deforestation in Eastern Indonesia (IUCN, 2022). This issue is particularly critical in Central Sulawesi, Southeast Sulawesi, and North Maluku, which are home to vast tracts of Indonesia’s rainforests. In 2001, primary forests covered 72% of Central Sulawesi, 64% of Southeast Sulawesi, and 74% of North Maluku, with 21%, 24%,

and 13%, respectively, classified as intact forests—essential for carbon storage and biodiversity. However, by 2023, these nickel-rich regions had lost 9.8%, 11%, and 7.5% of their primary forests, respectively (Global Forest Watch, 2024), though their overall forest coverage remains significant.

Nickel mining continues to encroach on forest zones, further accelerating deforestation. By 2022, nickel concessions spanned 1,037,435 hectares, with 765,237 hectares (68%) located within forested areas (Walhi, 2023; CNN Indonesia, 2023). The extensive overlap between nickel mining and protected forests is evident in Figure 2.16, where mining sites (red areas) and primary forests (green areas) frequently intersect. As a result, deforestation is occurring at an alarming rate, with global alert systems estimating that nickel mining has cleared 76,031 hectares of forest in Indonesia (Brown & Harris, 2024). However, a more detailed assessment using the University of Maryland’s (UMD) Global Forest Change (GFC) dataset, which tracks annual forest cover loss via Landsat imagery, indicates an even greater environmental impact. Between 2001 and 2022, a total of 153,364 hectares of forest was lost within 329 nickel concessions.

**FIGURE 2.16** Nickel Concession Overlap with Forest Area in Bar Chart (left hand side) and Map (right hand side)



Source: Walhi (2023); Nusantara Atlas (2024); Global Forest Watch (2024), author’s calculation

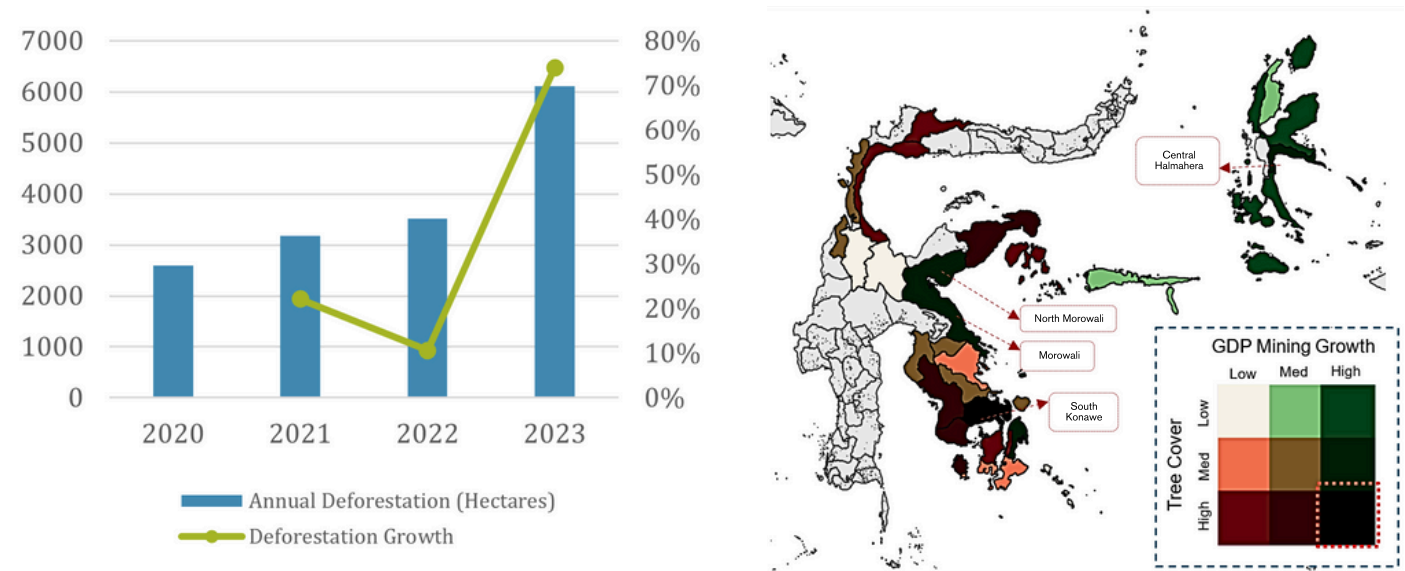
Further evidence underscores the link between nickel mining and deforestation trends. A study of 7,721 villages in Sulawesi found that deforestation rates in nickel-mining villages nearly doubled between 2011 and 2018 compared to villages without mining activity (Lo et al., 2024). Similarly, research by Brown and Harris (2024) revealed that RADD (radar) alerts detected forest clearing in 2023 at more than double the rate recorded in 2020, signaling a rapid acceleration of deforestation. As shown in Figure 2.17, regions experiencing high GDP growth from mining also exhibit higher rates of deforestation, reinforcing the correlation between economic expansion and environmental degradation. If this trend continues, over half a million additional hectares of Indonesian forest within nickel concessions will be at high risk of deforestation—not only from currently active mining sites but also due to newly issued concessions. In 2023, the number of nickel mining concessions surged to 373, marking a 23% increase from 2022 (Project Multatuli, 2024), further intensifying the threat to Indonesia’s forests.

TABLE 2.1 Three Estimation of Deforestation in Indonesia Nickel Mines

| Measurement Dataset | Years Measured                                    | Tree Cover Loss (hectares) |
|---------------------|---------------------------------------------------|----------------------------|
| Auriga's Dataset    | 2000 - 2022                                       | 24.811                     |
| UMD Forest Loss     | 2001 - 2022                                       | 153.368                    |
| Tree Map            | From latest license award/adjustment through 2022 | 76.031                     |

Source: Author's compilation

FIGURE 2.17 Deforestation in Nickel Mining Area (left hand side); Correlation between GDP Mining Growth and Deforestation in 2021 – 2023 (right hand side)



Source: Brown & Harris (2024); BPS (2024); Global Forest Watch (2024), author's calculation

Stanimirovora (2024) used UMD GFC data to estimate that between 2001 and 2020, global mining and related activities led to the loss of nearly 1.4 million hectares of tree cover. Indonesia accounted for 26.4% of this global mining-driven deforestation, losing 369,356 hectares, 36% of which were primary rainforests. Although direct comparisons are limited due to slight differences in the timeframes, the impact of nickel mining alone is particularly significant, representing 41.5% of Indonesia's total mining-related tree cover loss. This highlights the disproportionate role of nickel extraction in Indonesia's overall deforestation and its broader environmental implications.

Despite the estimation that already stated in Table 2.1, deforestation figures may be underestimated due to unacknowledged illegal mines that remain absent from Minerba One Data Indonesia (MODI) 2023's official concession records. Since these mines are unmapped, it is impossible to integrate them into deforestation assessments, creating a significant data gap (Hidayat & Hermawan, 2022). For instance, in East Halmahera alone, reports indicate that ten unrecognized mining sites exist (TI Indonesia, 2024), suggesting that actual deforestation is likely far greater than recorded estimates.

Additionally, while not as significant as mining, a new analysis of rainforest loss based on government data indicates that deforestation around smelters (nickel-processing plant) has risen significantly, increasing from an average of 33 km<sup>2</sup> to 63 km<sup>2</sup> per smelter (Milko et al., 2024). For example, Weda Bay Nickel (WBN), a tenant in the Indonesia Weda Industrial Park (IWIP) operating pyrometallurgical smelters, holds a concession area of 54,874 hectares, with nearly 65% of it situated within protected forests, mostly mangrove forest (Saturi, 2023). Over the past decade, the number of smelters in Indonesia has increased from 3 to 30, contributing to a 234% rise in deforestation by smelters (Hilterman, 2024).

Among the top 25 nickel mines<sup>5</sup> causing deforestation, many operate in regions classified as High Carbon Stock (HCS) forests. Six of these mines contain over 75% HCS forests, which are rich in carbon and intact ecosystems, while 14 consist of 50% or less, indicating degraded lands (Brown & Harris, 2024). These areas, which store significant amounts of carbon, face substantial risks due to nickel-driven deforestation, potentially releasing millions of metric tons of carbon dioxide into the atmosphere. For instance, deforestation within nickel concessions on Halmahera alone has resulted in an estimated loss of 2.04 million metric tons of stored carbon<sup>6</sup> (CRI, 2024). Preliminary findings suggest that regions with high GDP growth driven by mining activities may exhibit a significant positive forest carbon flux, indicating a reduced capacity of forests to absorb carbon (Figure 2.18).

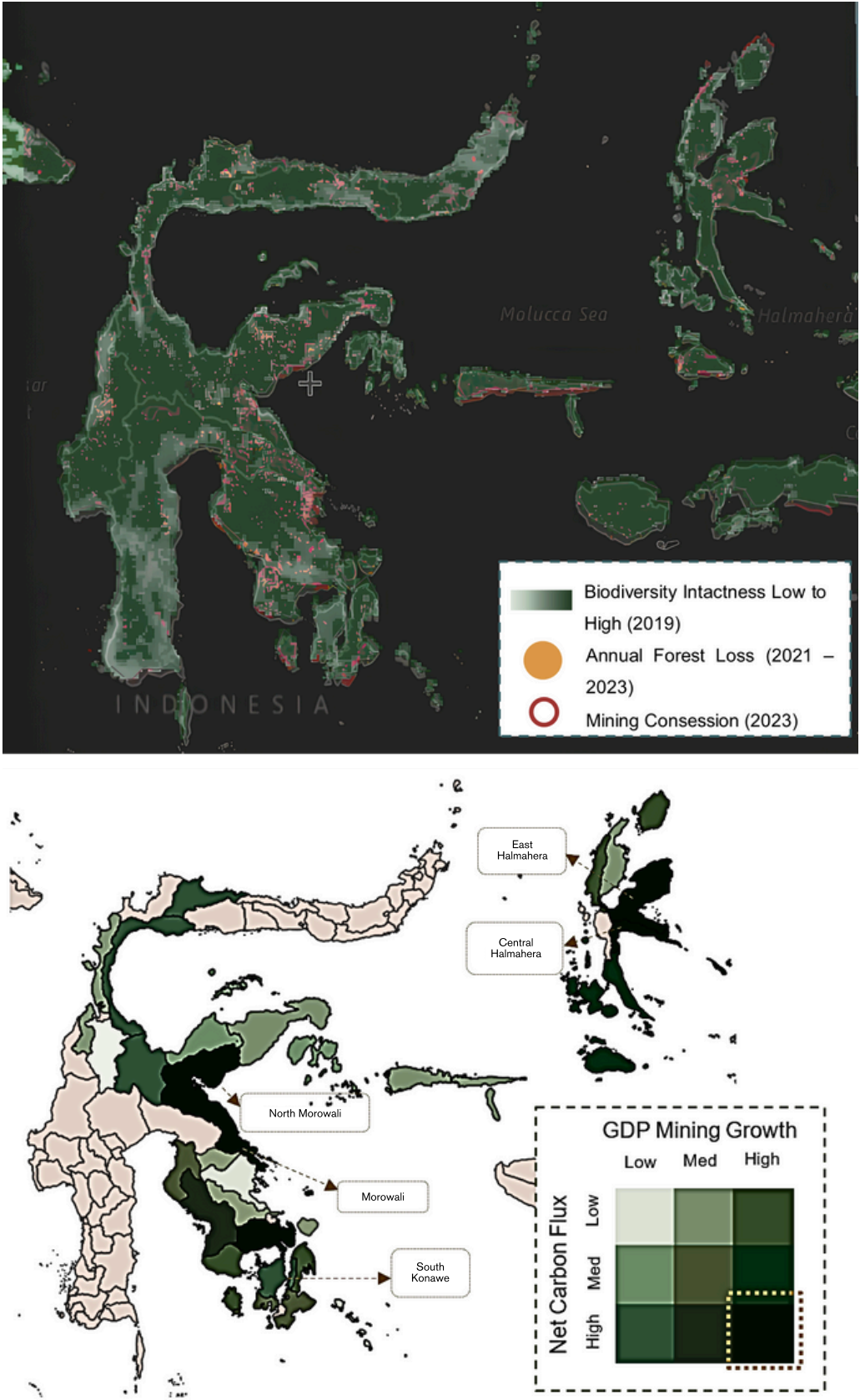
The expansion of nickel concessions poses significant threats to biodiversity, particularly in Sulawesi and Maluku, which are part of the Wallacea Region—a globally recognized biodiversity hotspot with numerous endemic species (Conservation International). As of 2022, the Ministry of Environment and Forestry reported that forest and marine conservation areas in Sulawesi, North Maluku, and Maluku span 5 million hectares, representing 18% of Indonesia's national conservation area (BPS, 2024). This region is also part of the Coral Triangle, a critical marine ecosystem vital for sustaining biodiversity. However, mining activities increasingly threaten the habitats of endemic bird species, including the Sulawesi hornbill, bilbong, and various pigeon species, particularly along routes from Kolaka City to Batu Putih District and extending to Morowali (Paskalis, 2024). Additionally, six of the top 25 nickel deforestation concessions overlap with Key Biodiversity Areas (KBAs), impacting a total of 27,693 hectares (Harris & Brown, 2024), further exacerbating ecological degradation in these sensitive regions.

5 Vale Indonesia - Sorako, Aneka Tambang - Kon Ut, Bintang Delapan Mineral, Vale Indonesia - Pomala, Vale Indonesia - Bahodopi, Mulia Makmur Perkasa, Bukit Makmur Istindo Nikel, Multi Dinar Karya, Pertambangan Bumi, Lawaki Tiar Raya, Weda Bay Nickel, Bahodopi Utara Block, Wana Kencana Mineral, Halmahera Sukses Mineral, Ceria Nugraha Indotama, Riota Jaya Lestari, Adhi Kartiko Pratama, Gemilang Mandiri Perkasa, Mulia Pacific Resources, Toshida Indonesia, Sulawesi Cahaya Mineral, Gemilang Bangun Perkasa, Cahaya Ginda Ganda, Anugerah Bumi Gemilang, and Sambaki Tambang Sentosa.

6 This value was calculated by multiplying the 5331 hectares of deforested forest by the emission factor of 382.75 CO<sub>2</sub>e per hectare.



**FIGURE 2.18** Biodiversity Intactness and Mining Concession Map (top) and Correlation between GDP Mining Growth and Forest Carbon Flux (bottom) between 2021 - 2023



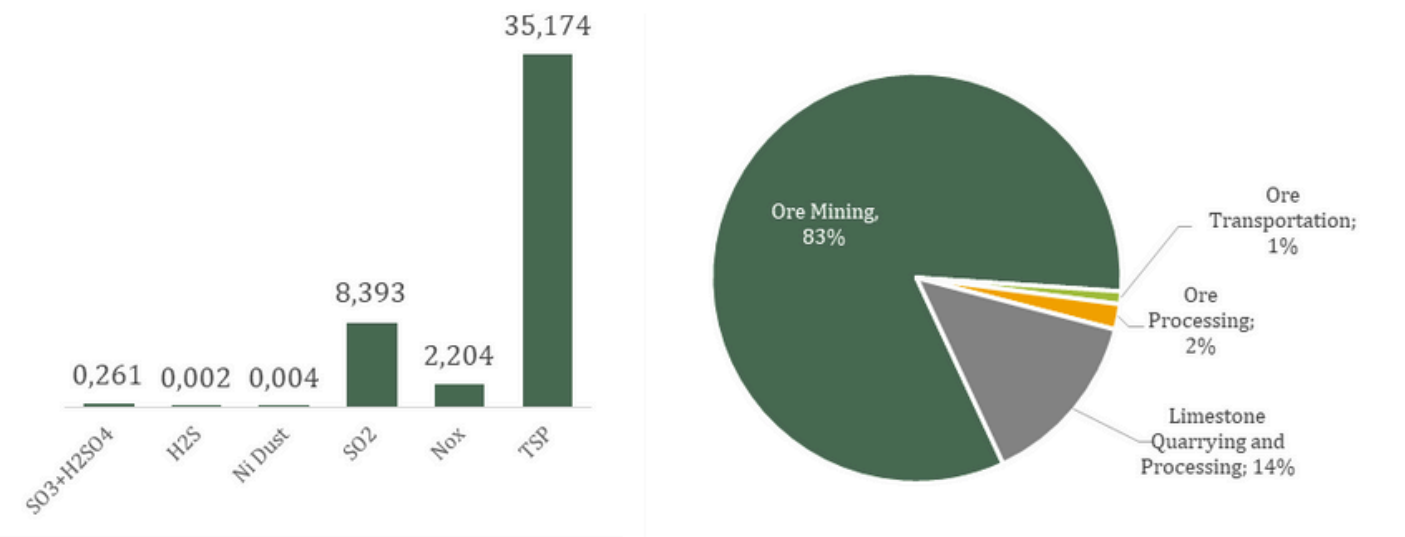
Source: BPS (2024); Nusantara Atlas (2024); Global Forest Watch (2024), author's calculation

The coastal expansion of nickel smelters poses a severe threat to marine biodiversity, particularly through mangrove deforestation and habitat degradation. Mangroves are vital breeding grounds for marine life, natural buffers against coastal erosion, and carbon sinks. A stark example is the loss of coral reefs and mangroves near the Indonesia Weda Industrial Park (IWIP) due to land reclamation for infrastructure projects like airports, office developments, and coal-fired power plants (AEER, 2024). This is especially concerning as Indonesia is part of the Coral Triangle, a biodiversity hotspot spanning 132,636 km<sup>2</sup> across six countries—Indonesia, Malaysia, Papua New Guinea, the Philippines, the Solomon Islands, and Timor-Leste (Gray, 2018). Despite covering just 1.5% of the world's ocean area, it holds 30% of global coral reefs, with 76% of the world's coral species, including 15 endemic species, and 37% of the world's coral reef fish species. The continued expansion of nickel smelting along Indonesia's coast threatens this critical marine ecosystem, highlighting the urgent need for stronger environmental protections.

2.2.2. Emission and Air Pollution

In addition to deforestation and habitat destruction, nickel downstream process in Indonesia contributes significantly to air pollution through three main sources: mining operations, captive coal-fired power plants, and smelting processes. The industry emits 58.6 tCO<sub>2</sub>e per ton of nickel, exceeding the global standard of 48 tCO<sub>2</sub>e per ton (Sani, 2024). The Institute for Essential Services Reform (IESR) reports that nickel production emissions in Indonesia are 30–40% higher than the global average (Hasiana, 2024).

**FIGURE 2.19** Example of Nickel Mining Emission Rates in Tons (left hand side) and Composition of TSP Emission Sources (right hand side) in Bukit Limber and Coastal Central Halmahera



Source: Kramawijaya (2017)

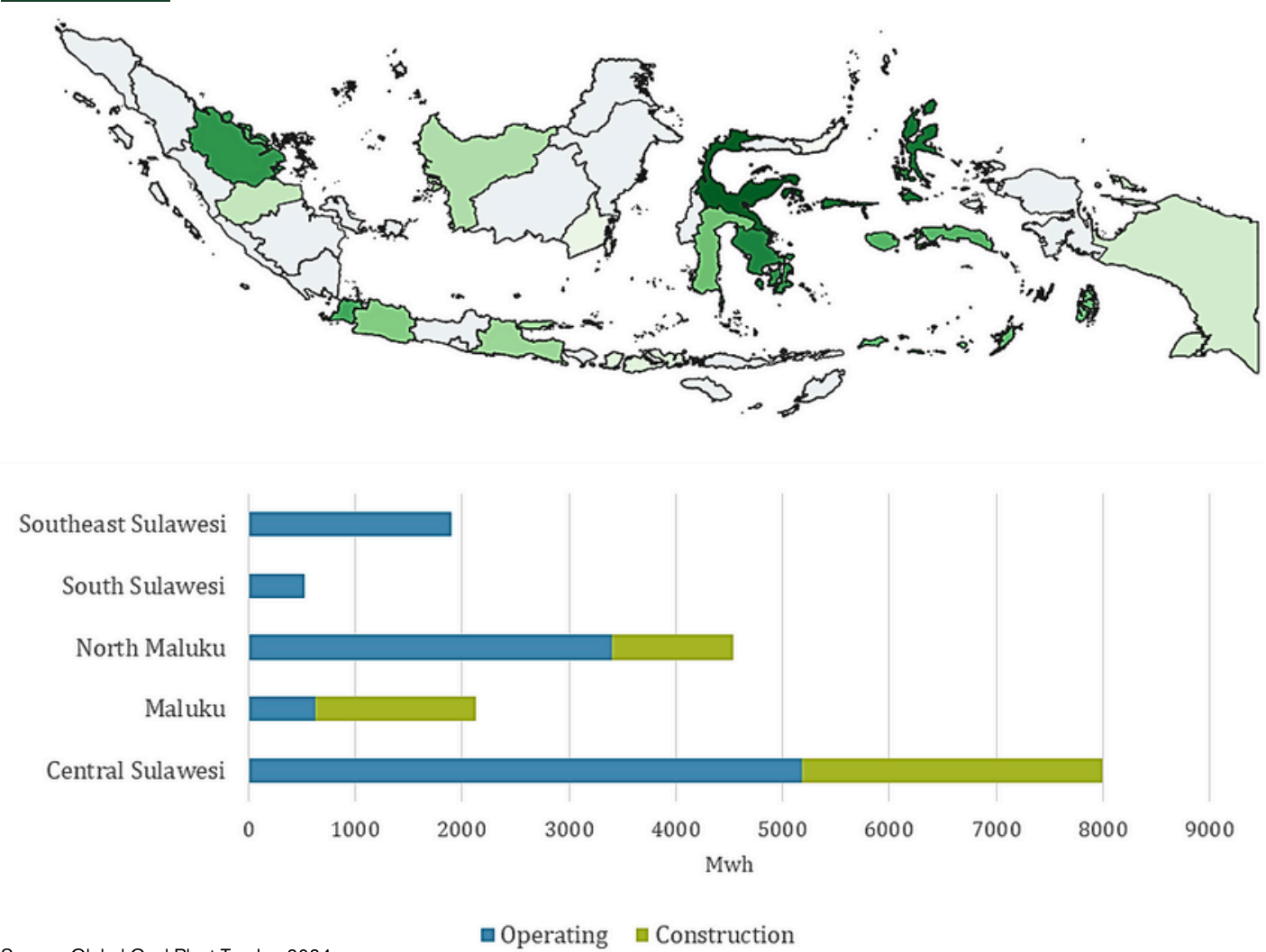
Additionally, nickel-based battery production generates higher greenhouse gas emissions than lithium iron phosphate (LFP) batteries. Research by Llamas-Orozco et al. (2023) projects that under an aggressive electric vehicle adoption scenario, cumulative emissions from nickel-based batteries could reach 8.1 GtCO<sub>2</sub>eq by 2050. However, switching to LFP batteries could reduce emissions by 1.5 GtCO<sub>2</sub>eq, as they produce fewer greenhouse gases, with an emission intensity of 55 kgCO<sub>2</sub>eq/kWh.

Nickel extraction generates various non-carbon emissions, including Total Suspended Particulate (TSP), NO<sub>x</sub>, SO<sub>2</sub>, SO<sub>3</sub>+H<sub>2</sub>SO<sub>4</sub>, nickel dust, and H<sub>2</sub>S, primarily from ore mining, transportation, processing, and limestone quarrying. A study by Kramawijaya (2017) examined two major nickel mining sites in Central Halmahera—Bukit Limber (Santa Monica) and Coastal, which contain an estimated 341 million tons of lateritic ore with an average nickel content of 1.49%.

Figure 2.19 indicates that the primary pollutant emitted is Total Suspended Particulate (TSP), with an annual emission rate of 35,173.96 tons, primarily originating from ore mining (83%), followed by limestone quarrying and processing (14%), ore processing (2%) and ore transportation (1%) (Figure 2.21). Among gaseous pollutants, SO<sub>2</sub> and NO<sub>x</sub> are the most significant, with annual emissions of 8,392.61 tons and 2,203.71 tons, respectively. SO<sub>2</sub> emissions mainly come from acid plants (72%), while NO<sub>x</sub> emissions are largely attributed to ore transportation (80%), highlighting the environmental impact of nickel mining operations.

In addition to mining activities, Indonesia's captive coal-fired power plants (CFPPs) play a major role in the country's high carbon emissions per ton of nickel. In a controversial decision, Indonesia revised its green investment taxonomy to classify captive CFPPs for nickel production as part of its green transition, drawing criticism from environmental groups. A 2023 report by CREA and GEM ranked Indonesia as the world's fifth-largest in captive CFPP capacity and third in new construction, with a total capacity of 9,815 MW. Captive CFPPs supply 23.7% of the nation's electricity, with 117 units operating at 10,821 MW, of which 76% (8,214 MW) powers the metal industry and 67% (7,273 MW) is allocated to nickel smelters (Figure 2.20).

**FIGURE 2.20** Coal Captive Power Plant Distribution (Top) and Coal Captive Power Plant Capacity for Nickel (Bottom) in Indonesia per 2023

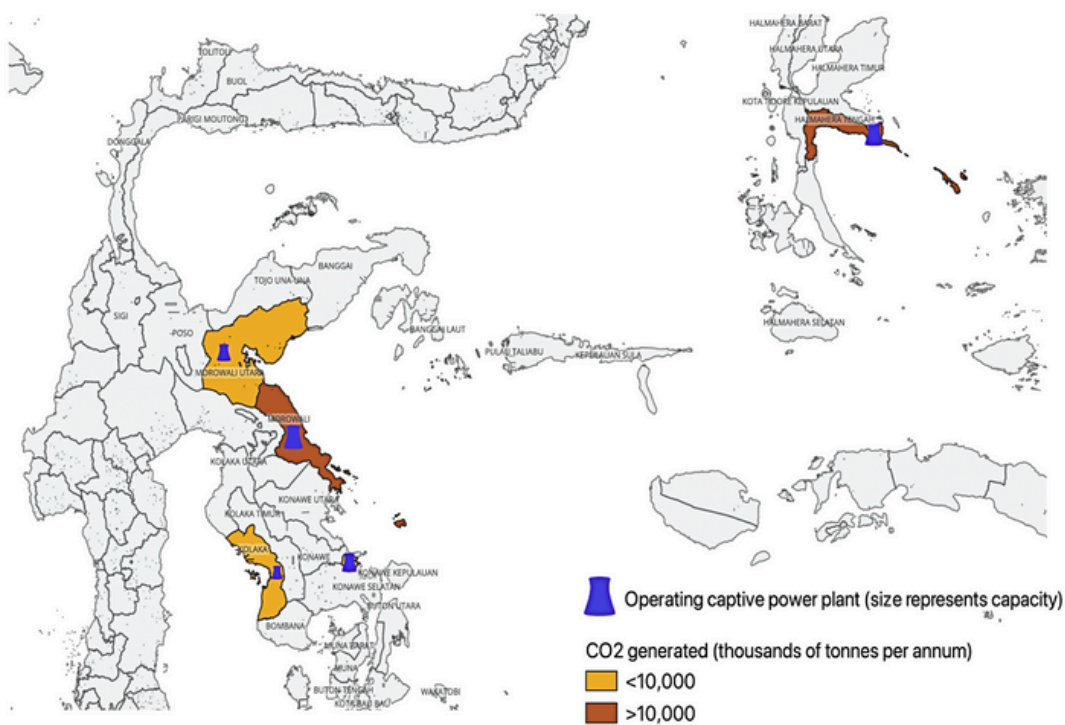


Source: Global Coal Plant Tracker 2024

Nickel industrial hubs in Sulawesi and Maluku, particularly Morowali Industrial Park and Weda Bay, host the highest concentration of captive CFPPs, significantly impacting regional emissions as visualize in Figure 2.21. Central Sulawesi's coal captive power plants emit approximately 23.68 million tons of CO<sub>2</sub> annually, while North Maluku produces 15.4 million tons and Southeast Sulawesi 8.93 million tons. In terms of emission intensity, Central Sulawesi leads with 2.4 million tons of CO<sub>2</sub> per terajoule (TJ), followed by Southeast Sulawesi (1.15 million tons) and North Maluku (1.06 million tons) (GEM, 2024).

If we convert these emissions into monetary value, Indonesia bears a substantial cost due to the emissions from captive coal-fired power plants (CFPPs). Rickel et al. (2018) calculated the country-level Social Cost of Carbon (SCC), which estimates the long-term economic damages caused by emitting one ton of carbon dioxide (CO<sub>2</sub>) into the atmosphere. As an archipelagic nation located along the equator, Indonesia faces a disproportionately high climate-related burden, incurring a significantly larger share of the global SCC relative to its current share of global emissions. The SCC for Indonesia is estimated at USD 10.9 per ton of CO<sub>2</sub>. Based on the emissions generated by nickel smelters, the social costs amount to approximately USD 258 million for Central Sulawesi, USD 167 million for North Maluku, and USD 97.3 million for Southeast Sulawesi.

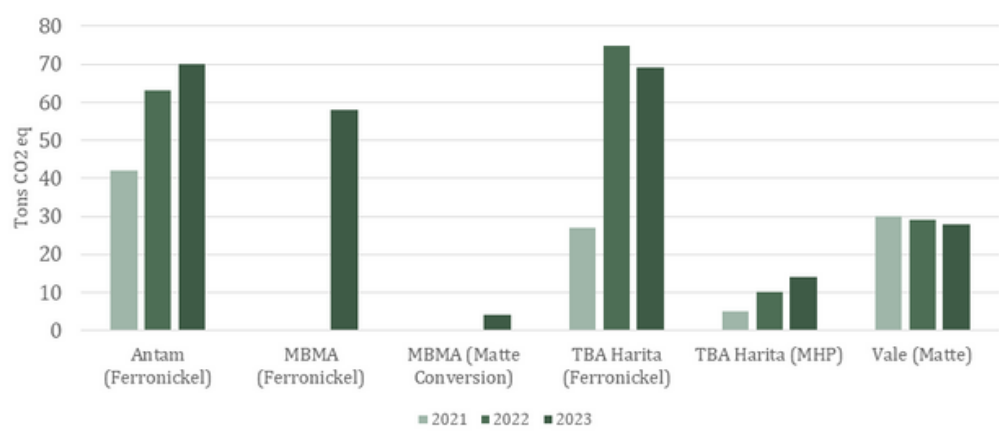
**FIGURE 2.21** Nickel Smelter Captive Coal Power Plants Annual Emission



Source: Global Energy Monitor (2024), author's calculation

Regarding emissions from nickel processing, the Institute for Energy Economics and Financial Analysis (IEEFA) has published a report examining the nickel-processing emissions of Indonesia's four largest publicly listed nickel companies. In 2023, PT Aneka Tambang (Antam), Merdeka Battery Materials (MBMA), Trimegah Bangun Persada (TBP Harita), and PT Vale Indonesia (Vale) collectively produced 353,000 tons of nickel, resulting in 15.3 million tons (mt) of CO<sub>2</sub> emissions (Figure 2.22). The greenhouse gas (GHG) intensity varied significantly among these companies, with Vale recording the lowest at 28.7 tCO<sub>2</sub> per ton of nickel due to its reliance on hydropower. In contrast, companies dependent on coal-fired power plants exhibited much higher emission intensities, ranging from 56.9 to 69.9 tCO<sub>2</sub> per ton.

**FIGURE 2.22** GHG Intensity per Tons of Nickel for Selected Indonesian Nickel Companies by Product



Source: IEEFA (2024)



The industry’s future growth trajectory raises concerns, as nickel production is projected to triple to 1.05 million tons by 2028, potentially escalating emissions to 38.5 mt, which would account for 4.5% of Indonesia’s total emissions in 2023. However, if all companies adopted Vale’s renewable energy model, emissions could be reduced by 43%, lowering them to 22.3 mt. It is important to note that Vale benefits from its strategic location near Lake Matano, one of Indonesia’s largest lakes, enabling it to utilize hydropower as a sustainable energy source. Directly comparing the cost of smelting using coal and hydropower is challenging, but it can be approximated using the Levelized Cost of Electricity (LCOE) for each energy source. On average, coal remains cheaper than large hydropower in Indonesia, with coal supercritical at 5.68 USD/kWh compared to hydropower at 7.91 USD/kWh (IESR, 2023). However, under low-investment scenarios with higher efficiency and lower financing costs (low WACC), hydropower can achieve a lower LCOE of 2.37 USD/kWh, compared to coal at 4.54 USD/kWh.

**FIGURE 2.23** Maximum Estimated Ambient Concentrations of NO<sub>2</sub>, PM<sub>2.5</sub>, and SO<sub>2</sub> from Nickel Smelters and Captive Power Plants (2015–2030)



Source: CREA & CELIOS (2024)

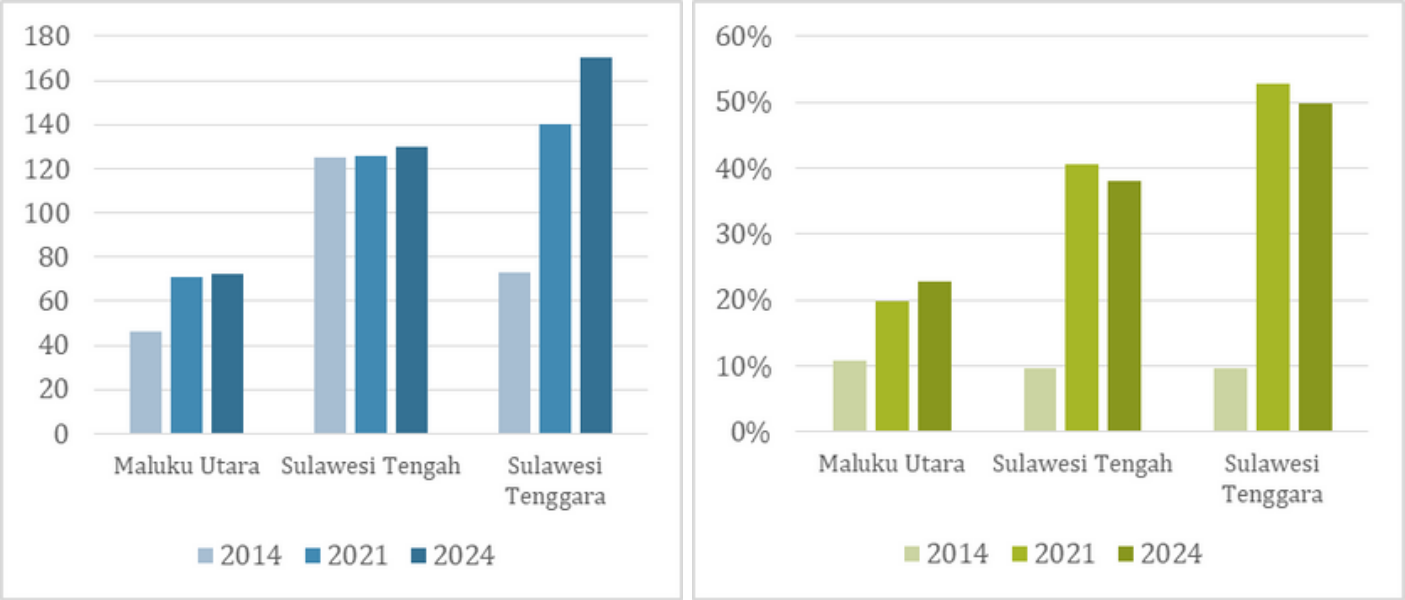
The nickel industry also significantly contributes to air pollution, emitting NO<sub>2</sub>, SO<sub>2</sub>, and PM<sub>2.5</sub>. Research by CREA & Celios (2024) projects sharp increases in these pollutants from 2015 to 2030 because of smelters facilities (Figure 2.23). NO<sub>2</sub> annual concentrations are expected to exceed 30 µg/m<sup>3</sup> by 2030, nearing the WHO guideline of 40 µg/m<sup>3</sup>, while daily levels could reach 180 µg/m<sup>3</sup>—triple the safe limit. PM<sub>2.5</sub> levels may rise to over 100 µg/m<sup>3</sup> annually and 2,500 µg/m<sup>3</sup> daily, far surpassing WHO recommendations. SO<sub>2</sub> emissions are projected to soar, with daily concentrations exceeding 4,000 µg/m<sup>3</sup> and hourly levels approaching 60,000 µg/m<sup>3</sup>. Additionally, toxic particle deposition, including mercury and heavy metals from coal combustion, poses severe threats to marine biodiversity. By 2030, mercury deposition could reach 3,700 kg annually in Sulawesi and North Maluku, risking bioaccumulation in marine life, particularly in protected areas like Aketajawe Lolobata National Park and Lasolo Bay Marine Park.



2.2.3. Water Pollution

The development of the nickel industry in the three major producing provinces has also impacted water quality. To proxy this, data from the statistics of Village Potential (Potensi Desa/Podes) can be used to assess environmental pollution at the village level in these provinces. Podes data is collected through field officers visiting each village or sub-district to gather information by interviewing key informants, such as village heads, local officials, or community leaders who have a deep understanding of local conditions. Villages are chosen as the unit of analysis because they are the smallest administrative units and are often directly affected by nickel mining and smelting activities. There is a noticeable upward trend in the number of villages experiencing water pollution, particularly in Southeast Sulawesi. This increase shows a significant spike in 2021, following the implementation of the export ban in 2020 (Figure 2.24). Additionally, the same figure indicates a rise in the percentage of villages affected by pollution from industrial activities, especially in 2021.

**FIGURE 2.24** Number of Villages Affected by Water Pollution (left hand side) and Percentage of Villages Affected by Industrial Water Pollution (right hand side)

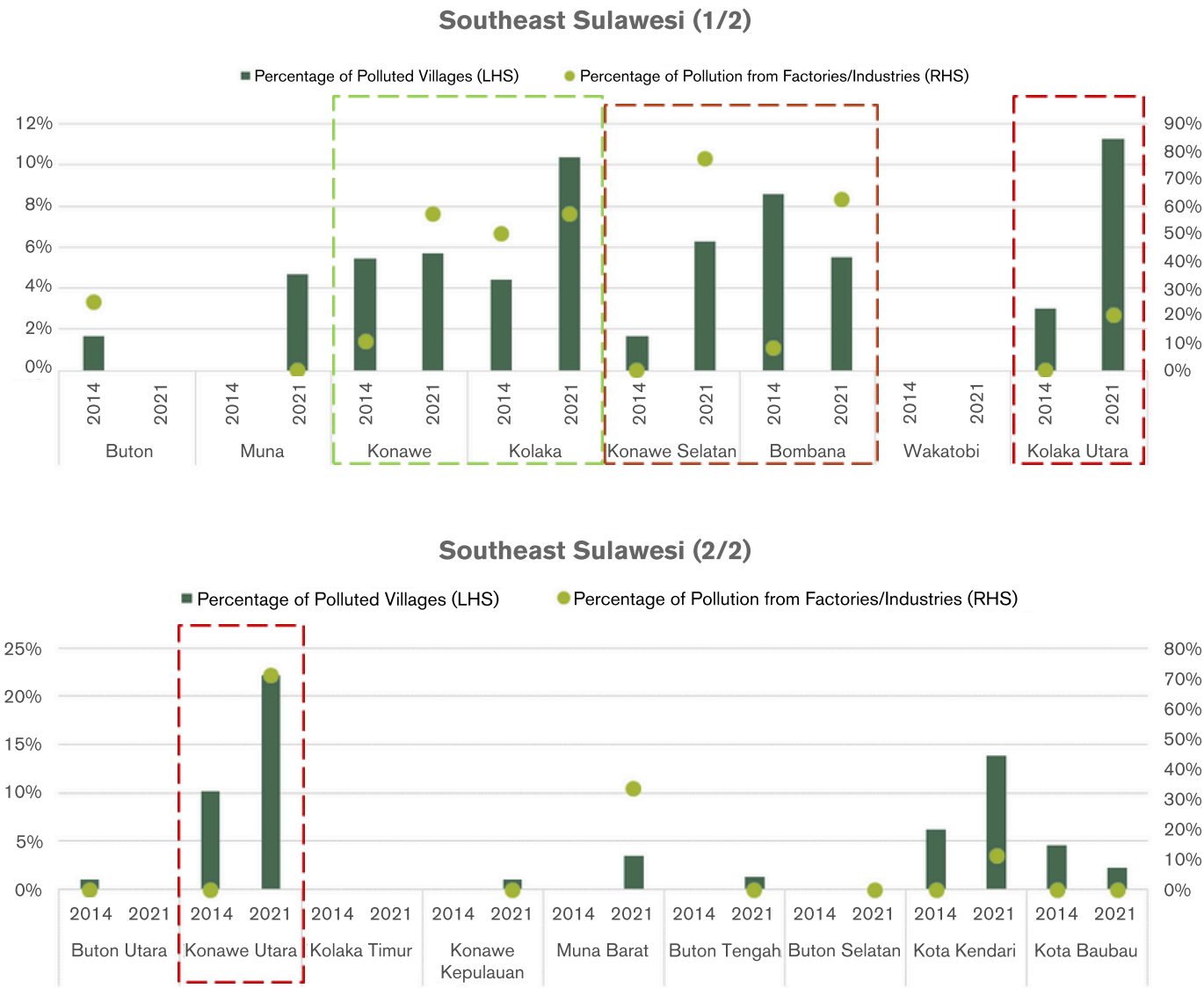


Source: BPS (2024)

When examining the data at the regency level in each province, it is evident that both the percentage of villages affected by water pollution and the percentage of pollution caused by industrial activities have increased in most areas. This trend is particularly prominent in regencies with nickel mines and smelters. Areas outlined with brown dashed rectangles indicate regions with both nickel and smelters, while those outlined in green indicate regions with nickel only. Most of these areas have experienced an increase in both percentages, especially in pollution caused by industrial activities.

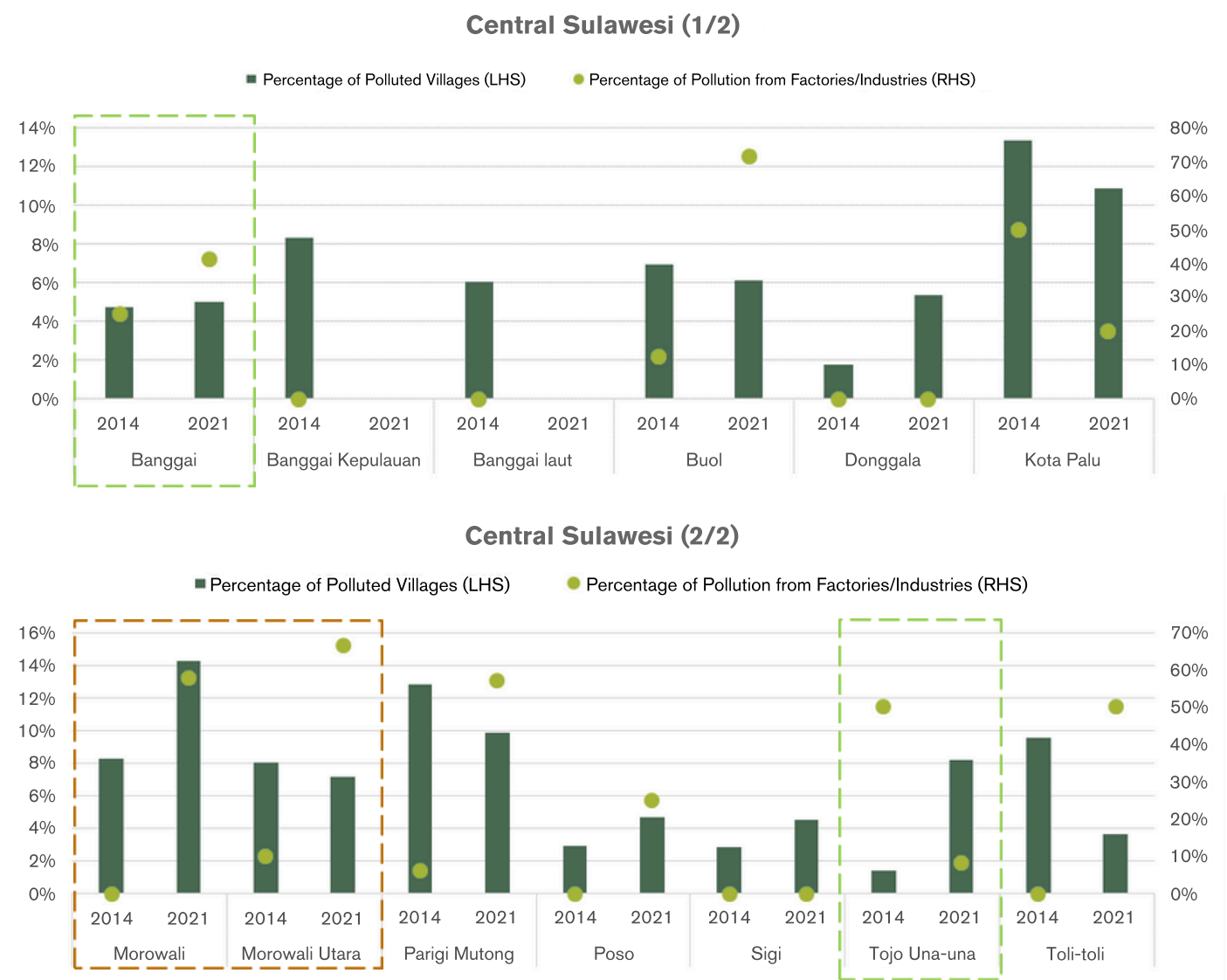
The village of North Konawe, in Southeast Sulawesi, which has both mining and smelting operations, shows the highest increase in the percentage of water-polluted areas and industrial-related pollution. The percentage of polluted areas increased by 12% in 2021, while pollution due to industrial activities rose sharply from 0% in 2014 to 71% in 2021 (Figure 2.25). In Central Sulawesi, at the provincial level, the number of water-polluted villages has slightly decreased, and at the regency/city level, seven areas experienced a decrease. However, only one of these areas is a nickel-producing region. Notably, there was a significant increase in the percentage of villages affected by industrial-related water pollution in Morowali, an industrial hub, rising from 0% in 2014 to 59% in 2021 (Figure 2.26).

**FIGURE 2.25** Villages Affected by Water Pollution (%) and Industrial Water Pollution (%) in Southeast Sulawesi, 2014 & 2021



Source: BPS (2022)

**FIGURE 2.26** Villages Affected by Water Pollution (%) and Industrial Water Pollution (%) in Central Sulawesi, 2014 & 2021

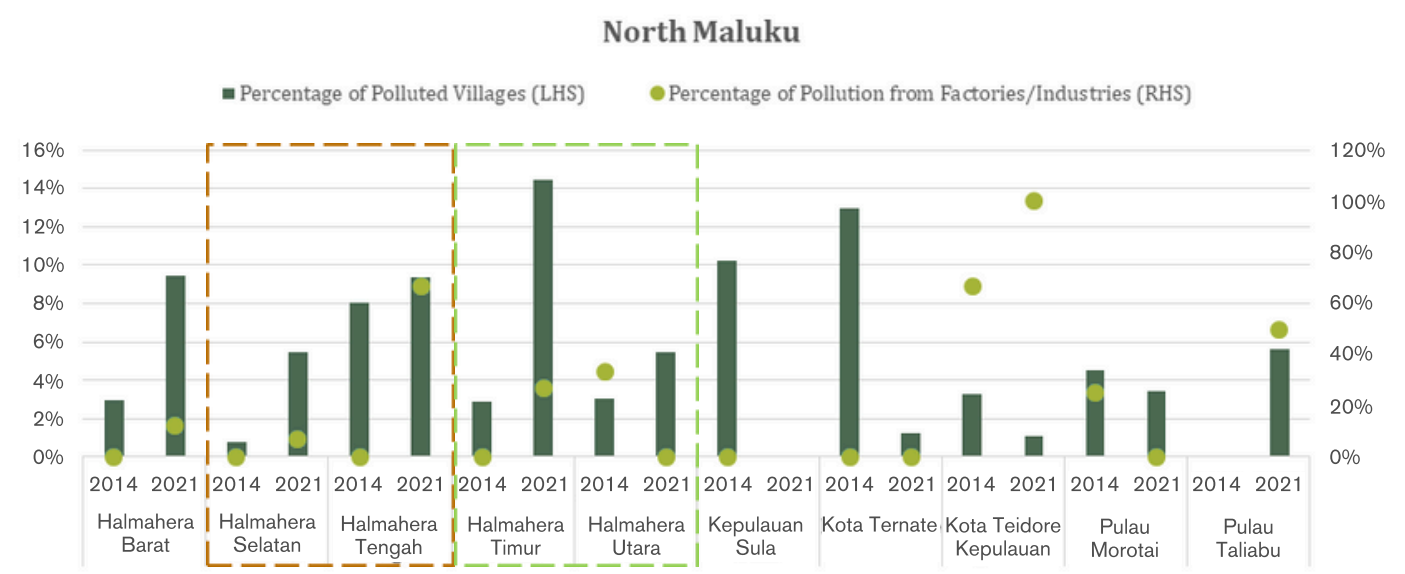


Source: BPS (2022)

In North Maluku, the highest increase in the percentage of water-polluted villages occurred in East Halmahera, while the highest percentage of industrial-related water pollution was in Central Halmahera, jumping from 0% in 2014 to 61% in 2021 due to the development of the Weda Bay Industrial Park (Figure 2.27). This data also shows that many villages, which previously had 0% industrial-related water pollution, experienced a sharp increase in 2021 following the nickel export ban and the aggressive push for downstream processing.

To provide a clearer picture of this water-related pollution, we can refer to reports from think tanks, NGOs, and news that offer more detailed, micro-level insights. Nickel mining and downstream activities in Indonesia have caused widespread water pollution, affecting groundwater, rivers, and marine ecosystems. Field research by the People’s Coalition for Fisheries Justice (Koalisi Rakyat untuk Keadilan Perikanan/KIARA) revealed that nickel mining projects have destroyed vital springs, which were the primary source of drinking water for communities in the highlands of Wawonii Island (ResponsiBank, 2024).

**FIGURE 2.27** Villages Affected by Water Pollution (%) and Industrial Water Pollution (%) in North Maluku, 2014 & 2021



Source: BPS (2022)

On Gebe Island, where mining should ideally be prohibited due to its small size, 20% of the land has been converted into mining concessions, threatening scarce groundwater resources. The island’s fragile hydrological conditions, dominated by limestone formations, make it highly vulnerable to water scarcity (Project Multatuli, 2024).

Rivers in nickel-producing areas have also suffered severe degradation. According to a press release by the Halmahera Studies Forum (Foshal), *Wahana Lingkungan Hidup Indonesia* (Walhi) North Maluku, and Trend Asia, nickel downstream activities have devastated the Akejira and Ake Kobe Rivers in Central Halmahera (Pristiandaru, 2024). Similarly, the Sageyan River in North Weda, Central Halmahera, has shown signs of contamination, with its waters frequently turning reddish-brown even without rainfall. An investigation by Forest Watch Indonesia (FWI) in September 2023 attributed this discoloration to soil erosion caused by deforestation in the upstream watershed area, which falls within mining concessions (FWI, 2023).

Marine ecosystems have been severely impacted by nickel operations, particularly due to the discharge of wastewater from hydrometallurgical processes using High-Pressure Acid Leach (HPAL) technology. This process generates around 32 million m³ of wastewater annually only in IWIP, containing high levels of hexavalent chromium and elevated temperatures, which threaten coral reefs, mangroves, and seagrass beds (AEER, 2024). In North Konawe, the coastal waters of Boedingi Village are now reddish-brown, with coral reefs at depths of 10 meters covered in nickel ore sediment. Similar conditions are observed in Obi Island, North Maluku, where seawater has become dense and reddish, and heavy metal contamination in marine life exceeds safe limits (Tamrin & Aris, 2022). Similarly, in Bahodopi, Morowali, the coastal waters have turned orange-brown and feel warmer due to industrial discharges, making it increasingly difficult for local communities to catch fish (Salman, 2024).

2.3. Result and Analysis of Social Impact

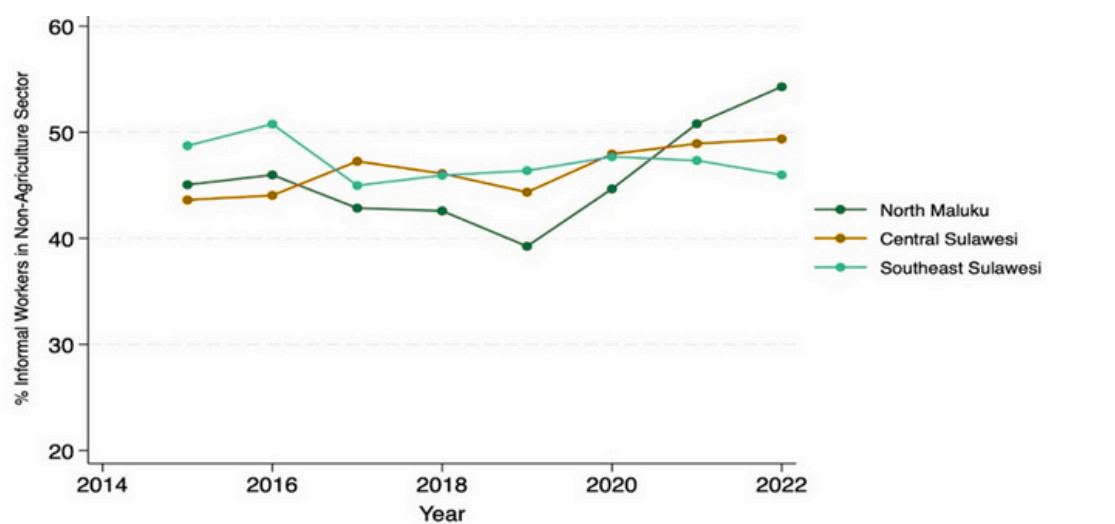
While the nickel industry may bring economic benefits, its social impact remains widely contested. This section will delve into the impacts of nickel industry expansion on informal employment, unemployment, poverty, occurrence of agrarian conflicts, and workers’ welfare.

2.3.1. Informality and Unemployment

Despite claims that nickel mining promotes local prosperity through job creation, its effectiveness in driving a structural shift towards formal sector employment remains uncertain. While preliminary data suggests promising prospects for formal employment within nickel-intensive regions, as indicated by an increase in manufacturing employment (see Figure 2.15), further examination of informal employment data raises questions. Percentage of informal employment in non-agriculture sector—defined as proportion of informal workers over total workers in non-agriculture sector—is used to assess whether manufacturing employment gains in the previous section manifest in the decline in proportion of informal workers.

Nonetheless, this trend did not hold true for Central Sulawesi and North Maluku between 2019 and 2022 (Figure 2.28). Both provinces experienced an increase in the proportion of informal workers in non-agricultural sectors during the period of the second export ban, which is perplexing. In North Maluku, there was a sharp rise in informal employment within the non-agricultural sector following the establishment of IWIP. By 2021 and 2022, over half of the region’s labor force engaged in non-agricultural activities was characterized by informal employment arrangements. Similarly, Central Sulawesi saw a slight increase in the proportion of informal employment in non-agricultural sectors throughout 2019 to 2022.

FIGURE 2.28 Informal Employment in Non-Agriculture Sector, 2014 - 2022

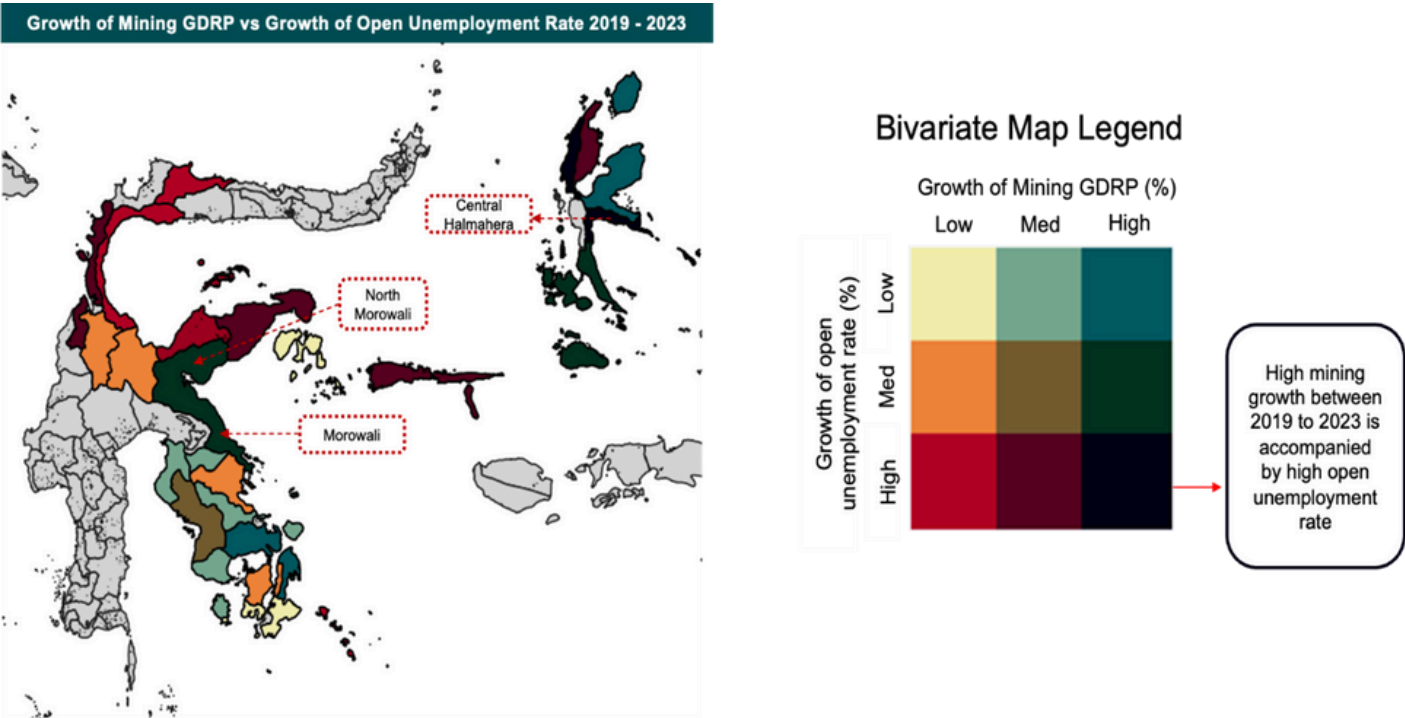


Source: Data from BPS (2024)

The increasing proportion of informal employment in the region presents a paradox, given the introduction of extensive manufacturing activities. Theoretically, the arrival of new manufacturing firms is expected to create formal job opportunities, thereby transitioning the regional employment landscape from one dominated by informality to a structure characterized by more formal employment. This anomaly may be indicative of two potential scenarios. First, a reallocation of local labor from agriculture to non-agricultural sectors could be occurring, driven by externalities associated with the entry of new mining and manufacturing enterprises that adversely affect agricultural productivity, compelling workers to seek alternative employment. Moreover, nickel mining and manufacturing tend to attract significant in-migration, leading to increased demand for goods and services that displaced agricultural workers might supply through informal employment in non-agricultural sectors.

Furthermore, the potential economic benefits derived from the consumption patterns of new residents may incentivize previously economically inactive individuals to enter the labor market as participants in the informal non-agricultural sector. Consequently, it can be inferred that the introduction of nickel mining and manufacturing may concurrently reduce employment in agriculture while fostering novel opportunities within the informal non-agricultural domain. Recent empirical evidence supports this hypothesis; for instance, Tritto (2022) reported that IMIP's industrial complex directly employed approximately 43,000 workers and indirectly supported around 30,000 small service providers and local businesses, which are pivotal in labor-intensive, informal economic activities in Indonesia.

**FIGURE 2.29** Villages Affected by Water Pollution (%) and Industrial Water Pollution (%) in North Maluku, 2014 & 2021



Source: BPS (2024), author's calculation



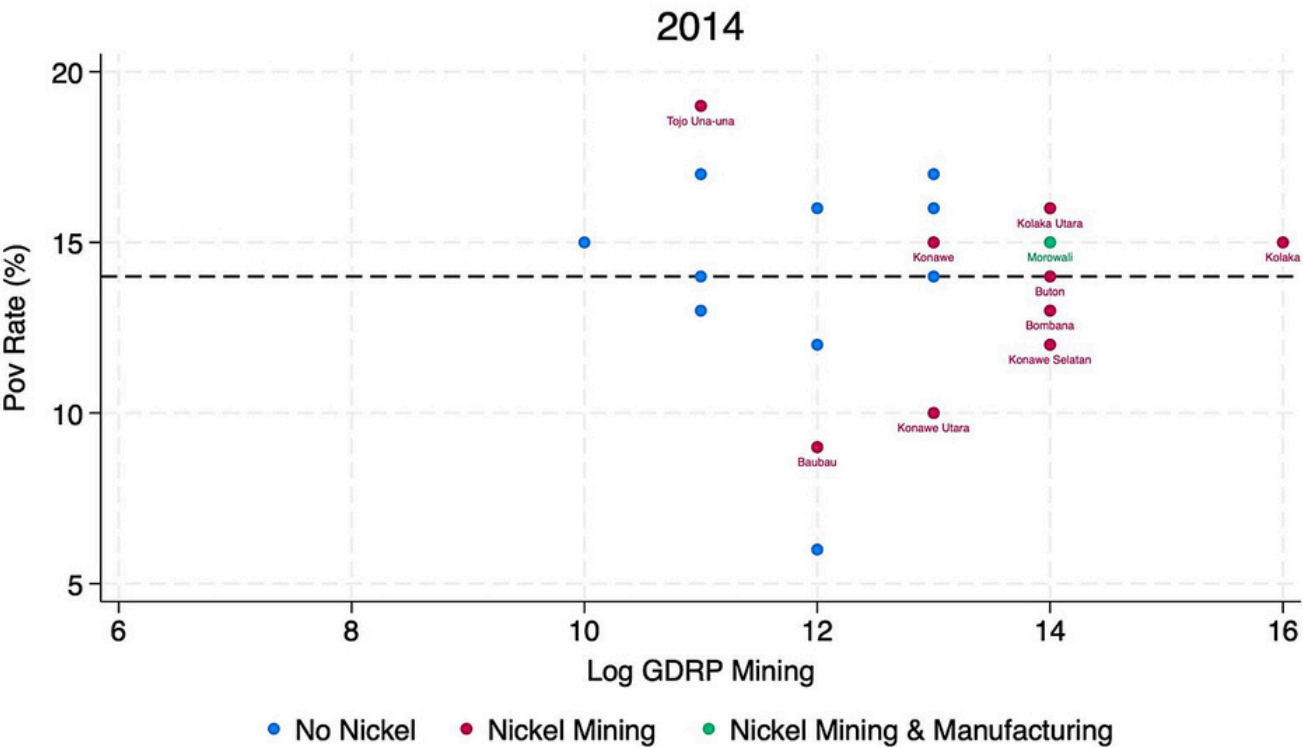
Despite the growth of the mining sector, open unemployment in nickel-rich regions remained high between 2019 and 2023 (Figure 2.29). This situation necessitates further resource-based industrialization that is more labor-intensive and can provide more jobs. The perceived employment opportunities in the mining sector may attract a large influx of workers from other areas, increasing the labor pool and potentially saturating the market. This influx can limit job availability, even as the mining sector expands, resulting in continued high unemployment rates.

### **2.3.2. Poverty**

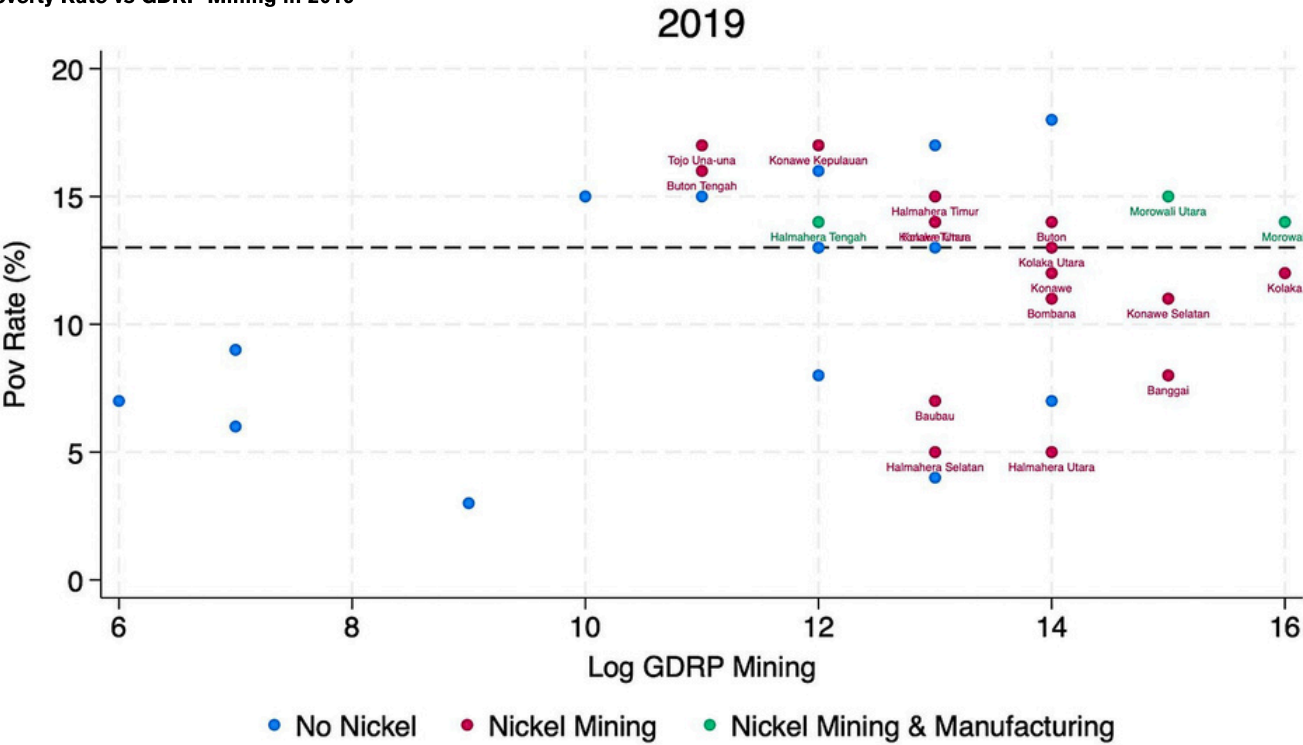
Regions within nickel-rich provinces generally have poverty rate higher than the national average. In 2014 and 2019, there was no significant improvement in median poverty rates, which remained around 14%. Compared to national average of 11% in 2014 and 9% in 2019, the median poverty rate in regions within the nickel-rich provinces are higher (Figure 2.30). This raises questions whether downstream activities following the implementation of nickel export restrictions in 2014 had any effect on poverty alleviation. Specifically, during the initial five years post-regulation, only a limited number of nickel-mining regions experienced a reduction in poverty rates below the median, whereas the majority demonstrated minimal or no improvement. Even in Morowali, a primary hub for nickel manufacturing, the impact was marginal., a primary hub for nickel manufacturing, the impact was marginal.

**FIGURE 2.30** Poverty Rate Post First Nickel Ore Export Ban, 2014 & 2019

Poverty Rate vs GDRP Mining in 2014



Poverty Rate vs GDRP Mining in 2019

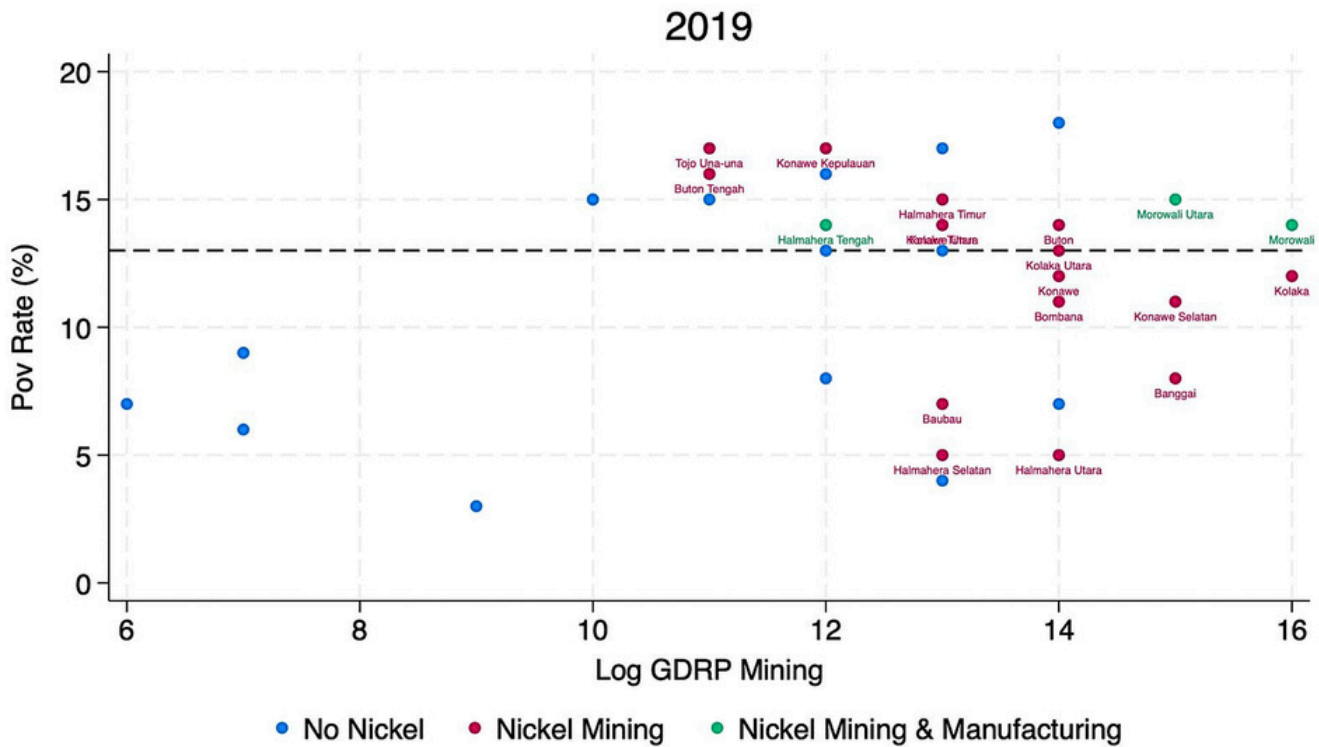


Source: Data from BPS (2024)

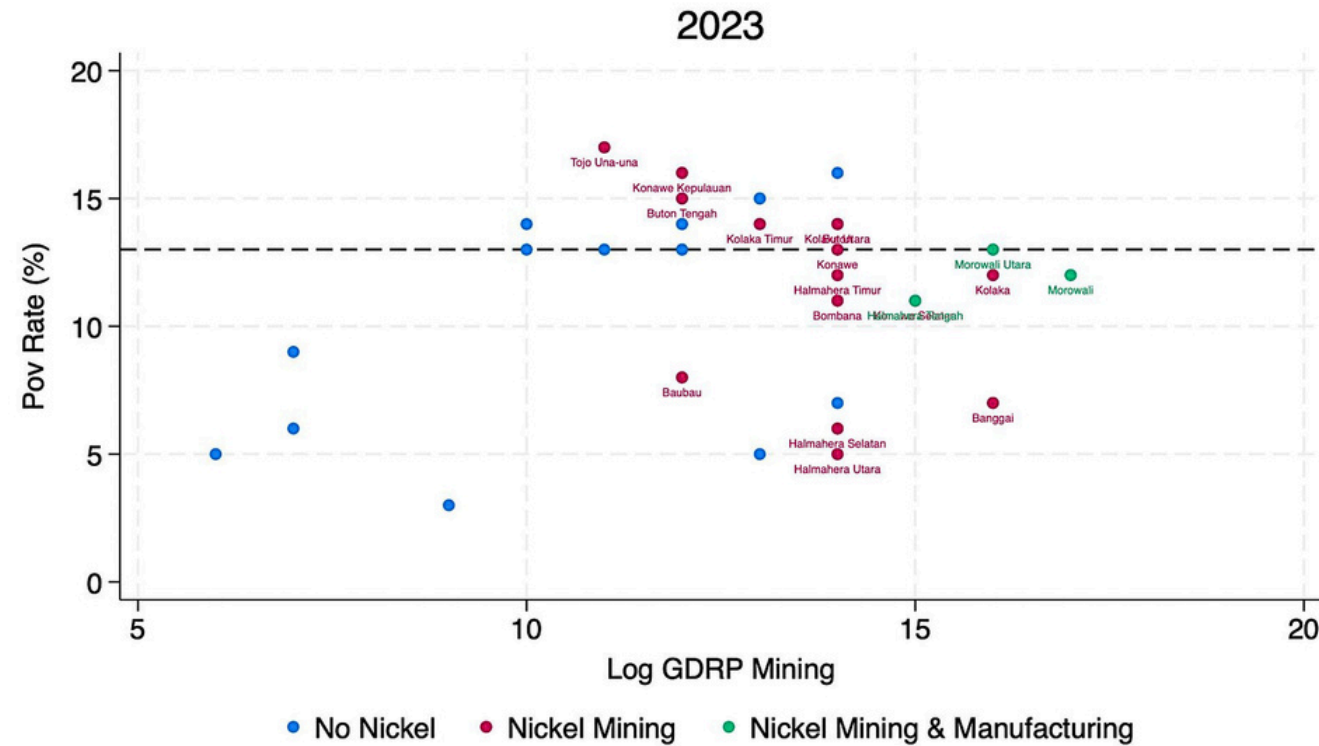
Following the second export ban, initial assessments indicate a more substantive 2023, showing no improvement (Figure 2.31)

**FIGURE 2.31** Poverty Rate Post Second Nickel Ore Export Ban, 2019 & 2023

Poverty Rate vs GDRP Mining in 2019



Poverty Rate vs GDRP Mining in 2023



Source: Data from BPS (2024)

Our findings suggest two principal insights. First, nickel mining in isolation appears insufficient to alleviate poverty; while it provides some local employment effect. Within the first three years of the policy, regions with substantial nickel mining and manufacturing outputs, such as Morowali, North Morowali, and Central Halmahera, experienced a decline in regional median poverty rates. Conversely, areas with limited or no nickel manufacturing activity did not show similar improvements, except for East Halmahera. Despite these changes, the median poverty rates remain higher than the national average of 9% in, its negative externalities on agricultural livelihoods may offset its benefits. Conversely, a proactive nickel manufacturing sector could leverage a multiplier effect on the local economy, thereby contributing more substantially to poverty reduction. Second, the initial export restriction has exerted a greater influence on regions dominated by mining activities compared to those with significant nickel manufacturing, likely because the continued exportation of certain ores maintains incentives for mining output and employment. In contrast, the comprehensive second ban compels domestic manufacturing firms to sustain operations and employment, particularly in regions characterized by the presence of both nickel mining and manufacturing.

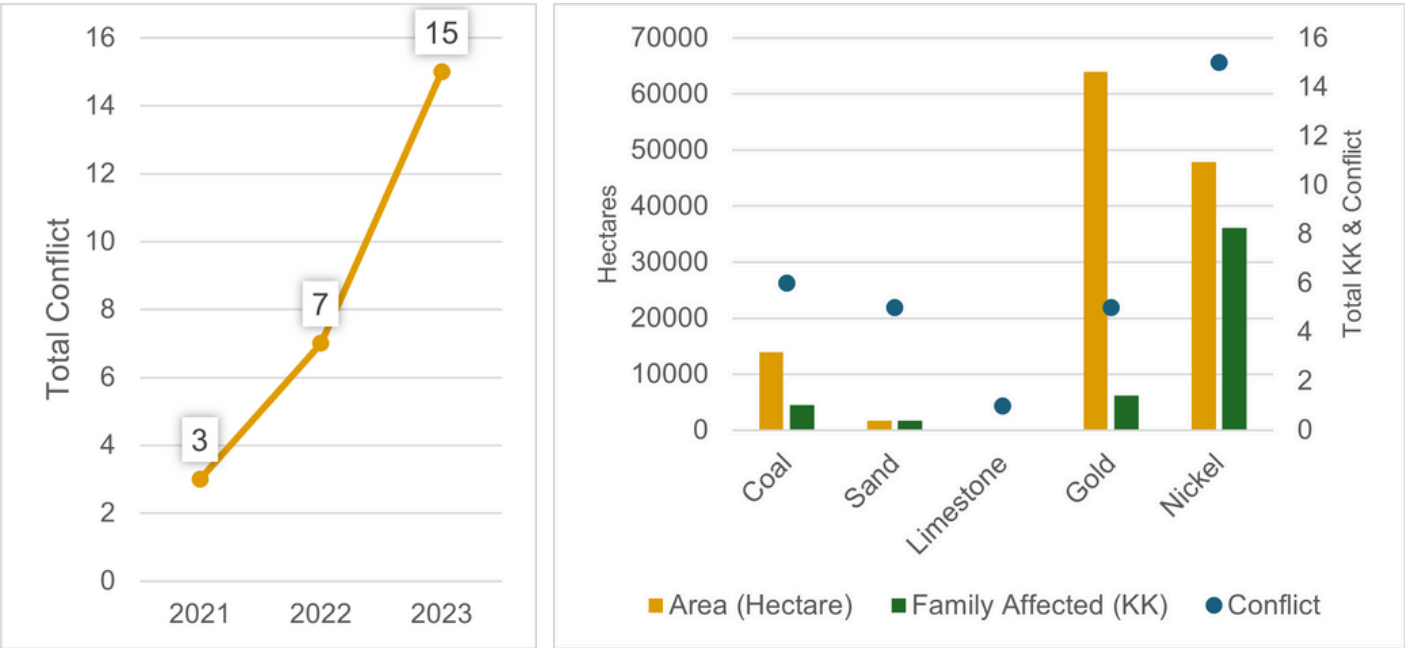
While our findings diverge somewhat from those reported by Mandala (2014), who observed an increase in poverty within nickel-rich provinces, we posit that these differences may be attributable to variations in data granularity. Our analysis leverages district-level data, which offers a more granular perspective on the localized impacts of nickel mining and manufacturing activities on poverty rates, in contrast to the province-level data utilized by Mandala (2014). This distinction may account for the observed differences in findings, and further research could usefully explore the nuances of these approaches to provide a more comprehensive understanding of the relationship between nickel mining and poverty dynamics.

### **2.3.3. Agrarian Conflict**

A significant concern regarding nickel mining and downstream processing in Indonesia is the social costs associated with potentially unethical and unsafe land acquisition and operational practices. Numerous reports indicate that land-grabbing practices infringe upon the rights of Indigenous communities through forced displacement from settlements and farms, or by failing to secure their free, prior, and informed consent (FPIC). A recent report by Climate Rights International (2024) reveals that nickel industry development in Halmahera, North Maluku, has been characterized by inequitable land sales, intimidation of local populations, and compulsory land excavation. The report also documents instances where local communities were not informed of the intended purpose of land acquisition, indicating a lack of compliance with legal requirements mandating FPIC.

In 2023, the Agrarian Reform Consortium (2023) documented a rising trend of agrarian conflicts within nickel extraction areas, marking this sector as a substantial contributor to overall mining-related disputes (Figure 2.32). Specifically, the number of conflicts reported in these regions escalated from 3 in 2021 to 15 in 2023. Moreover, in the same year, conflicts linked to nickel extraction spanned an area of almost 50,000 hectares, impacting 36,137 families. Although the total conflict counts and land area affected may be comparatively lower than those in industries such as palm oil, the significant number of families impacted underscores that nickel concessions frequently occur in proximity to local settlements.

**FIGURE 2.32** Number of Agrarian Conflicts Reported (left hand side) and Agrarian Conflicts in Mining Areas in 2023 (right hand side)



Source: Agrarian Reform Consortium (2023)

The report attributes these conflicts to regulatory modifications that have streamlined the process for mining companies and nickel processing industries to clear land. The classification of nickel mining companies under National Strategic Projects (*Proyek Strategis Nasional/PSN*) status further facilitates expedited land clearance. In combination with inadequate enforcement of responsible land acquisition practices, these factors have fostered contentious land procurement processes that subsequently trigger conflicts between local communities and nickel companies.

2.3.4. Workers' Welfare

Beyond the construction phase, significant concerns have emerged regarding unsafe working conditions within nickel manufacturing companies. Between 2015 and 2023, these companies recorded 93 recurring workplace incidents that resulted in the fatalities of 91 workers and injuries to 152 others (Trend Asia, 2024). Predominant accidents, including toxic gas leaks and exposure to SO<sub>2</sub>, underscore the systemic vulnerabilities in occupational safety practices (Solidar, 2024). Such recurrent failures not only reflect compromised Occupational Safety and Health (OSH) standards but also point to shortcomings in regulatory safety audits, despite persistent demands for improved working conditions.

Moreover, problematic labor arrangements—characterized by inadequate wages, excessive working hours, and arbitrary dismissals—further exacerbate the situation (INKRISPENA, 2022). Trend Asia's 2024 findings indicate that workers are subjected to sub-minimum wage compensation and are provided with insufficient equipment in high-risk work environments. Furthermore, female employees encounter specific discriminatory practices, including bureaucratic impediments to accessing legally mandated menstrual leave, a heightened risk of workplace harassment, and a considerable pay gap compared to their male colleagues.

These issues collectively undermine worker welfare and pose a significant deterrent to prospective investors concerned about reputational damage, as evidenced by the reported withdrawal of two multinational corporations from a multi-billion-dollar nickel-cobalt refinery project, driven by ethical and sustainability apprehensions linked to nickel sourcing in Indonesia (Shennum, 2024). Consequently, these dynamics underscore an urgent need for comprehensive reforms to mitigate the profound social and human costs associated with the nickel mining and downstream industries in Indonesia.

Our preliminary conclusions in this chapter are derived from secondary data and qualitative assessments, suggesting a potentially skewed balance where the economic advantages of these activities are less pronounced than their considerably detrimental environmental and social consequences. While environmental concerns are addressed as a distinct category, we recognize their inherent interconnectedness with both economic and social spheres. The environmental impacts stemming from nickel mining and processing constitute a significant portion of our impact assessment, reflecting both the extensive real-world effects and the comprehensive documentation available.

Regarding economic factors, further investigation is warranted to ascertain the equitable distribution of benefits derived from nickel mining and processing. Our preliminary findings indicate an imbalance, as regional growth predominantly fuels manufacturing and informal employment sectors without fostering a transition towards more formalized employment structures. This observation is particularly salient given the government's frequent reliance on job creation as a primary justification for promoting these activities. While manufacturing employment is used as a proxy for formal employment, future research should aim to disaggregate employment data into formal and informal sectors for a more nuanced analysis.

Given the projected depletion of nickel reserves in Indonesia, the temporal constraints on realizing the full potential of nickel mining and processing are critical. We contend that current practices are not conducive to long-term sustainability, potentially leading to a negative net welfare effect. Consequently, substantive policy reforms are imperative to optimize the benefits derived from nickel mining and processing activities and to ensure a more sustainable trajectory.



## CHAPTER 3

### III. Policy Making

We posit that the adverse consequences arising from nickel mining and processing operations may stem from the erosion of policy safeguards. Consequently, in this chapter, we attempt to establish a connection between the impacts detailed in the Chapter 2 and the policies governing three pertinent dimensions: economic, environmental, and social. It is important to recognize the potential intercorrelation among policies, whether within the same dimension or across different dimensions, as policy measures are often interconnected and can influence one another.

While we acknowledge that our policy mapping might not capture every single policy that has influenced the observed outcomes, our analysis focuses on the policies that are both relevant and exert the most substantial potential impact on the findings presented in Chapter 2. The policy mapping results presented in this chapter will serve as the groundwork for formulating policy recommendations in a subsequent chapter, enabling us to propose informed and targeted adjustments to mitigate negative impacts and promote sustainable practices.

#### 3.1. Economic Policy

##### 3.1.1 Ease of Doing Business

Indonesia, as a developing country, needs to attract more productive investment to expand its economy and advance the technological progress. Several factors, including bureaucratic inefficiency, rigid labor market, and regulatory overlap, contribute to Indonesia's lack of domestic investment. To address this issue, the Indonesian government has formulated an omnibus bill—Job Creation Law or *Undang-undang Cipta Kerja* (Law No. 6 of 2023)—aiming to streamline regulations, reduce the investment barriers, and increase business certainty. This regulation is expected to attract more investment and eventually may create more jobs in Indonesia.

The Job Creation Law (Kemenkomarves, 2023) has introduced several measures to attract more investment. These include improvements in the investments climate and business activities; employment; protection of micro, small, and medium enterprises (MSMEs); support for research and innovation; securing land procurement; regulating economic zones and national strategic project; managing government administration; and ensuring law enforcement. Chapter III of the Law focuses on improving the investment climate and business activities. It discusses the simplification of basic and sectoral business permits and streamlining investment prerequisites, particularly in the mining and manufacturing sectors. While the mining sectors permits cover oil, coal, geothermal, and electricity, the manufacturing sector regulations simplify the rules for the resource-based industries. Chapter VI addresses the ease of doing business, introducing more flexible measures related to immigration, patents, taxation, and market regulation.

In addition to its other provisions, this law also aims to restructure land regulations through the establishment of Indonesia Land Bank Authority. This authority is responsible for the land acquisition, land management, and land distribution for public interest, including the industrial and technological development zones (Tempo, 2021). This initiative, along with the strengthening of land management rights, may help reduce institutional uncertainties and transaction costs in land provision, thereby improving the investment climate and ease of doing business.

Although the Job Creation Law does not thoroughly regulate the ease of doing business in the downstream process, it may also impact the nickel mining and industries. Despite its importance in attracting more investment, the Law poses risks not only to nickel-related industries and communities but also to the national economy. Increased investment inflow in Indonesia raises the risk of overexploitation, which may lead to economic sustainability issues. Besides, this law also may jeopardize the protection and welfare of workers in the nickel-related industries. Considering Indonesia's decentralized system, there may be an emerging principle-agent problem between national and local government, potentially creating inefficiencies in investment coordination and policy implementation.

### **3.1.2 Nickel Ore Export Ban**

Indonesia has historically exported its nickel ore as a raw material. Given its substantial nickel reserves, Indonesia is determined to secure more value-added creation from its nickel resources. In 2014, Indonesia imposed its first export ban on nickel ore based on Law No.4 of 2009 on Mineral and Coal, which was then relaxed in 2017. The government implemented a stronger measure by imposing Law No.3/2020 on Mineral and Coal. This law also generally regulates the requirement for nickel-related industries to prioritize local people in industrial and mining activities, and to establish and fund the community development program. Several regulations (MEMR Regulation No.11/2019; Presidential Regulation No.109/2020; GR No.25/ 2024) have also mandated the investors to build domestic smelters to secure the industrial value-added.

The nickel ore export ban is a bold initiative by the Indonesian government and may be perceived as a resource nationalism. This resource can become Indonesia's comparative advantage to build a green industry ecosystem domestically. Based on Law No.3/2020 on Mineral and Coal, this resource downstream was followed by the bauxite export ban after three years of nickel export ban implementation. However, the progress in bauxite downstream has not been as successful as nickel downstream, due to more limited bauxite downstream products (i.e. aluminum) and higher smelter costs (Muliawati, 2024). In addition, Indonesia has a relatively weaker market power in the global bauxite market compared to nickel, with contributions of 5% and 3,3% to global bauxite production and reserves in 2023, respectively (USGS, 2024). Furthermore, potential bauxite alternatives, including laterites for the aluminum-making process (Dissanayake et al., 2021), may pose challenges to the bauxite industry. Henceforth, Indonesia may consider allocating more resources into the nickel downstream process rather than bauxite.

Despite nearly a decade since the second nickel ore export ban was implemented, the nickel downstream process still faces some challenges. First, the global nickel-related industries, including those in Indonesia, are predominantly controlled by a few countries, resulting in an unfair monopolistic market structure. Second, trade disputes involving strategic products from Indonesia's trading partners, such as the EU's palm oil ban, remain unresolved. Although Indonesia won in the palm oil dispute in the WTO (Baskoro, 2025), the status of the nickel ban dispute is still unclear. Third, Indonesia's nickel downstream process is limited to relatively low value-added stage, such as smelting and steelmaking.

### 3.1.3 Industrial Policies

To achieve higher growth, Indonesia should consider engaging in a higher value-added industry, such as in the EV supply chain. The Indonesian government has formulated several industrial strategies to advance the downstream agenda to higher value-added process, including EV manufacturing, and Local Content Requirements (LCR). Through Presidential Regulation No. 79/2023 (*Peraturan Presiden/PP No.79/2023*), which revises the more ambitious targets set in PR No. 55/2019 (*Peraturan Presiden/PP No.55/2019*), the government has established LCR targets for EV cars at 40% of in 2026, 60% in 2029, and 80% afterward. However, this policy may not be effective if it is not supported by domestic production and infrastructure readiness. Therefore, there should be an alignment between LCR timeline and domestic production capacity. The government should also ensure the provision mechanism of the required infrastructure (e.g. charging stations, renewable energy supply). Additionally, this initiative also requires an in-depth cost-benefit analysis, to determine the cost-effective and most feasible production stage for Indonesia.

Indonesia has also attempted to foster industrialization by formulating regulations for Special Economic Zones (SEZ) in Government Regulation No. 40/2021 (*Peraturan Pemerintah/PP No.40/2021*). The Job Creation Law (Law No. 6/2023) also includes regulations for SEZ (Chapter IX), covering various business activities such as manufacturing, logistics, technological development, tourism, education, health, energy, and others. Based on The Job Creation Law, several incentives offered in the SEZ, including import facilitation, tax facilitation, business permits, land acquisition, and immigration measures. This law also regulates the national strategic project—or widely known as the *Proyek Strategis Nasional* (PSN) in Chapter X. PSN is also regulated in the GR No.42/2021 (*Peraturan Pemerintah No.42/2021*), which emphasizes the institutional mechanism and incentives to provide more certainty for business activities.

### 3.1.4 Revenue Sharing Funds

Resource revenue distribution is crucial to ensure that regional governments benefit from mining and industries activities. Indonesia has established a mechanism to share the resource revenue to regional governments through the Law No. 1/2022 (*Undang-Undang 1/2022*). This law specifically regulates the revenue sharing fund for resource-rich provincial governments and resource-rich cities/regencies-level governments, and their neighboring regions. Currently, the revenue sharing fund for mineral resources consists of land-rent (*iuran tetap*) and royalty (*iuran produksi*). Given the high revenue sharing mechanism to provincial and cities/regencies-level government currently (80%), this privilege should be utilized productively and efficiently. There should be a nickel revenue management in the regional government, considering the dynamic price of commodities and the fact that nickel is a nonrenewable resource.

The role of regional government is crucial in enhancing local welfare, and the utilization of resource revenue is vital to fund development initiatives. The allocation of revenue should be regulated for several purposes. First, it can be used to expand productive sectors, thereby diversifying the nickel-rich regions. This is important to secure high and sustainable growth for the region. Second, resource revenue can be allocated to compensate affected sectors and individuals, as nickel mining and industrial activities may negatively impact certain groups. Third, the use of resource revenue is also important to prepare for future transitions, considering the possibility of resource depletion or substitution by the technological advancements.

## 3.2. Environmental Policy

Indonesia has established environmental safeguard policies to regulate resource extraction and industrial activities. However, recent legal amendments have weakened these protections in an effort to attract investment, particularly in sectors classified under National Strategic Projects (PSN), including nickel mining and smelting. Our assessment indicates that permitting and environmental assessment regulations are at the core of environmental challenges in nickel downstream. Additionally, the relaxation of forest protection laws has exacerbated deforestation and biodiversity loss, while lower standards in emission and pollution control regulations have contributed to air and water pollution. The following analysis provides a detailed examination of Indonesia's legal framework and its implications for the nickel industry.

### 3.2.1 Permission and Environmental Assessment (AMDAL)

The concept of National Strategic Projects (PSN) in Indonesia has become increasingly controversial due to its vague criteria. Law No. 6/2023 on Job Creation fails to clearly define what constitutes "strategic" and "national" in the context of PSN. This unclear definition leading to arbitrary classifications where numerous projects are labeled as PSN without delivering significant public benefits, especially for the people around the project. Currently, 280 projects are designated as PSN, including nine industrial zones related to nickel, with nickel smelter development positioned as a key program.<sup>7</sup> This raises critical concerns, as many PSN initiatives, particularly those driven by the private sector, prioritize economic growth and investment over equitable development and environmental sustainability.

The lack of clear guidelines, limited public involvement, and disregard for constitutional values, sociocultural dynamics, and environmental ethics undermine the legitimacy of PSN as a tool for genuine national progress.

The legal framework governing the acceleration of PSN, particularly as outlined in Government Regulation No. 42/2021 and Presidential Regulation No. 109/2020, provides an array of privileges to PSN projects. These include government guarantees for technically and financially feasible initiatives, streamlined access to financing through Public-Private Partnerships (PPP), and extensive political risk mitigation. While these measures are designed to attract investors, they also create an environment where projects, such as nickel downstream operations, can proceed with minimal scrutiny.

One of the most concerning aspects is the government's expedited permitting process mandated by the government, which significantly reduces the time and rigor typically required for environmental assessments and business licensing. The government strictly applies the principle of administrative priority in resolving legal issues related to PSN as stipulated in Presidential Regulation No. 3/2016 and Presidential Instruction No. 1/2016. In cases where irregularities are suspected in the implementation of PSN, the government overrides the penal *primum remedium* principle, which prioritizes criminal sanctions through the judicial system (Yakup, 2024). Moreover, a legal loophole exists as the government still lacks a mechanism to sanction or address stalled PSN projects or those that have adverse impacts on communities and are deemed unfit to continue.

Government Regulation No. 5/2021 on Risk-Based Business Licensing indicates an over-centralization of business licensing. Although local governments, through the Investment and One-Stop Service Agency (DPMPTSP), also have the authority to grant licenses, this authority is removed for foreign investments, as stated in Article 22. This is particularly concerning for the nickel downstream industry, as approximately 90% of nickel smelting operations are foreign-invested (Kemenko Marves, 2024).

The regulation also centralizes risk assessments at the ministerial level in Article 19, excluding local governments from the decision-making process. Additionally, Article 21 establishes that the norms, standards, procedures, and criteria (NSPK) for risk-based business licensing are determined solely by the central government. This one-size-fits-all approach ignores the geographical and socio-environmental factors that affect the risks associated with specific industries. Moreover, the Law can serve as a shield for businesses, allowing them to proceed with construction even if they have not yet met full compliance with environmental assessments, with regulatory requirements deferred to later stages, as seen in the case of PT KFI (Ardan, 2024).

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7 Industrial Park: Morowali, Konawe, Pulau Obi, Weda Bay, Pomala, Kendari, Motui, Huabao Industry Park, Dahuaxing Industry Park; Smelter Development Program: Construction of eight nickel processing and refining facilities, two integrated with mining operations, and one integrated with both mining and industrial areas as part of the National EV Battery Development Program.



Government Regulation No. 22/2021 on Environmental Protection and Management Implementation introduces several provisions that simplify the environmental licensing process and significantly reduce the effectiveness of Environmental Impact Assessments (AMDAL). In addition, the Omnibus Law along with the GR No.22/2021 reduces the strictness of absolute liability (Sucahyo, 2022). The revision of Article 88 of Law No. 32/2009 in the Omnibus Law changes the liability standard from a “strict liability” framework to a more lenient “fault-based” approach. Article 501(1) in conjunction with Article 500(3)t states that absolute liability must be proven through civil law enforcement if the environmental supervision report shows that the business is not in compliance. In GR No. 22/2021, strict liability only applies if monitoring finds non-compliance, meaning there must be proof of fault. This creates a disorientation in the meaning of strict liability, as it implies that liability requires proof of fault, which contradicts the original principle of strict liability.

Moreover, the Omnibus Law has centralized AMDAL approvals under the Ministry of Environment. For example, in the mining sector, local governments no longer have licensing authority. The AMDAL Review Commission has been replaced by an Environmental Feasibility Assessment Team, which is now formed by the central government, whereas previously, governors, regents, or mayors could also form such teams based on their authority.

The Omnibus Law has removed Article 26(2) of Law No. 32/2009 (UU PPLH), which previously mandated that public participation in the preparation of Environmental Impact Assessments (AMDAL) be conducted with transparent and comprehensive information provided before project implementation. Additionally, Article 26(4), which granted the public the right to object to AMDAL documents, has also been eliminated. While other legal principles, such as the general legal right to file a lawsuit if harmed, Law No. 14/2008 on Public Information Disclosure, or the authority of environmental NGOs to file lawsuits under Article 92 of Law No.32/2009, may still apply, this removal signals a further weakening of environmental accountability and limits community oversight on potentially harmful projects.

This reduction of transparency is alarming, especially since the public disclosure has been a persistent issue for the Commission on Information, particularly in the mining sector. Between 2017 and 2022, 88% of the 25 information disputes related to AMDAL, Environmental Approval, or IUP involved requests for mining sector AMDAL documents,<sup>8</sup> underscoring the growing lack of transparency in environmental governance. Many institutions still consider licensing documents and contracts in the natural resources and environmental sectors as non-public and exempt from transparency regulations (Thea, 2023).

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8 ICEL Presentation at the Focused Group Discussion on "Towards a More Responsible Nickel Supply Chain to Support Indonesia's Energy Transition" on Wednesday, February 12, 2025.



The transparency of mining permits and environmental assessments in Indonesia remains highly limited. Although platforms like MODI (Minerba One Data Indonesia) provide access to IUP data and AMDALnet lists environmental approvals, the available information is minimal—only displaying permit numbers, company names, and validity periods. The public cannot access copies of permits or precise site coordinates, limiting oversight and accountability. Additionally, mitigation action plans in AMDAL reports are poorly understood, with no clear mechanisms for evaluation or enforcement. Institutional capacity to monitor compliance remains weak, and grievance mechanisms are either inaccessible or ineffective.

The Job Creation Law has weakened criminal accountability for environmental violations by prioritizing administrative sanctions over criminal penalties. Key provisions in Law No. 32/2009 (Articles 98, 99, 102, 104, 109, and 110) have been amended to replace imprisonment with fines, making it more difficult to prosecute corporate offenses. By emphasizing fines over imprisonment, enforcement disproportionately favors businesses with financial resources, allowing wealthier violators to avoid prosecution while only those unable to pay face legal consequences. This misinterpretation of the *ultimum remedium* principle weakens deterrence, as penalties are based on financial capacity rather than environmental harm, effectively reducing corporate responsibility (FH UI, 2024).

The enactment of Law No. 3/2020 on the Amendment to the Mineral and Coal Mining Law has centralized permitting authority and environmental assessments under the central government, similar to the Omnibus Law and GR No. 22/2021. This shift means that local governments no longer have jurisdiction to take action if communities are harmed by mining companies. Alarming, communities opposing mining activities could face criminal charges with fines of up to IDR100 million, further suppressing grassroots resistance, as outlined in Article 162. This stands in stark contrast to Article 66 of Law No. 32/2009, which protects the public fighting for environmental rights from criminalization.

The Mineral and Coal Mining Law No. 3/2020 facilitates the extension of operational permits for all forms of licenses, including Contract of Work (*Kontrak Kerja/KK*), Mining Business License (*Izin Usaha Pertambangan/IUP*), and Special Mining Business License (*Izin Usaha Pertambangan Khusus/IUPK*). In the previous Mineral and Coal Mining Law (*Undang-Undang Mineral dan Batubara/UU Minerba*), permit extensions were stated with the clause "may be extended", which has now been replaced with "guaranteed" in the revised law. This change can be seen in Articles 47, 83, 169, 169A, and 169B. According to Article 83, the extension period is even longer if the operation involves an integrated metallic mineral production facility with processing and/or refining facilities, with a guaranteed extension of 30 years for the operational period, and an additional 10 years for each subsequent extension. Furthermore, Article 83 removes the previous maximum size of IUPK areas for the operational phase, which was previously set at 25,000 hectares, and now it is undefined as it is determined based on the Minister's evaluation. Additionally, according to Article 96b, mining companies are given the choice between reclamation or post-mining restoration, with minimal enforcement, allowing environmental degradation to persist.

Despite weakened permitting processes and relaxed environmental assessments, corruption remains rampant in Indonesia's nickel sector. Research by the China Global South Project (Timmerman, 2024) found that nearly 30% of identified nickel projects—including both mining and processing—were linked to corruption or illegal mining. Among 332 documented projects, 98 were associated with corruption, with 90 cases specifically tied to corrupt mining operations, underscoring the failure of regulatory safeguards and the prevalence of unlawful practices in the industry.

The government's pro-mining stance is also evident in its policies granting concessions on small islands. Although Law No. 1/2014 on Coastal Zone and Small Islands Management explicitly bans mining activities on islands smaller than 2,000 square kilometers, its enforcement has proven to be severely lacking. In March 2024, the Constitutional Court through Decision No.35/PUU-XXI/2023 upheld this legal protection by rejecting a mining company's attempt to challenge the ban. Despite this definitive ruling, mining operations continue unchecked on several small islands, such as Kabaena (873 sq km), Wawonii (715 sq km), and Gebe (224 sq km), leading to significant environmental destruction and jeopardizing the livelihoods of local communities (Bhawono, 2024). The situation is particularly alarming in North Maluku, where 84 islands have effectively been privatized, granting corporations control over land and natural resources in clear violation of existing laws (Grahadyarini, 2024).

### 3.2.2. Forest Protection

The enactment of Law No. 6/2023 on Job Creation and its derivative regulations has significantly weakened forest protection, accelerating deforestation, especially due to nickel downstream activities. Under Law No. 41/1999 (Article 38), the use of forest areas for mining required a borrow-to-use permit granted by the Minister, and if the impact was significant, widespread, and strategically valuable, approval from the DPR (Indonesian Parliament) was also required. This is further reflected in Article 19, which states that any changes to forest designation that have significant impacts and strategic value require DPR approval. This process ensured legislative oversight for changes to forest designation or function. However, Article 36 of the Omnibus Law amends these two articles, removing the DPR approval clause. Additionally, while Article 19 of Law No. 41/1999 stated that changes to forest designation could be determined by the government, the Omnibus Law limits this authority exclusively to the central government.

Under Law No. 41/1999 on Forestry, the use of forest areas for protected forests is categorized into three types under Article 26, while production forests are categorized into six types under Article 28, based on their function and purpose. However, the Omnibus Law simplifies this into a single "business permit", which streamlines business processes but risks oversimplifying the permits for various activities with different objectives. Furthermore, this business permit expands its beneficiaries to include not only individuals and cooperatives, as stipulated in Articles 27 and 29 of Law No. 41/1999, but also state-owned enterprises (*Badan Usaha Milik Negara/BUMN*), regional state-owned enterprises (*Badan Usaha Milik Daerah/BUMD*), and private companies. This deregulation favors businesses with capital and influence, making it significantly easier for large corporations to obtain forest concessions.

The term "Indonesian Private Companies" has also been removed, replacing it with "Private Companies", which allows foreign companies easier access to obtain concessions.

Previously under Article 18(2) of Law No. 41/1999 on Forestry, there is a requirement to maintain at least 30% forest cover within watersheds and islands served as an essential environmental safeguard. However, this provision was removed under the Omnibus Law, making it easy to open forests near watersheds. Furthermore, the inclusion of Articles 110A and 110B in the Omnibus Law effectively provides amnesty for companies that have illegally operated in forest areas for years. By prioritizing administrative sanctions over criminal penalties, the law undermines the deterrent effect of environmental regulations, encouraging corporations to exploit forests.

Article 84 of GR No. 23/2021 further undermines forest protection by allowing the conversion of protected and conservation forests, particularly for National Strategic Projects (PSN). These projects only requires proposals from the Minister or the head of the relevant institution, bypassing the involvement of local governments or forest area managers. This regulation permits projects to proceed even in provinces already experiencing forest deficits, where existing forest cover is insufficient to support ecological balance, watershed functions, and biodiversity conservation (Maqoma, 2021).

The permitting process has also been drastically accelerated, reducing approval times from 104 to just 34 days, enabling project activities to begin almost immediately after permit issuance. This rushed approval system severely limits environmental assessments, increasing the risks of deforestation and biodiversity loss while prioritizing economic expansion over sustainability. Under Article 94 of GR No. 23/2021, project developers are even exempt from paying non-tax state revenues (PNBP) for forest area usage and compensation fees, reducing the financial obligations typically associated with exploiting forest resources.

Article 10 of GR No. 43/2021 (Article 10) further weakens spatial planning by permitting PSN licensing and concessions to proceed even if they conflict with district and provincial land-use plans. This legal loophole has led to numerous cases of nickel concessions overlapping with critical forest areas, as companies no longer need to comply with regional spatial planning regulations. Furthermore, Article 19 of PR No. 58/2017 allows PSN locations that do not align with district/city spatial plans to receive a spatial alignment recommendation from the Minister of Agrarian Affairs and Spatial Planning (*Agraria dan Tata Ruang*/ATR), without requiring revisions at the local level.

These legal changes collectively create a regulatory environment that prioritize economic development and industrial expansion over forest conservation. This explains why nickel downstream practices in Indonesia have become a significant driver of deforestation, as companies exploit these legal gaps to secure concessions in previously protected forest areas with minimal environmental accountability.

### 3.2.3. Emission and Pollution Control

Despite Indonesia's commitment to achieving Net Zero emissions, the country continues to permit the construction of new coal-fired captive power plants, particularly to support downstream industries such as nickel processing. This policy is outlined in Article 3(4)b of Presidential Regulation (PR) No. 112/2022 on the Acceleration of Renewable Energy Development for Electricity Supply, which permits the establishment of new coal power plants if deemed essential for industries that add value to natural resources. Although these plants are required to reduce greenhouse gas emissions by at least 35% over the next 10 years compared to the average emissions of coal power plants in Indonesia in 2021, and must operate until 2050, this approach is still inadequate to address the negative environmental impact and meet Nationally Determined Contributions (NDCs). The number of captive coal power plants is not limited, and their rapid increase, as discussed in Chapter 2, exacerbates the issue. Furthermore, the emission reduction target is based on emission intensity, rather than focusing on reducing the absolute amount of emissions. Additionally, while Indonesia's NDCs aim to be achieved five years later, only about 35% of the target has been met by 2023 (MEMR, 2024).

Adding to the concern, Indonesia's emission standards for thermal power plants are considerably weaker compared to international benchmarks. According to Minister of Environment and Forestry Regulation No. 15/2019, the emission limits for coal-fired power plants are set at 200 mg/Nm<sup>3</sup> for Sulfur Dioxide (SO<sub>2</sub>), 200 mg/Nm<sup>3</sup> for Nitrogen Oxides (NO<sub>x</sub>), and 50 mg/Nm<sup>3</sup> for Particulate Matter 2.5 (PM 2.5). In contrast, countries like China enforce far stricter standards with limits of 35 mg/Nm<sup>3</sup> for SO<sub>2</sub>, 50 mg/Nm<sup>3</sup> for NO<sub>x</sub>, and 10 mg/Nm<sup>3</sup> for PM. Similarly, Japan and South Korea maintain stringent emission thresholds, both capping SO<sub>2</sub> at 60 mg/Nm<sup>3</sup> and 50 mg/Nm<sup>3</sup>, NO<sub>x</sub> at 50 mg/Nm<sup>3</sup>, and PM at 10 mg/Nm<sup>3</sup>.

When comparing Indonesia's water pollution control regulations, particularly GR No. 22/2021, with the Initiative for Responsible Mining Assurance (IRMA) standards, significant differences emerge in terms of stringency and environmental safeguards. Indonesia's regulation sets relatively lenient thresholds for key parameters, such as pH (6–9), Total Suspended Solids (TSS) at 50 mg/L, and Chemical Oxygen Demand (COD) at 100 mg/L. In contrast, IRMA imposes stricter limits to better protect aquatic ecosystems, requiring a narrower pH range of 6.5–8.5, TSS capped at 30 mg/L, and COD limited to 50 mg/L. For heavy metals critical to mining activities, IRMA sets far stricter limits—nickel at 0.1 mg/L compared to Indonesia's 1 mg/L, and mercury at 0.0005 mg/L versus Indonesia's 0.002 mg/L—reflecting a stronger focus on mitigating mining-related environmental risks.

IRMA also mandates the monitoring of additional pollutants not covered under Indonesian law, such as trivalent chromium, selenium, vanadium, and persistent organic pollutants (POPs), alongside tighter controls on organic pollution, with Biochemical Oxygen Demand (BOD) capped at 20 mg/L compared to Indonesia's 50 mg/L. Unlike Indonesia's fixed thresholds, IRMA requires continuous monitoring and adaptive management based on real-time environmental conditions, particularly for parameters like ammonia, ensuring more responsive environmental protection. For a more detailed comparison, refer to the report "*Dilema Halmahera di Tengah Industri Nikel*" available [here](#).

Indonesia's ability to control air and water pollution has been further compromised by the removal of Article 102 of Law No. 32/2009 in the Omnibus Law. This change eliminates criminal sanctions for individuals handling hazardous and toxic waste (B3) without permits, thereby reducing accountability for environmental violations. Additionally, while marine water quality monitoring is crucial for pollution control, it remains limited to selected areas such as Jakarta Bay, Semarang Bay, and Benoa Bay (AEER, 2023). Meanwhile, nickel processing industrial zones in Central Sulawesi, Southeast Sulawesi, and North Maluku, despite their significant pollution risks, are not included in the priority monitoring areas. To mitigate these environmental threats, KLHK must extend its monitoring efforts to cover these high-risk regions.

Deep Sea Tailing Placement (DSTP) is widely recognized as a high-risk method for managing mining waste due to its severe environmental impacts. Although it is a cost-effective solution, only 0.6% of the world's mines utilized DSTP as of 2015 (Naturvernforbundet, 2022), reflecting a global trend toward banning this practice. Countries like the United States, Canada, and Australia have implemented strict bans or heavy restrictions on DSTP because of its irreversible damage to marine ecosystems, including sediment smothering, disruption of benthic habitats, and bioaccumulation of heavy metals in marine food chains. These countries have shifted to land-based tailings management systems, which, while more expensive, offer greater environmental safeguards.

In Indonesia, the risks associated with DSTP are significantly amplified by the country's archipelagic geography and seismic activity, as it lies at the convergence of three major tectonic plates. This makes the region highly vulnerable to earthquakes and underwater landslides, increasing the likelihood of tailing pipeline failures and leaks. Additionally, the fluctuating thermocline layers in Indonesia's tropical seas can cause tailings to resurface, spreading contaminants to shallower waters and impacting coral reefs, mangroves, and fishery resources. The lack of comprehensive scientific studies on Indonesia's deep-sea environments further exacerbates these risks, limiting the country's ability to predict or mitigate potential ecological disasters (Tapilatu, 2023).

Despite the significant risks, Indonesia continues to permit deep-sea tailings placement (DSTP) under GR No. 22/2021 and MoEF Regulation No. 6/2021. While Article 200 of the MoEF Regulation and Article 246 of the GR impose restrictions, including the requirement for technical approval and adherence to designated disposal locations, they also mandate the avoidance of sensitive marine areas such as coral reefs, mangrove forests, and fish spawning zones. Article 395 of the GR No.22/2021 outlines specific criteria for dumping sites, including the need for them to be located at a permanent thermocline on the ocean floor, or, if absent, at depths of 100 meters or more, with topography that directs tailings to 200-meter depths and no upwelling phenomena. Additionally, Article 391 imposes strict requirements for dumping B3 waste, mandating neutralization or detoxification of hazardous substances before disposal.



Despite the regulatory measures, the precautionary principle, which emphasizes prevention over remediation, suggests that such waste disposal practices should not be permitted at all (AEER, 2023). Research by WALHI North Maluku indicates that seawater quality around Obi Island exceeds government-set thresholds, with murky waters and fish contaminated with heavy metals. Similarly, DSTP activities at PT Indonesia Morowali Industrial Park (IMIP) have caused severe environmental degradation, including reduced fish catches and damage to coral reefs, seagrass beds, and mangrove forests (Rachmawati, 2024).

### **3.3. Social Policy**

The rapid expansion of Indonesia's nickel downstream industry has brought significant economic transformations, but it also raises critical social challenges. As large-scale investments drive sectoral growth, the social dynamics of workforce composition, labor rights, and local community engagement come under scrutiny. This section explores key social policies shaping the nickel industry, particularly foreign worker permits, and evaluates their implications on employment, local communities, and Indonesia's long-term industrial development.

#### **3.3.1. Foreign Workers Permit**

The lack of impact of nickel manufacturing on the increase in proportion of formal employment indicates a need for a detailed examination of corporate workforce recruitment strategies. Given the considerable investments by Chinese companies in the nickel downstream sector, it is unsurprising that these firms experience a significant influx of foreign labor from China to operate within nickel industrial areas predominantly managed by Chinese entities. In the context of IMIP, it is estimated that these companies employed 5,000 foreign workers from China as of 2020 (Tritto, 2022). The preference for foreign workers may be attributed to potential language barriers between management and employees, as well as the superior operational expertise they offer.

Recognizing the imperative to expedite operational processes, the Indonesian Government enacted reforms to streamline the permitting procedure for foreign labor employment through the implementation of Presidential Regulation (PR) No. 20/2018 on Foreign Worker Utilization, which is later replaced by GR No. 34/2021 on Foreign Worker Utilization, a direct implementing regulation of the Omnibus Law on Job Creation in 2021, further revised in 2023. Under the new regulation, the government removes the obligation for companies planning to employ foreign workers to obtain Permit to Employ Foreign Workers (Izin Mempekerjakan Tenaga Kerja Asing/IMTA) and instead only requires companies to obtain legalization of Foreign Workers Utilization Plan (Rencana Penggunaan Tenaga Kerja Asing/RPTKA) (Zainuddin et al., 2023).



Zainuddin et al. (2023) highlight a key difference in the legal standing of IMTA and RPTKA, noting that the RPTKA is part of the administrative procedure while the IMTA is the resulting output. This distinction suggests a potentially weaker regulatory structure for the RPTKA, which could diminish the effectiveness of screening out lower-skilled foreign workers. The IMTA also functions as a crucial document for inspecting foreign worker entry. Therefore, the absence of mandatory IMTA could lead to an increased number of undocumented foreign workers, potentially intensifying competition for local employment and worsening existing social tensions between companies and local communities.

### **3.3.2. Land Dispute**

Developing domestic nickel downstream industry facilities has always been the government's priority. In the Presidential Decree (PD) No. 109/2020 on Acceleration of National Strategic Project Implementation, the government listed the development of nickel related facilities such as IMIP, IWIP, and smelters as national strategic projects (PSN), which refers to projects with high potential for job creation and welfare improvement. PSNs are provided ample facilities by the state, which according to the Government Regulation (GR) No.42/2021, includes government facilitation on the resolution of land acquisition dispute with the affected landowners. However, given the relatively high number of reported land dispute, it is likely that the implementation is inconsistent with the mandate.

Further complicating the issue, GR No. 64/2021 on Land Banking strengthens state control over land by adopting the *domein verklaring* principle, which expands the government's authority over land ownership (Berenschot, 2020). This regulation has been criticized for easing foreign corporate access to land and potentially enabling land grabbing under the pretext of public interest land procurement (WALHI, 2023). These policies risk exacerbating land conflicts, displacing local communities, and undermining equitable land governance.

### **3.3.3. Occupational Safety and Health Standard**

The state requires both nickel mining and processing industry to adhere to the MoEMR Regulation No. 26/2018 regarding safety provision, which stipulates the requirement for the companies to provide safety equipment for workers, establish clear safety governance and perform regular audit. However, as explained in the previous section, the persistent incidence of workplace hazards, coupled with multiple reports of substandard work environments within nickel manufacturing companies, suggests a critical deficiency in the implementation of Occupational Safety and Health (OSH) standards, as well as inadequate regulatory enforcement and monitoring mechanisms.

In this chapter, we conclude that Indonesia is currently at a crossroads in its efforts to enhance the investment climate. The introduction of the Omnibus Law and National Strategic Projects (PSN) were intended to stimulate investment, economic growth, and job creation. However, our analysis reveals that these policies have significantly reshaped the landscape, often resulting in adverse social and economic consequences in the pursuit of investment and value-added policies. While some regulations showed promise, their implementation has not met expectations.

Our desk research reveals that the environmental impact has emerged as the most notable concern compared to other aspects. Consequently, environmental safeguards occupy a substantial portion of our policy mapping. This highlights a crucial issue with the implementation of these environmental safeguards and raises an important question: Does improving the investment climate necessarily require weakening environmental protections? Are there alternative strategies that can enhance the investment climate without compromising these essential safeguards?

As the demand for sustainable commodities continues to grow, the importance of strong safeguards becomes even more crucial. The global shift towards energy transition highlights the need for robust protections to ensure long-term sustainable growth in Indonesia's mining and industrial sectors. The challenge lies in finding a balanced approach that fosters investment while safeguarding both the environment and society.

CHAPTER 4

IV. POLICY RECOMMENDATIONS

In the previous chapters, this policy paper has assessed the impact of nickel downstream activities from economic, environmental, and social aspects, complemented by the policy mapping. This chapter integrates all findings to synthesize relevant policy recommendations, drawing on good practices from other mineral-rich countries. It is important to note that the recommendations are limited to the research scope, including the geographical selection (Southeast Sulawesi, Central Sulawesi, and North Maluku), and data availability. These recommendations are still based on early findings and require further examination and assessment by relevant stakeholders—such as government, industries, academics, NGOs, etc.

4.1. Good Practices in Transition Minerals Policy Implementation

To develop effective policies that address the triple bottom line challenges of the nickel industry and ensure a more sustainable downstream process, Indonesia must learn from the experiences of other countries. These case studies serve as benchmarks, enabling Indonesia to adopt best practices tailored to the specific challenges and bottlenecks within its current regulatory framework. The following examples highlight good policy practices from various countries concerning the economic, social, and environmental aspects of the nickel industry.

| Sector      | Good Practice                                                                                                                                                                                                                                                                                                                                                                                       |
|-------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Economic    | <b>United States:</b> The US Infrastructure Investment and Jobs Act provides funding to support energy and minerals research in collaboration with academic partners as part of the strategies for developing domestic supply chains for critical minerals.                                                                                                                                         |
|             | <b>Australia:</b> Australia is tackling skill shortages in its mining industry with an apprenticeship program that supports 10,000 apprentices. This program helps cover living expenses while providing industry mentors, peer support, and networking opportunities. Each apprentice can receive up to USD10,000 in assistance, helping them build the skills needed for the clean energy sector. |
| Environment | <b>Canada:</b> The Canadian Industry Program for Energy Conservation (CIPEC) funds mining companies that meet high standards in energy efficiency and emissions reduction under the Towards Sustainable Mining (TSM) Climate Change Protocol, covering up to 50% of eligible costs <sup>9</sup> up (max USD40,000).                                                                                 |

9 Expenses related to implementing energy management best practices that qualify for funding under Natural Resources Canada's (NRCan) ecoENERGY efficiency program.

| Sector        | Good Practice                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|---------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|               | <p><b>Mexico:</b> In 2023, Mexico reformed its Mining Law, National Waters Law, LGEEPA, and LGPGIR, tightening regulations. Mining concessions were reduced from 50 to 30 years, require prior permits, and no longer grant automatic land access. Companies must pay 5% of net profits to affected communities and face criminal liability for violations. Mining in protected areas is banned, industrial water use can be restricted, and concession holders now bear permanent responsibility for mining waste.</p>                                                              |
|               | <p><b>South Korea:</b> The Waste Control Act regulates hazardous waste management, disposal, and recycling to protect health and the environment. Proposed amendments promote battery recycling by recognizing black mass as a recyclable resource, reducing regulatory burdens and aligning with EU and US practices.</p>                                                                                                                                                                                                                                                           |
| <b>Social</b> | <p><b>Germany:</b> The Supply Chain Act in Germany obligates companies to implement due diligence by identifying, assessing and prioritizing risks in their supply chain. Companies are required to publish policy statements, set out measures to minimize violations in human rights, and establish channels for complaints. The legislation seeks to foster fair and responsible production by obligating companies to uphold robust occupational safety and health standards, guarantee fair wages and the right to unionize, and prevent unjust land acquisition practices.</p> |

## 4.2. Policy Recommendation

The recommendations provided are based on observed impacts and an analysis of existing policies. They encompass a range of actions, including the revocation of outdated regulations, the introduction of new ones, and the enforcement of current policies. Below are some recommendations derived from our analysis.

### 4.2.1. Policy Recommendation

- Accelerating Resource-based Industrialization toward Higher Value-Added Products:** Indonesia's nickel is predominantly used in the steel-making production, which generates substantial value-added benefits. However, Indonesia could explore diversifying its nickel-based industries by leveraging its nickel resources to develop an electric vehicle (EV) supply chain ecosystem. This approach offers higher value-added creation and incorporates more advanced technologies. Nevertheless, it is essential to consider the feasibility of this diversification in the short, medium, and long-term.

- **Improving the Domestic Benefit from Downstream Activities:** At this stage, Indonesia still relies on foreign investment to support its industrialization. However, it should reassess the benefits gained from the downstream activities by establishing comprehensive international cooperation with other nickel-rich countries. This cooperation should aim to create a fair market structure and ensure transparency in international dealings. Additionally, Indonesia needs to implement risk mitigation measures, such as trade policies, to anticipate and address any unfair market structures or trade disputes with other trading partners.
- **Preparing for Future Transition:** Nickel-rich regions should establish a transition framework to mitigate various risks, including the potential effects of nickel depletion and the emergence of nickel substitutes. At the regional level, most of nickel-rich provinces heavily rely on nickel-related sectors. To address the risk of future economic downturns due to declining nickel production, it is crucial to prepare for economic diversification in these nickel-rich regions.
- **Regulating the Spending from Resource Revenue:** The role of local fiscal spending is crucial for enhancing local economic welfare. In addition, an equitable distribution of benefits from nickel downstream can contribute to the poverty reduction. Additional local fiscal revenue gained from the nickel downstream should be allocated to the following priorities: (1) expanding productive sectors; (2) compensating the impacted sectors and people; and (3) preparing the future transitions.
- **Establishing a Local Investment Fund from Nickel-related Sectors:** The government should proportionately manage economic rents to ensure effective distribution. To optimize the use of resource revenue, sound revenue management is essential. This is crucial to mitigate commodity price volatility, and the depletion of nickel reserves. This can be achieved through the establishment of a local investment fund, which may be allocated in the form of long-term intergenerational savings and/or productive spending.
- **Conducting a Strategic Coordination:** Considering its encompassing sectors and stakeholders, downstream strategies should be well managed and coordinated. A strong coordination between national and local government is required in managing the downstream activities. Additionally, coherence in the policy-making process is essential, such as aligning local content requirement (LCR) requirements, infrastructure readiness and domestic production capacity).

#### 4.2.2. Environmental Aspect

##### 4.2.2.1. Permit and Environmental Assessment

- **Limiting Mining Concession Duration and Ensure Transparent Bidding Processes:** Mexico reduced its mining concession period from 50 to 30 years and implemented an open bidding system with strict environmental, social, and labor compliance requirements. Indonesia should adopt a similar approach by reducing the duration of Mining Business Licenses with the possibility of extension based on robust evaluation.

The government must also prevent corruption and bribery to ensure that permits are granted to those who adhere to best practices. Additionally, stricter regulations should be enforced to prevent IUP grantees from subcontracting mining activities to third parties that undermine ESG compliance through cost-cutting and poor oversight.

- **Strengthening Environmental Impact Assessment (AMDAL)**  
**Procedures:** AMDAL assessments must be conducted before project approvals to prevent environmental damage. A more rigorous screening process should ensure that only companies with sustainable practices receive permits. Public participation should be reinstated, allowing communities and civil society organizations to challenge AMDAL approvals and increase transparency. Besides, penalties for individuals or communities opposing environmentally harmful mining activities should be removed, ensuring legal protection for environmental defenders. Empowering local governments, as the authorities closest to mineral operations, is essential for strengthening oversight and ensuring compliance with environmental regulations. Enhancing the capacity and quality of supervisory bodies is also critical, given the increasing number of permits issued that require effective monitoring.
- **Providing Transparent Data Related to Permit and Environmental Assessment:** To enhance transparency and accountability in mining permits and environmental assessments, Indonesia must implement Right-to-Know laws to AMDAL and permit data, including project coordinates, environmental impact assessments, and mitigation plans. Moreover, some AMDAL indicators—such as emission tracking—should be accessible to the public in real time. Strengthening real-time monitoring systems, public review mechanisms, and transparent grievance platforms will ensure regulatory compliance and empower communities to hold nickel operations accountable.
- **Revaluating National Strategic Projects (PSN):** All projects under PSN status should undergo a comprehensive review, and the special privileges granted to PSN projects should be assessed and revised. Stricter evaluation and monitoring of nickel-related projects must be enforced to ensure compliance with environmental regulations and sustainability objectives.
- **Collaborating with the Chinese Government to Implement Green BRI:** Given China's significant role in Indonesia's nickel industry under the Belt and Road Initiative (BRI), the Indonesian government should work with China to ensure the adoption of stricter environmental standards for BRI-funded mining and industrial projects. This collaboration should focus on implementing green technologies, minimizing deforestation, and reducing industrial pollution to align nickel development with global sustainability commitments.



#### 4.2.2.2. Forest Protection

- **Strengthening Law Enforcement and End Corporate Amnesty:** Articles 110A and 110B of the Omnibus Law, which effectively provide amnesty for illegal deforestation, must be repealed. Instead, stricter criminal penalties and liability measures should be reintroduced to deter corporate misconduct. Companies engaged in illegal deforestation should face severe financial penalties, operational suspensions, and legal prosecution to ensure accountability.
- **Restoring Local Government Oversight and Public Participation:** Current laws exclude local governments and forest managers from decision-making, allowing uncontrolled expansion of nickel concessions. To address this, the authority over forest management should be decentralized. This would give local governments, indigenous communities, and environmental organizations a greater role in monitoring and decision-making. Strengthening local participation will enhance accountability and improve enforcement of forest conservation policies.
- **Reinstating and Strengthening Legal Limits on Forest Conversion:** The 30% forest cover rule for watersheds and islands, which was removed under Law No. 6/2023 on Job Creation, must be reinstated and strictly enforced to prevent large-scale deforestation. Additionally, land conversion of protected forests and High Carbon Stock (HCS) areas must be explicitly prohibited, even for National Strategic Projects (PSN), to ensure that ecological integrity is maintained.
- **Enforcing Reclamation and Post-Mining Restoration:** Although nickel mining currently does not have abandoned sites, we can learn from the coal mining practices. Stronger enforcement mechanisms must be implemented to ensure that mining companies fulfill their reclamation and post-mining restoration obligations. Harsher penalties should be imposed on companies that fail to rehabilitate deforested areas, making ecosystem restoration a mandatory and enforceable condition of mining permits.

#### 4.2.2.3. Emission and Pollution Control

- **Promoting the Use of Cleaner Energy for Smelters:** The Indonesian government should mandate and incentivize the transition to cleaner energy sources for nickel smelting operations to reduce reliance on coal-fired captive power plants (PLTU Captive) and minimize carbon emissions from the nickel industry. Shifting energy sources, though not fully renewable, is a practical and achievable solution. For example, the use of gas and LNG can serve as a cleaner alternative to coal, significantly reducing carbon emissions. Besides, implementing Carbon Capture, Utilization, and Storage (CCUS) technologies can help mitigate emissions from existing fossil fuel-based operations. In the long term, hydrogen energy offers a promising zero-emission solution. While these alternatives may not be fully renewable, they represent a critical step toward reducing the environmental impact of smelting operations while ensuring energy reliability.

- Banning Deep Sea Tailing Placement (DSTP) and Ocean Dumping of Mining Waste:** Despite Government Regulation No. 22/2021 setting specific requirements for DSTP, leakages and violations of water quality standards have repeatedly occurred in practice, threatening marine ecosystems and local livelihoods. Numerous cases of pipeline leaks and contamination exceeding legal limits have been recorded, proving the risks of DSTP. Given that most countries have already banned this practice and following the precautionary principle, Indonesia should prohibit DSTP entirely to prevent further environmental damage. Government can focus on enhancing the practice of tailing management by confining tailings in tailings storage facilities (TSFs) in the form of dams. Solid-liquid recovery of mine tailings is an excellent approach to improve mining operations' sustainability and circularity and ensure cleaner environments.
- Strengthening and Enforcing Environmental Standards:** Indonesia must not only tighten emission and water quality standards but also strengthen monitoring and enforcement to ensure compliance. Increasing regulatory limits will be ineffective without consistent oversight, regular inspections, and preventative measures. To improve regulatory effectiveness, Indonesia's pollution standards should be aligned with stricter global benchmarks, such as IRMA and other environmentally friendly countries standards. Strengthening enforcement mechanisms will ensure that companies are held accountable for environmental violations, protecting both ecosystems and public health.
- Enhancing Enforcement and Transparency in PROPER:** The voluntary nature of PROPER allows non-participating companies to avoid oversight and sanctions, diminishing weakening its effectiveness as an environmental compliance tool. Moreover, the lack of enforcement means that companies found in violation often face no consequences, while limited transparency prevents public access to information on black-rated companies. To improve PROPER participation should be mandatory for high-risk industries, ensuring consistent monitoring and accountability. KLHK must also enhance transparency by publicly disclosing company ratings and violations, enabling greater corporate responsibility and public oversight.

#### 4.2.3. Social Aspect

- Preventing Local Job Displacement by Regulating Low-Skilled Foreign Workers:** To mitigate the risk of local job displacement caused by the influx of low-skilled foreign workers, it is essential to introduce additional provisions alongside the RPTKA (Foreign Workers Utilization Plan). These measures could include setting quotas for low-skilled foreign workers and implementing, monitoring and evaluation mechanisms to ensure compliance and effectiveness. By reducing social friction, this restriction may also indirectly encourage firms to prioritize hiring higher skilled workers, thereby fostering knowledge spillover.

- **Enhancing Land Dispute Resolution Transparency:** To improve the state's role in resolving land acquisition disputes between nickel manufacturing facilities listed as PSNs and affected landowners, it is recommended to utilize a publicly accessible dashboard. This dashboard would provide real-time information on ongoing conflicts, their resolution status, the number of households affected, and outcomes, thereby bolstering accountability.
- **Enforcing Occupational Safety and Health (OSH) Standards:** To incentivize compliance with proper OSH standards in nickel mining and manufacturing companies, clear enforcement mechanisms should be enacted. These mechanisms can include impartial investigations of potential violations and the imposition of escalating penalties in cases of suspected non-compliance. Besides, the government should consider limiting companies' image-improvement activities, particularly when there is evidence of OSH violations or attempts to cover workplace accidents.

Based on the impact and policy mapping gap analysis on the nickel downstream, this policy paper formulates a set of recommendations from the economic, social, and environmental perspectives. These recommendations are not exhaustive, as this paper focuses only on the most pressing and relevant issues. The policy recommendations are presented in three aspects separately, yet cross-cutting issues among all of three sectors were identified in the impact analysis and policy mapping, necessitating the formation of a comprehensive set of policy reform.

Considering the intersection of economic, social, and environmental aspects, the comprehensive set of policy reforms must address each objective without creating trade-offs with other aspects. Long-term reforms should be institutionalized to ensure sustainable growth in the nickel-rich regions. Given the existing harm to environmental and social aspects, coupled with the current suboptimal economic value-added creation from nickel downstream, it is necessary to initiate policy reforms immediately before Indonesia enters its nickel depletion phase. Further research avenues can involve efforts to enhance transparency and accountability in nickel-related mining and manufacturing industries, supported by more robust data. In addition, current studies on nickel downstream are still incomplete; therefore, further studies can be extended to encompass many other aspects.

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## Balancing Act: Complementary Strategies for Indonesia’s Sustainable Nickel Downstream Industry